

# **Rewiring Immune Regulation Through Acute Exercise**

**A Systematic Review and Meta-Analysis on TH1 and TH2 Cell Responses -  
Documentation**

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# Background

This site documents the complete analysis pipeline behind our systematic review and meta-analysis of T-helper-cell responses to acute exercise. For each outcome - **Th1**, **Th2**, and the **Th1/Th2 ratio** - the chapters reproduce the random-effects meta-analysis across three post-exercise time windows (immediate, ~2 h, and ~17 h), together with outlier and influence diagnostics, small-study/publication-bias checks, and moderator and meta-regression analyses. Study quality is appraised with ROBINS-I, and the underlying effect-size computations are documented throughout. The raw data and the full analysis script are available below for reproducibility.

## Downloads

| Resource                           | File                                        |
|------------------------------------|---------------------------------------------|
| Raw data - effect-size computation | <code>effect_computation_Th1Th2.xlsx</code> |
| Analysis script                    | <code>meta_Th1Th2.R</code>                  |
| Summary Formulae                   | <code>formulae.pdf</code>                   |



# 1. Summary of Formulae

## 1.1. Introduction

Only formulas that were actually used in the calculations are highlighted by placing them in a box to improve readability and to make the key equations easy to identify. Other formulas are included for completeness and conceptual clarity.

### Notation

Per study and time point

- $n$ : sample size
- $\tilde{x}, \tilde{y}$ : estimated means (approximated from published summary statistics)
- $\tilde{s}_x, \tilde{s}_y$ : estimated standard deviations (approximated from published summary statistics)
- $m$ : median
- $q_1, q_3$ : 1st and 3rd quartile
- $a$ : minimum,  $b$ : maximum
- $\Phi^{-1}$ : inverse standard normal cumulative distribution function (e.g., R: `qnorm`)
- $\rho$ : Correlation between Th1 and Th2 measurements

## 1.2. Data Extraction and Computation Strategy

### 1.2.1. Overview

This analysis requires extracting or computing the mean Th1/Th2 ratio and its associated standard deviation for each study and time point. However, the included studies reported their descriptive statistics in heterogeneous formats. To enable uniform computation of effect sizes, we developed a systematic approach:

1. **Classify how each study reported data** (see Section 1.2.2)
2. **Estimate means and SDs** when not directly reported (see Section 1.3)
3. **Compute the Th1/Th2 ratio** and its variance (see Section 1.4)
4. **Calculate standardized effect sizes** with small-sample correction (see Section 1.5)

The **computation table** (see Supplementary Materials) documents which approach (A1, A2, C1, C2, or C3) was applied to each study, providing full methodological transparency.

### 1.2.2. Data Reporting Types

Studies fell into two categories based on how Th1 and Th2 values were reported:

**A1: Summary Statistics Directly Reported** - Article reported means and standard deviations (or sufficient summary statistics to estimate them) - No access to individual participant data

## 1. Summary of Formulae

**A2: Raw Data Available** - Individual participant-level data were available (either directly from the article or upon request from authors) - We computed means, SDs, and participant-level correlations directly from the raw data

### 1.2.3. Summary Statistics Estimation Scenarios

When articles reported summary statistics other than means and SDs, we estimated means and SDs using sample-size-adjusted formulas (Wan et al. 2014) depending on which statistics were available:

**Table 1.1.:** Summary Statistics Estimation Scenarios and Corresponding Formulas

| Scenario | Available Data      | $\bar{x}_{estimated}$ | $s_{estimated}$ |
|----------|---------------------|-----------------------|-----------------|
| C1       | Min, Max, Med., $n$ | (1)                   | (2)             |
| C2       | Q1, Med., Q3, $n$   | (3)                   | (4)             |
| C3       | Mean, SE, $n$       | Direct                | (5)             |

## 1.3. Estimating mean and SD from median/IQR/range

Table 1.1 above summarizes the estimation scenarios.

The detailed formulas for each scenario are presented below, following the methodology of Wan et al. (2014).

### 1.3.1. Scenario C1: Min + Max + Median + $n$ ( $a, m, b, n$ )

$$\bar{x} \approx \frac{a + 2m + b}{4} \quad (1)$$

$$s \approx \frac{b - a}{2\Phi^{-1}\left(\frac{n-0.375}{n+0.25}\right)} \quad (2)$$

### 1.3.2. Scenario C2: Median + IQR + $n$ ( $q_1, m, q_3, n$ )

$$\bar{x} \approx \frac{q_1 + m + q_3}{3} \quad (3)$$

$$s \approx \frac{q_3 - q_1}{2\Phi^{-1}\left(\frac{0.75n-0.125}{n+0.25}\right)} \quad (4)$$

### 1.3.3. Scenario C3: Mean, SE

$$s = SE\sqrt{n} \quad (5)$$

## 1.4. Mean and Variance of a Ratio

Because several studies did not report raw data, the Th1/Th2 ratio could not be calculated directly. For these studies, we first approximated the mean and standard deviation from the published summary statistics (see Section 1.3). Using the estimated mean and SD, we then computed the Th1/Th2 ratio for studies that did not publish the ratio but provided sufficient information to derive it.

### 1.4.1. Calculation of the mean ratio

Let  $\tilde{x}_R$  denote the mean Th1/Th2 ratio, let  $\tilde{x}_{\text{Th1}}$  and  $\tilde{y}_{\text{Th2}}$  denote the means of Th1 and Th2, respectively, obtained either by estimation from published summary statistics or by calculation using raw data.

$$\tilde{x}_R = \frac{\tilde{x}_{\text{Th1}}}{\tilde{y}_{\text{Th2}}} \quad (\tilde{y}_{\text{Th2}} \neq 0) \quad (6)$$

### 1.4.2. Calculation of the variance (and standard deviation) of a ratio

It is important to understand that Th1 and Th2 are two dependent variables. Taking this into account, we have used the following formula, published by Van Kempen and Van Vliet (2000):

Their Formula 7 approximates the variance of a ratio using a first-order Taylor expansion (Van Kempen and Van Vliet 2000, 301).

$$\text{var}\left(\frac{\tilde{x}}{\tilde{y}}\right) \approx \frac{\text{Var}(\tilde{x})}{\tilde{y}^2} + \frac{\tilde{x}^2 \text{Var}(\tilde{y})}{\tilde{y}^4} - 2 \frac{\tilde{x}}{\tilde{y}^3} \text{cov}(\tilde{x}, \tilde{y}) \quad (7)$$

Since  $\text{var}() = s^2$  and  $\text{cov}(\tilde{x}, \tilde{y}) = \rho s_{\tilde{x}} s_{\tilde{y}}$  (by rearranging) this can be written as:

$$\text{var}\left(\frac{\tilde{x}}{\tilde{y}}\right) \approx \frac{s_{\tilde{x}}^2}{\tilde{y}^2} + \frac{\tilde{x}^2 s_{\tilde{y}}^2}{\tilde{y}^4} - 2 \frac{\tilde{x} \rho s_{\tilde{x}} s_{\tilde{y}}}{\tilde{y}^3} \quad (8)$$

which can be rewritten as:

$$\text{var}\left(\frac{\tilde{x}}{\tilde{y}}\right) \approx \left(\frac{s_{\tilde{x}}}{\tilde{y}}\right)^2 + \left(\frac{\tilde{x} s_{\tilde{y}}}{\tilde{y}^2}\right)^2 - 2 \rho \frac{\tilde{x} s_{\tilde{x}} s_{\tilde{y}}}{\tilde{y}^3} \quad (9)$$

The corresponding standard deviation is obtained by

$$\boxed{\begin{aligned} SD\left(\frac{\tilde{x}}{\tilde{y}}\right) &\approx \sqrt{\left(\frac{s_{\tilde{x}}}{\tilde{y}}\right)^2 + \left(\frac{\tilde{x} s_{\tilde{y}}}{\tilde{y}^2}\right)^2 - 2 \rho \frac{\tilde{x} s_{\tilde{x}} s_{\tilde{y}}}{\tilde{y}^3}} \\ \rho &= 0.5 \end{aligned}} \quad (1.1)$$

## 1.5. Effect Size Calculation

### 1.5.1. Unstandardized mean difference

- $\tilde{x}_{\text{pre}}, s_{\text{pre}}$
- $\tilde{x}_{\text{post}}, s_{\text{post}}$
- $r$ : Pre-Post correlation (outcome-specific)
- $n$ : is the number of pairs

*Note: A fixed pre-post correlation of  $r = 0.5$  was applied uniformly across all effect size calculations.*

Mean Change

$$D = \tilde{x}_{\text{post}} - \tilde{x}_{\text{pre}} \quad (11)$$

(Borenstein 2013, 22)

Standard deviation of  $D$

$$SD_{\text{diff}} = \sqrt{s_{\text{pre}}^2 + s_{\text{post}}^2 - 2r s_{\text{pre}} s_{\text{post}}} \quad (12)$$

(Borenstein 2013, 24)

Variance of  $D$

$$V_D = \frac{SD_{\text{diff}}^2}{n} \quad (13)$$

(Borenstein 2013, 24)

Standard error of  $D$

$$SE_D = \sqrt{V_D} \quad (14)$$

(Borenstein 2013, 24)

Within-subject SD

$$S_{\text{within}} = \frac{SD_{\text{diff}}}{\sqrt{2(1-r)}} \quad (15)$$

(Borenstein 2013, 29)

### 1.5.2. Standardized mean difference

$$d = \frac{D}{S_{\text{within}}} \quad (16)$$

(Borenstein 2013, 29)

Variance and standard error of  $d$

$$V_d = \left( \frac{1}{n} + \frac{d^2}{2n} \right) 2(1-r) \quad (17)$$

(Borenstein 2013, 29)

$$SE_d = \sqrt{V_d} \quad (18)$$

(Borenstein 2013, 29)

### 1.5.3. Small-sample correction (Hedges' $g$ )

Degrees of freedom for pre-post measures ( $df$ ):

$$df = n - 1 \quad (19)$$

(Borenstein 2013, 29)

Correction factor ( $J$ )

$$J = 1 - \frac{3}{4df - 1} \quad (20)$$

(Borenstein 2013, 27)

Hedges'  $g$ :

$$g = J d \quad (21)$$

(Borenstein 2013, 27)

Variance and SE of  $g$

$$V_g = J^2 V_d \quad (22)$$

(Borenstein 2013, 27)

$$SE_g = \sqrt{V_g} \quad (23)$$

(Borenstein 2013, 27)

## 1.6. References



## 2. Workspace Setup

All chapters load a shared setup file (`_common.R`) that installs/loads the required packages, sets a consistent global `ggplot2` theme, defines the leave-one-out outlier-detection functions, and exposes small helpers for the standard funnel, forest, and bubble plots used throughout. This keeps every analysis chapter self-contained and reproducible regardless of render order.

### 2.1. Packages

### 2.2. Outlier Detection Functions

**Custom functions — required for all outlier analyses below.**

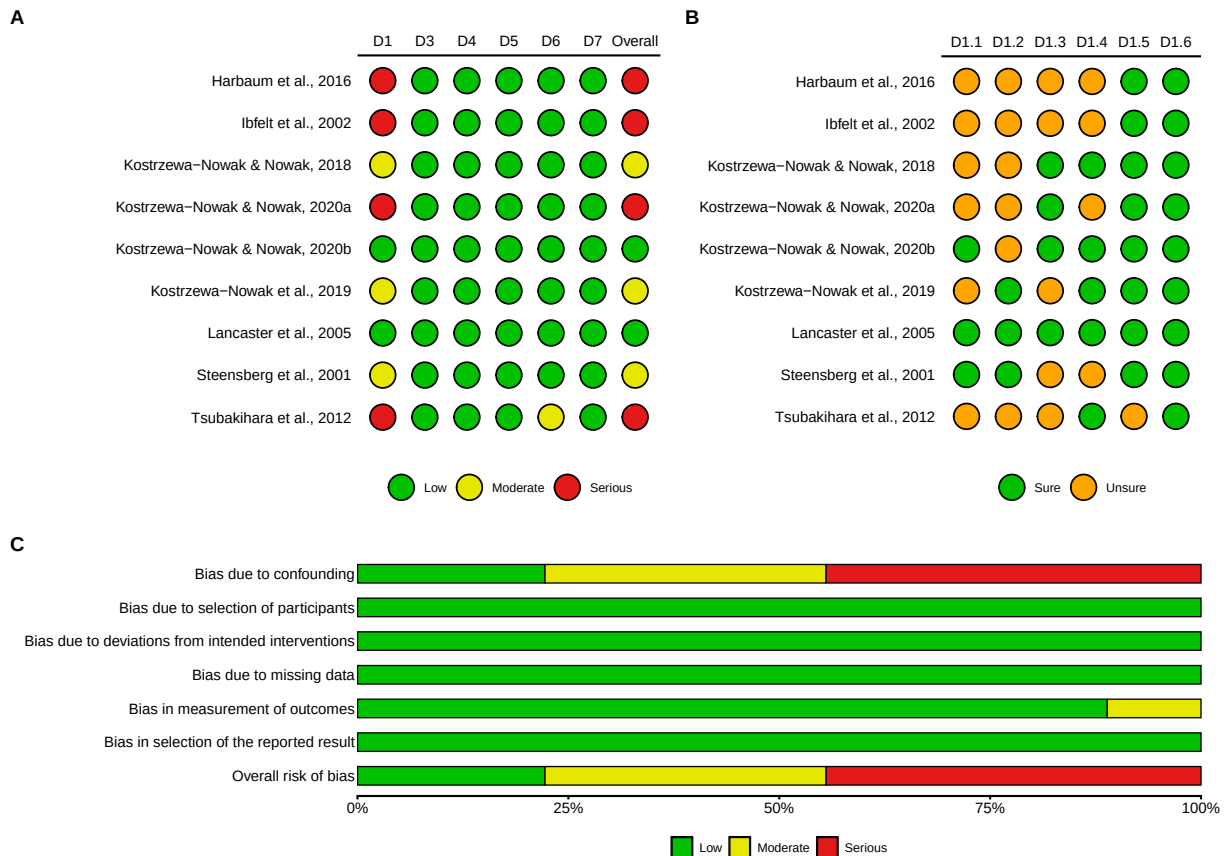
The following four functions implement the leave-one-out standardised residual approach described in Hedges and Olkin (1985) and Viechtbauer and Cheung (2010). A study is classified as an outlier if its standardised residual exceeds  $|3|$ ; values above  $|1.96|$  warrant closer inspection. The model (fixed- vs. random-effects) is selected automatically from the robust heterogeneity index  $I_r^2$  ( $< 30\% \rightarrow$  fixed-effects, otherwise random-effects). The outlier routine is always passed the **variance** (`seTE2`) as its `s2` argument.

The full definitions live in `_common.R` and are sourced automatically at the top of every chapter.



### 3. Risk of Bias Assessment (ROBINS-I)

Downloads: ROBINS-I Data (xlsx) · ROBINS-I R-Script



#### Confounding-factor items (Panel B).

- **D1.1** - No exercise 24-48 h pre-intervention?
- **D1.2** - Standardized food intake among participants (e.g. all fasted or standardized breakfast)?
- **D1.3** - No caffeine, acute medication, or alcohol 24 h before exercise?
- **D1.4** - Identical warm-up between participants?
- **D1.5** - Good way to assess the controlled intensity (HR control, VO<sub>2</sub>, VT)?
- **D1.6** - Homogeneity of the group (sex, age, BMI, training status)?



**Part I.**

**Th1**



## 4. Th1 — Immediate Post-Exercise

### 4.1. Data Import

The base analysis uses the pre-computed Hedges'  $g$  file for Th1 at  $r = 0.5$  (`effect_size/Th1/pre_post/Th1_pre_post_r_050`). Sensitivity analyses can substitute the  $r = 0.3 / 0.4 / 0.6 / 0.7$  files in the same folder. The pipeline column names are mapped to the analysis conventions (VE\_g, Sex, Age, BMI) by `load_es()`.

### 4.2. Primary Meta-Analysis

`metagen()` fits a random-effects model on the pre-computed effect sizes ( $g$ ,  $SE_g$ ) using the **Hedges-Olkin (HE)** <sup>2</sup> estimator and the **Knapp-Hartung** small-sample correction (`hakn = TRUE`). After inspecting the full set, **Kostrzewa-Nowak et al. (2019)** was detected as very different across meta-analysis and while not influential on the overall effect, its values were unrealistic and brought substantial misleading variance.

#### 4.2.1. Full Modell {meta}

Number of studies:  $k = 18$

|                           | SMD     | 95% CI            | t    |
|---------------------------|---------|-------------------|------|
| Random effects model (HK) | 0.8982  | [ 0.3316; 1.4649] | 3.34 |
| Prediction interval       |         | [-1.5159; 3.3124] |      |
|                           | p-value |                   |      |
| Random effects model (HK) | 0.0038  |                   |      |
| Prediction interval       |         |                   |      |

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 1.2328$  [0.5533; 2.8692];  $\tau = 1.1103$  [0.7438; 1.6939]  
 $I^2 = 86.8\%$  [80.7%; 91.0%];  $H = 2.76$  [2.27; 3.34]

Test of heterogeneity:

| Q      | d.f. | p-value  |
|--------|------|----------|
| 129.23 | 17   | < 0.0001 |

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model ( $df = 17$ )
- Prediction interval based on t-distribution ( $df = 17$ )

### 4.2.2. Adapted Model {meta}

#### Caution

Kostrzewa-Nowak et al. (2019) removed (unrealistic values / misleading variance); re-fit

Number of studies:  $k = 17$

|                           | SMD     | 95% CI            | t    |
|---------------------------|---------|-------------------|------|
| Random effects model (HK) | 0.9980  | [ 0.4524; 1.5436] | 3.88 |
| Prediction interval       |         | [-1.2488; 3.2448] |      |
|                           | p-value |                   |      |
| Random effects model (HK) | 0.0013  |                   |      |
| Prediction interval       |         |                   |      |

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 1.0535$  [0.4647; 2.5537];  $\tau = 1.0264$  [0.6817; 1.5980]

$I^2 = 86.4\%$  [79.8%; 90.9%];  $H = 2.72$  [2.22; 3.32]

Test of heterogeneity:

| Q      | d.f. | p-value  |
|--------|------|----------|
| 117.96 | 16   | < 0.0001 |

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model (df = 16)
- Prediction interval based on t-distribution (df = 16)

### 4.2.3. Adapted Model {metafor}

Random-Effects Model ( $k = 17$ ;  $\tau^2$  estimator: HE)

$\tau^2$  (estimated amount of total heterogeneity): 1.0535 (SE = 0.4224)

$\tau$  (square root of estimated  $\tau^2$  value): 1.0264

$I^2$  (total heterogeneity / total variability): 90.76%

$H^2$  (total variability / sampling variability): 10.82

Test for Heterogeneity:

$Q(df = 16) = 117.9586$ ,  $p\text{-val} < .0001$

Model Results:

| estimate | se     | tval   | df | pval   | ci.lb  | ci.ub  |    |
|----------|--------|--------|----|--------|--------|--------|----|
| 0.9980   | 0.2574 | 3.8774 | 16 | 0.0013 | 0.4524 | 1.5436 | ** |

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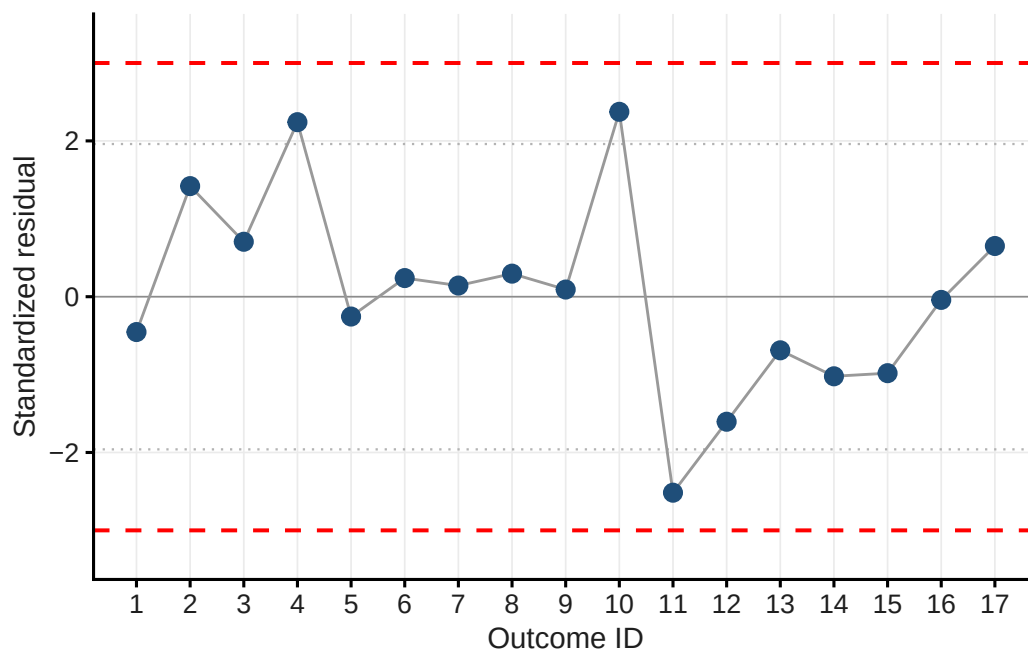
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

### 4.3. Outlier Detection

Leave-one-out standardised residuals are computed via `metaoutliers()`, which is passed the **variance** (`m1$seTE^2`). Studies with  $|z| > 3$  are flagged as outliers;  $|z| > 1.96$  warrants inspection.

**Table 4.1.:** Standardised residuals

| Short_reference                  | g      | std_res |
|----------------------------------|--------|---------|
| Ibfelt et al. (2002)             | 0.577  | -0.454  |
| Kostrzewa-Nowak & Nowak (2018)a  | 2.289  | 1.420   |
| Kostrzewa-Nowak & Nowak (2018)b  | 1.613  | 0.705   |
| Kostrzewa-Nowak & Nowak (2020a)a | 2.781  | 2.241   |
| Kostrzewa-Nowak & Nowak (2020a)b | 0.761  | -0.255  |
| Kostrzewa-Nowak & Nowak (2020b)a | 1.195  | 0.239   |
| Kostrzewa-Nowak & Nowak (2020b)b | 1.111  | 0.143   |
| Kostrzewa-Nowak & Nowak (2020b)c | 1.245  | 0.296   |
| Kostrzewa-Nowak & Nowak (2020b)d | 1.066  | 0.092   |
| Kostrzewa-Nowak & Nowak (2020b)e | 3.282  | 2.374   |
| Steensberg et al. (2001)         | -1.092 | -2.516  |
| Tsubakihara et al. (2012)        | -0.276 | -1.605  |
| Lancaster et al. (2005)a         | 0.373  | -0.688  |
| Lancaster et al. (2005)b         | 0.091  | -1.022  |
| Lancaster et al. (2005)c         | 0.124  | -0.982  |
| Harbaum et al. (2016)a           | 0.949  | -0.041  |
| Harbaum et al. (2016)b           | 1.584  | 0.651   |

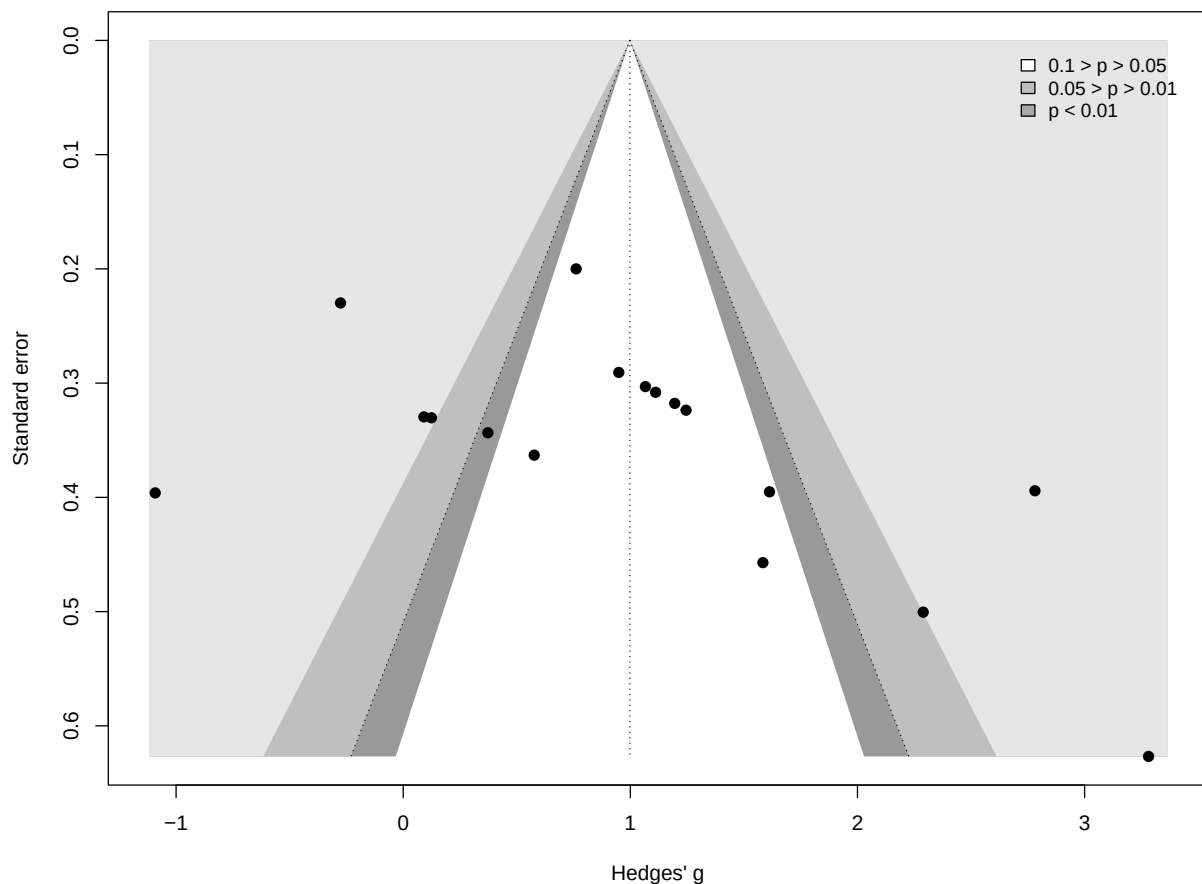


**! Important**

No outcome has standardized residuals above 3 or below -3.

## 4.4. Funnel Plot & Influential Cases

The contour-enhanced funnel plot shows study estimates against their standard errors; shaded regions mark significance zones ( $\alpha = 0.10, 0.05, 0.01$ ). The `dfbetas` diagnostic from `metafor::rma.mv()` identifies **influential cases** — distinct from statistical outliers and not necessarily excluded.



**Table 4.2.:** DFBETAS influence diagnostics

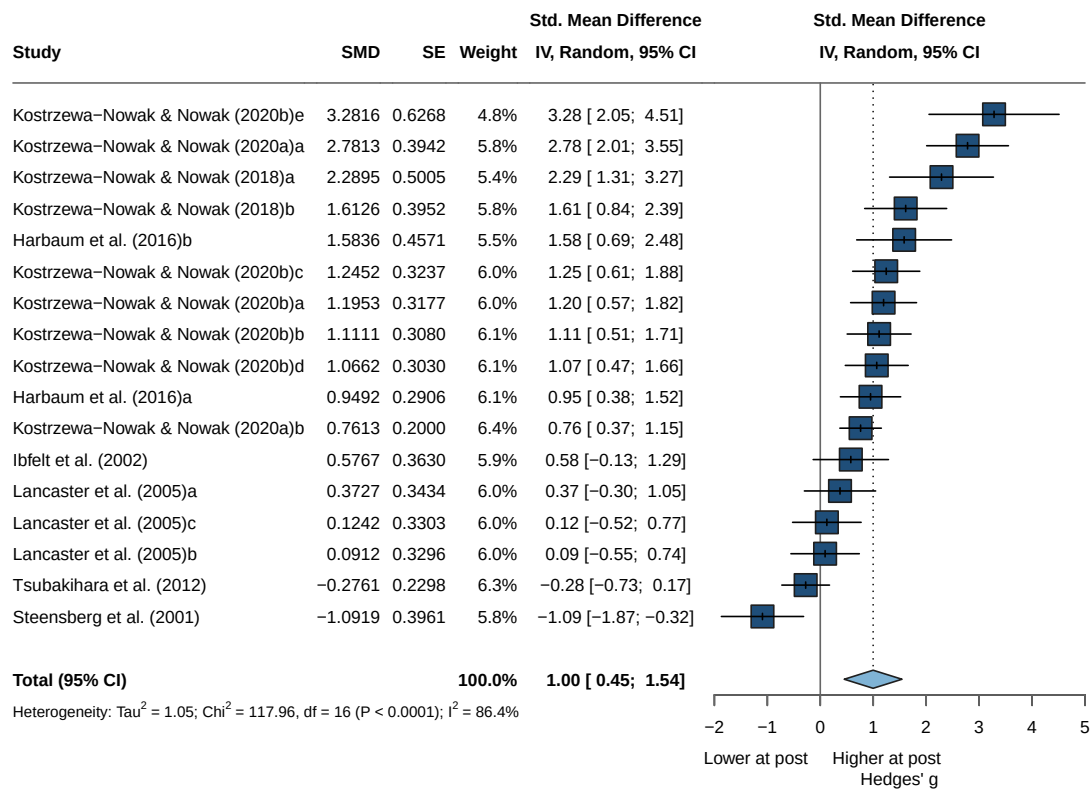
| Short_reference                    | g     | dfbetas     |
|------------------------------------|-------|-------------|
| Ibfelt et al. (2002)               | 0.577 | -0.13459692 |
| Kostrzewska-Nowak & Nowak (2018)a  | 2.289 | 0.48230850  |
| Kostrzewska-Nowak & Nowak (2018)b  | 1.613 | 0.43065936  |
| Kostrzewska-Nowak & Nowak (2020a)a | 2.781 | 1.04875284  |
| Kostrzewska-Nowak & Nowak (2020a)b | 0.761 | -0.07030521 |
| Kostrzewska-Nowak & Nowak (2020b)a | 1.195 | 0.33513405  |
| Kostrzewska-Nowak & Nowak (2020b)b | 1.111 | 0.28334594  |
| Kostrzewska-Nowak & Nowak (2020b)c | 1.245 | 0.36199846  |
| Kostrzewska-Nowak & Nowak (2020b)d | 1.066 | 0.25226569  |

**Table 4.2.:** DFBETAS influence diagnostics

| Short_reference                    | g      | dfbetas     |
|------------------------------------|--------|-------------|
| Kostrzewska-Nowak & Nowak (2020b)e | 3.282  | 0.50656391  |
| Steensberg et al. (2001)           | -1.092 | -0.98309619 |
| Tsubakihara et al. (2012)          | -0.276 | -1.80121819 |
| Lancaster et al. (2005)a           | 0.373  | -0.29482591 |
| Lancaster et al. (2005)b           | 0.091  | -0.53785734 |
| Lancaster et al. (2005)c           | 0.124  | -0.50993030 |
| Harbaum et al. (2016)a             | 0.949  | 0.15837850  |
| Harbaum et al. (2016)b             | 1.584  | 0.30720822  |

One value was below  $-1$  and one above  $1$ , so both were identified as influential cases. However, neither is an outlier (Kostrzewska-Nowak & Nowak (2020a)a and Tsubakihara et al. (2012)). Accordingly, all outcomes are retained, and the sources of heterogeneity are examined in the moderator analysis.

## 4.5. Forest Plot



## 4.6. Asymmetry Test

Egger's regression test (`method = "linreg"`) evaluates funnel asymmetry as a proxy for small-study/publication effects (one-tailed,  $p < .10$ ). Where asymmetry is suggested, a trim-and-fill analysis estimates potentially missing studies.

### 4.6.1. Egger's regression test

Linear regression test of funnel plot asymmetry

Test result:  $t = 2.11$ ,  $df = 15$ ,  $p\text{-value} = 0.0523$

Bias estimate: 4.9260 (SE = 2.3369)

Details:

- multiplicative residual heterogeneity variance ( $\tau^2 = 6.0667$ )
- predictor: standard error
- weight: inverse variance
- reference: Egger et al. (1997), BMJ

### 4.6.2. Trim and Fill

Estimated number of missing studies on the left side: 0 (SE = 2.5907)

Random-Effects Model ( $k = 17$ ;  $\tau^2$  estimator: HE)

$\tau^2$  (estimated amount of total heterogeneity): 1.0535 (SE = 0.4224)

$\tau$  (square root of estimated  $\tau^2$  value): 1.0264

$I^2$  (total heterogeneity / total variability): 90.76%

$H^2$  (total variability / sampling variability): 10.82

Test for Heterogeneity:

$Q(df = 16) = 117.9586$ ,  $p\text{-val} < .0001$

Model Results:

| estimate | se     | zval   | pval   | ci.lb  | ci.ub  |     |
|----------|--------|--------|--------|--------|--------|-----|
| 0.9980   | 0.2643 | 3.7765 | 0.0002 | 0.4800 | 1.5159 | *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** No missing studies

## 4.7. Moderator Analyses

### 4.7.1. Risk of Bias

Subgroup analysis by overall RoB rating tests whether study quality moderates the pooled effect; the between-subgroup statistic ( $Q_b$ ) indicates whether subgroup means differ.

#### 4.7.1.1. Model

Number of studies:  $k = 17$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 1.0535$  [0.4647; 2.5537];  $\tau = 1.0264$  [0.6817; 1.5980]

$I^2 = 86.4\%$  [79.8%; 90.9%];  $H = 2.72$  [2.22; 3.32]

Test of heterogeneity:

| Q      | d.f. | p-value  |
|--------|------|----------|
| 117.96 | 16   | < 0.0001 |

Results for subgroups (random effects model (HK)):

|                | k | SMD    | 95% CI            | $\tau^2$ | $\tau$ |
|----------------|---|--------|-------------------|----------|--------|
| RoB = Serious  | 6 | 1.0287 | [-0.0523; 2.1097] | 0.9592   | 0.9794 |
| RoB = Moderate | 3 | 0.9237 | [-3.5234; 5.3709] | 3.0131   | 1.7358 |
| RoB = Low      | 8 | 0.9996 | [ 0.2061; 1.7932] | 0.8995   | 0.9484 |

|                | Q     | $I^2$ |
|----------------|-------|-------|
| RoB = Serious  | 50.50 | 90.1% |
| RoB = Moderate | 35.85 | 94.4% |
| RoB = Low      | 31.00 | 77.4% |

Test for subgroup differences (random effects model (HK)):

|                | Q    | d.f. | p-value |
|----------------|------|------|---------|
| Between groups | 0.01 | 2    | 0.9952  |

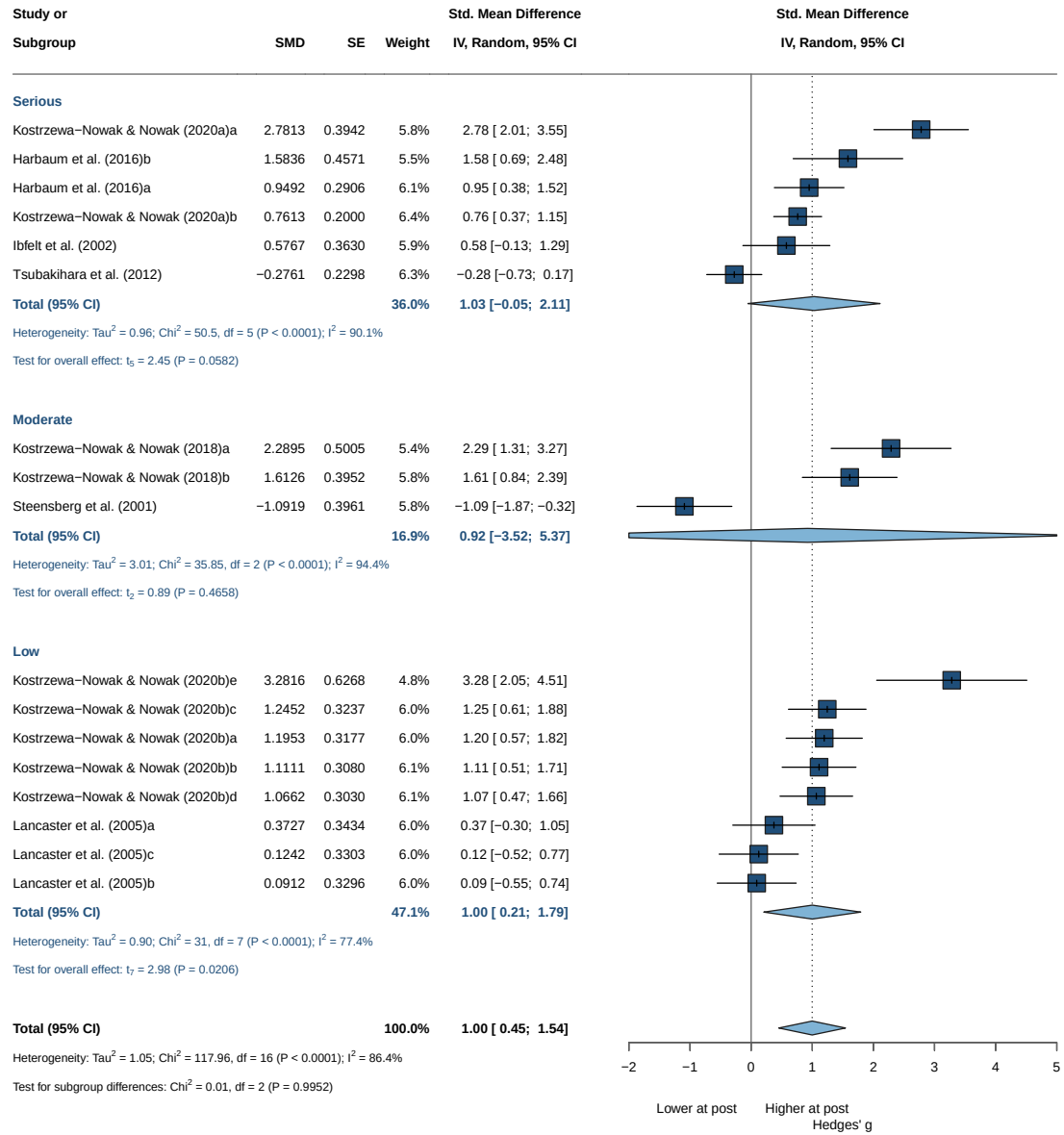
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

#### Note

Not significant results

## 4.7.1.2. Forrestplot



## 4.7.2. Staining

Number of studies:  $k = 17$

Quantifying heterogeneity (with 95% CIs):

```
tau^2 = 1.0535 [0.4647; 2.5537]; tau = 1.0264 [0.6817; 1.5980]
I^2 = 86.4% [79.8%; 90.9%]; H = 2.72 [2.22; 3.32]
```

Test of heterogeneity:

```
Q d.f. p-value
117.96 16 < 0.0001
```

Results for subgroups (random effects model (HK)):

|                  | k  | SMD    | 95% CI            | tau^2  | tau    |
|------------------|----|--------|-------------------|--------|--------|
| staining = intra | 14 | 1.0632 | [ 0.4218; 1.7046] | 1.1630 | 1.0784 |
| staining = extra | 3  | 0.7028 | [-1.6445; 3.0501] | 0.7783 | 0.8822 |

|                  | Q     | I^2   |
|------------------|-------|-------|
| staining = intra | 91.40 | 85.8% |
| staining = extra | 18.92 | 89.4% |

Test for subgroup differences (random effects model (HK)):

```
Q d.f. p-value
Between groups 0.34 1 0.5617
```

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

#### **i** Note

Not significant results

### 4.7.3. Sex

#### 4.7.3.1. Model

Number of studies: k = 17

Quantifying heterogeneity (with 95% CIs):

```
tau^2 = 1.0535 [0.4647; 2.5537]; tau = 1.0264 [0.6817; 1.5980]
I^2 = 86.4% [79.8%; 90.9%]; H = 2.72 [2.22; 3.32]
```

Test of heterogeneity:

```
Q d.f. p-value
117.96 16 < 0.0001
```

Results for subgroups (random effects model (HK)):

|            | k  | SMD    | 95% CI            | tau^2  | tau    |
|------------|----|--------|-------------------|--------|--------|
| sex = male | 14 | 1.0632 | [ 0.4218; 1.7046] | 1.1630 | 1.0784 |

#### 4. Th1 — Immediate Post-Exercise

```
sex = female    1 -0.2761 [-0.7265; 0.1743]    --    --
sex = mixed     2  1.1682 [-2.6647; 5.0012]  0.0545 0.2335
               Q    I2
sex = male     91.40 85.8%
sex = female    0.00    --
sex = mixed     1.37 27.1%
```

Test for subgroup differences (random effects model (HK)):

```
               Q d.f.  p-value
Between groups 19.89    2 < 0.0001
```

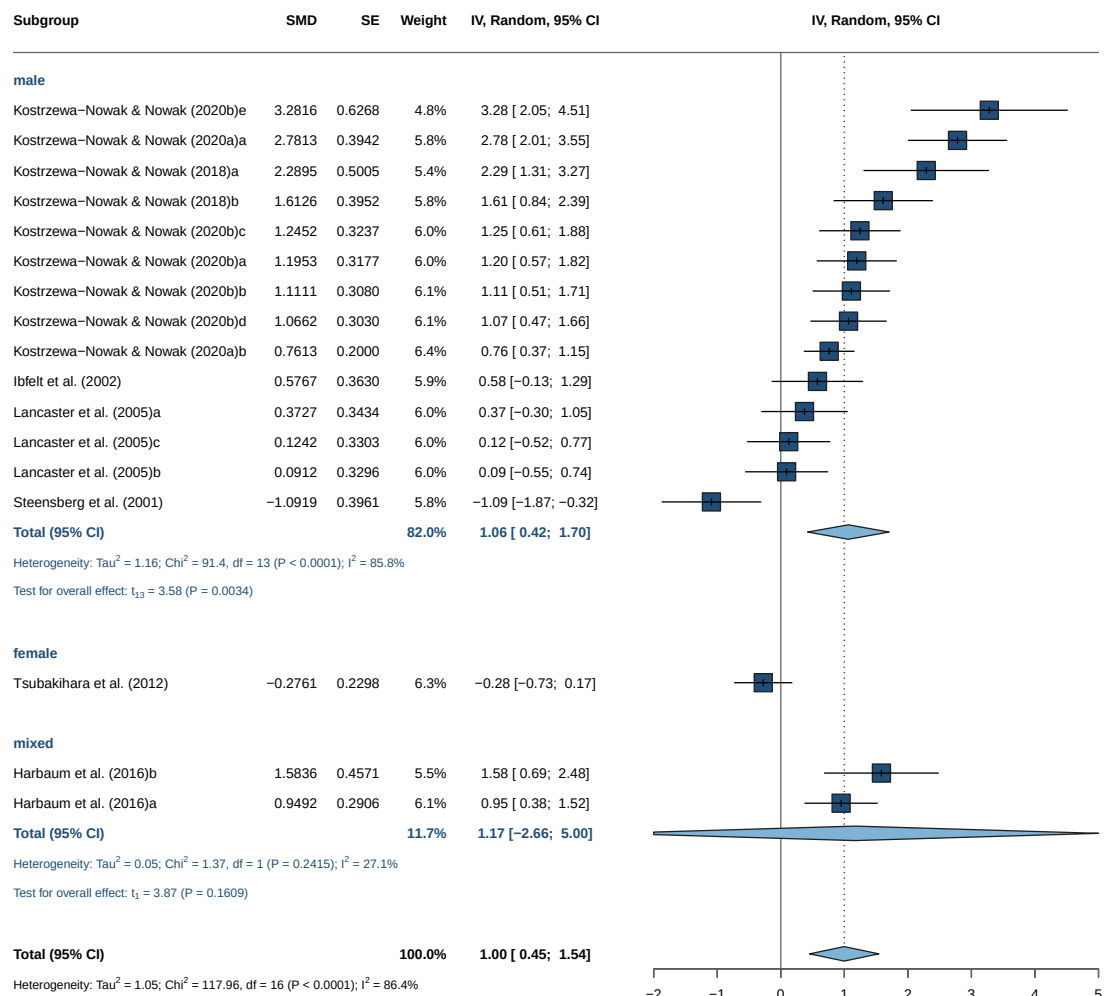
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

#### ! Important

Significant results but 14 vs 1 vs 2

## 4.7.3.2. Forrest Plot



## 4.7.4. Intensity

Number of studies:  $k = 16$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 1.0050$  [0.4207; 2.5358];  $\tau = 1.0025$  [0.6486; 1.5924]

$I^2 = 84.0\%$  [75.3%; 89.6%];  $H = 2.50$  [2.01; 3.10]

Test of heterogeneity:

| Q     | d.f. | p-value  |
|-------|------|----------|
| 93.51 | 15   | < 0.0001 |

Results for subgroups (random effects model (HK)):

|                      | k  | SMD    | 95% CI            | $\tau^2$ |
|----------------------|----|--------|-------------------|----------|
| intensity = Vigorous | 5  | 0.0310 | [-0.7527; 0.8147] | 0.2967   |
| intensity = Maximal  | 11 | 1.5400 | [ 1.0222; 2.0579] | 0.5104   |

|                      | tau    | Q     | $I^2$ |
|----------------------|--------|-------|-------|
| intensity = Vigorous | 0.5447 | 11.36 | 64.8% |
| intensity = Maximal  | 0.7145 | 38.82 | 74.2% |

#### 4. Th1 — Immediate Post-Exercise

Test for subgroup differences (random effects model (HK)):

|                | Q     | d.f. | p-value  |
|----------------|-------|------|----------|
| Between groups | 17.03 | 1    | < 0.0001 |

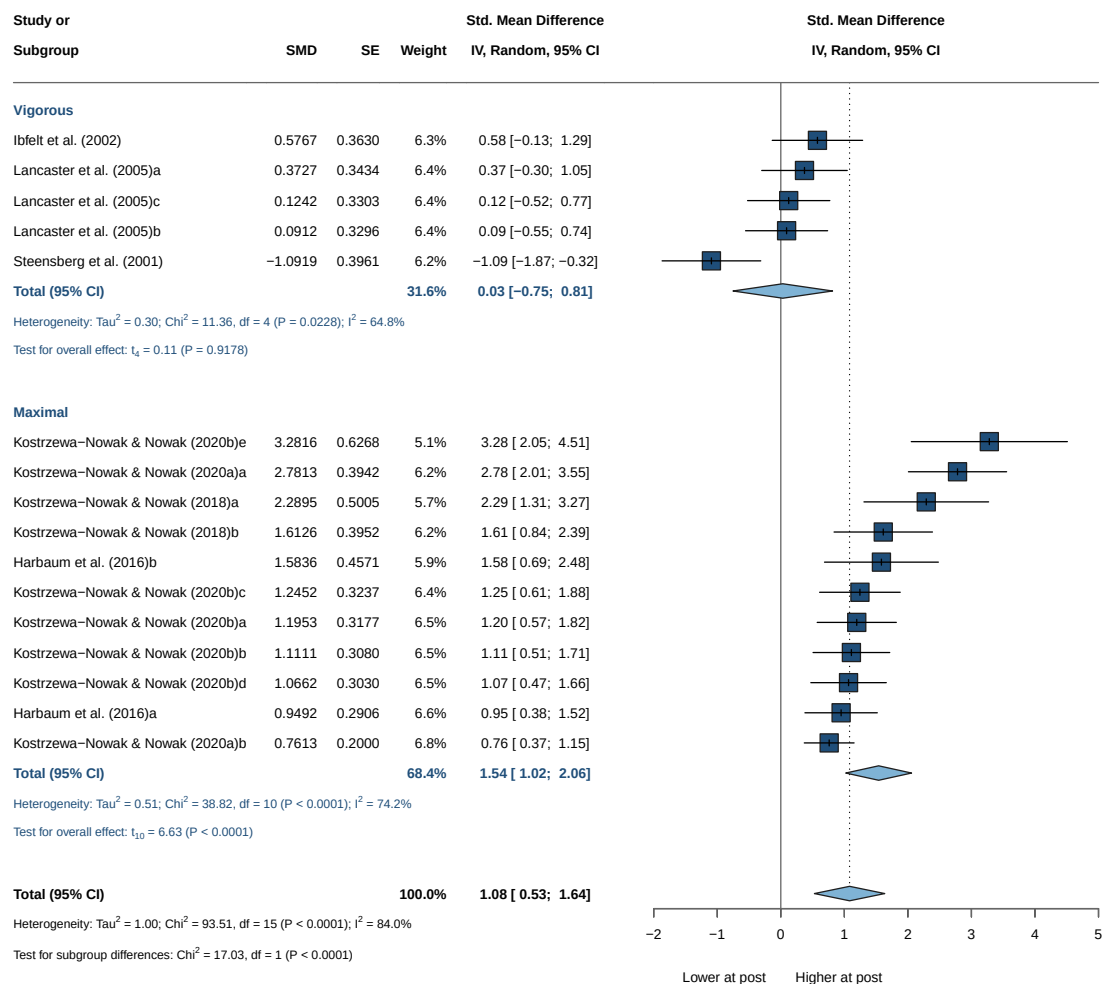
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

### ! Important

Significant results

#### 4.7.4.1. Forrest Plot



#### 4.7.5. Continuous Moderators (Meta-Regression)

Continuous moderators are tested via weighted meta-regression (`metareg()`). A significant slope indicates the moderator explains part of the between-study heterogeneity ( $\tau^2$ ); bubble plots show the fitted line with

95% CI.

#### 4.7.5.1. Age

Mixed-Effects Model (k = 16; tau<sup>2</sup> estimator: HE)

| logLik   | deviance | AIC     | BIC     | AICc    |
|----------|----------|---------|---------|---------|
| -22.6865 | 49.0041  | 51.3730 | 53.6908 | 53.3730 |

|                                                                    |                      |
|--------------------------------------------------------------------|----------------------|
| tau <sup>2</sup> (estimated amount of residual heterogeneity):     | 1.0410 (SE = 0.4487) |
| tau (square root of estimated tau <sup>2</sup> value):             | 1.0203               |
| I <sup>2</sup> (residual heterogeneity / unaccounted variability): | 89.97%               |
| H <sup>2</sup> (unaccounted variability / sampling variability):   | 9.97                 |
| R <sup>2</sup> (amount of heterogeneity accounted for):            | 0.00%                |

Test for Residual Heterogeneity:

QE(df = 14) = 91.8671, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 14) = 0.4880, p-val = 0.4963

Model Results:

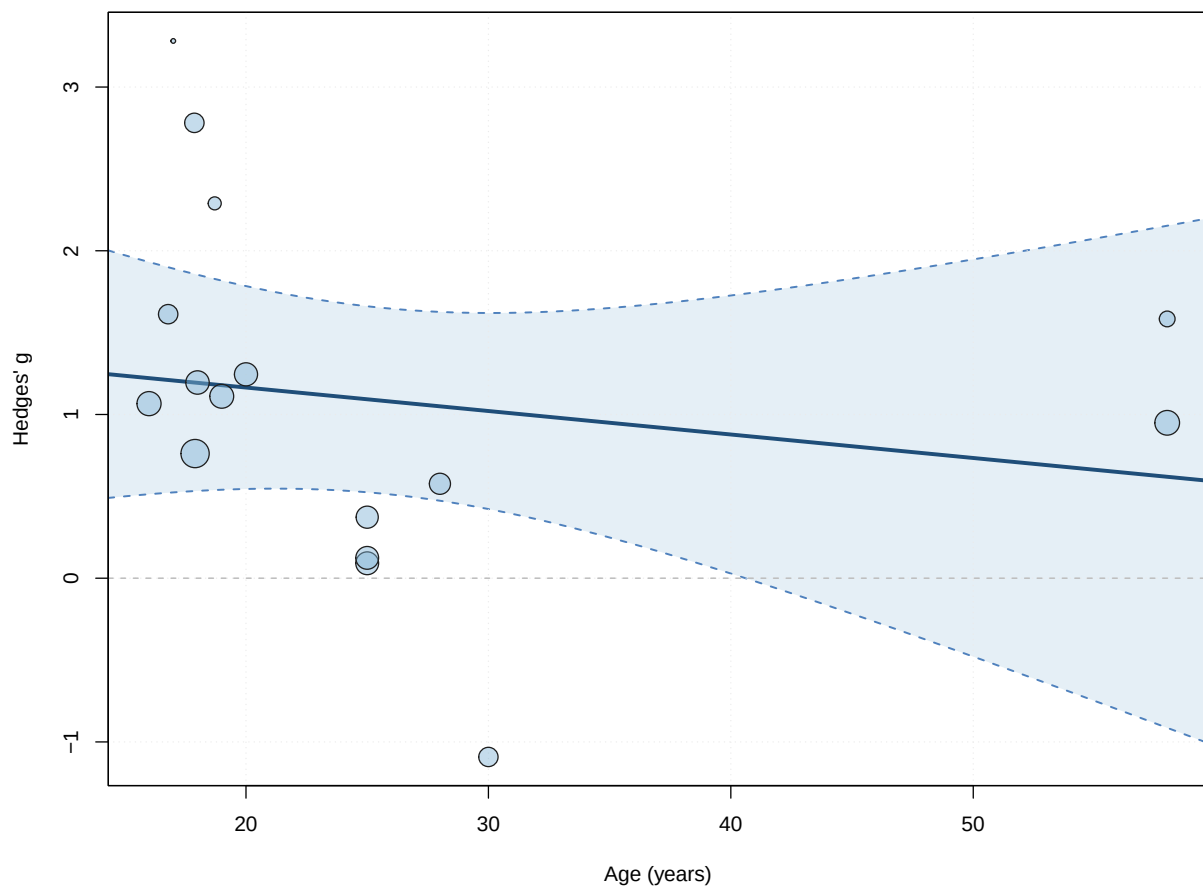
|         | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|---------|----|--------|---------|-------|
| intrcpt | 1.4515   | 0.5896 | 2.4621  | 14 | 0.0274 | 0.1871  |       |
| age     | -0.0143  | 0.0205 | -0.6986 | 14 | 0.4963 | -0.0584 |       |
| intrcpt | 2.7160   |        |         |    |        |         | *     |
| age     | 0.0297   |        |         |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant



#### 4.7.5.2. BMI

Mixed-Effects Model (k = 12; tau<sup>2</sup> estimator: HE)

| logLik   | deviance | AIC     | BIC     | AICc    |
|----------|----------|---------|---------|---------|
| -15.7587 | 34.9893  | 37.5174 | 38.9721 | 40.5174 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.8458 (SE = 0.4461)  
 tau (square root of estimated tau<sup>2</sup> value): 0.9197  
 I<sup>2</sup> (residual heterogeneity / unaccounted variability): 89.09%  
 H<sup>2</sup> (unaccounted variability / sampling variability): 9.17  
 R<sup>2</sup> (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 10) = 77.5843, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 10) = 0.0134, p-val = 0.9102

Model Results:

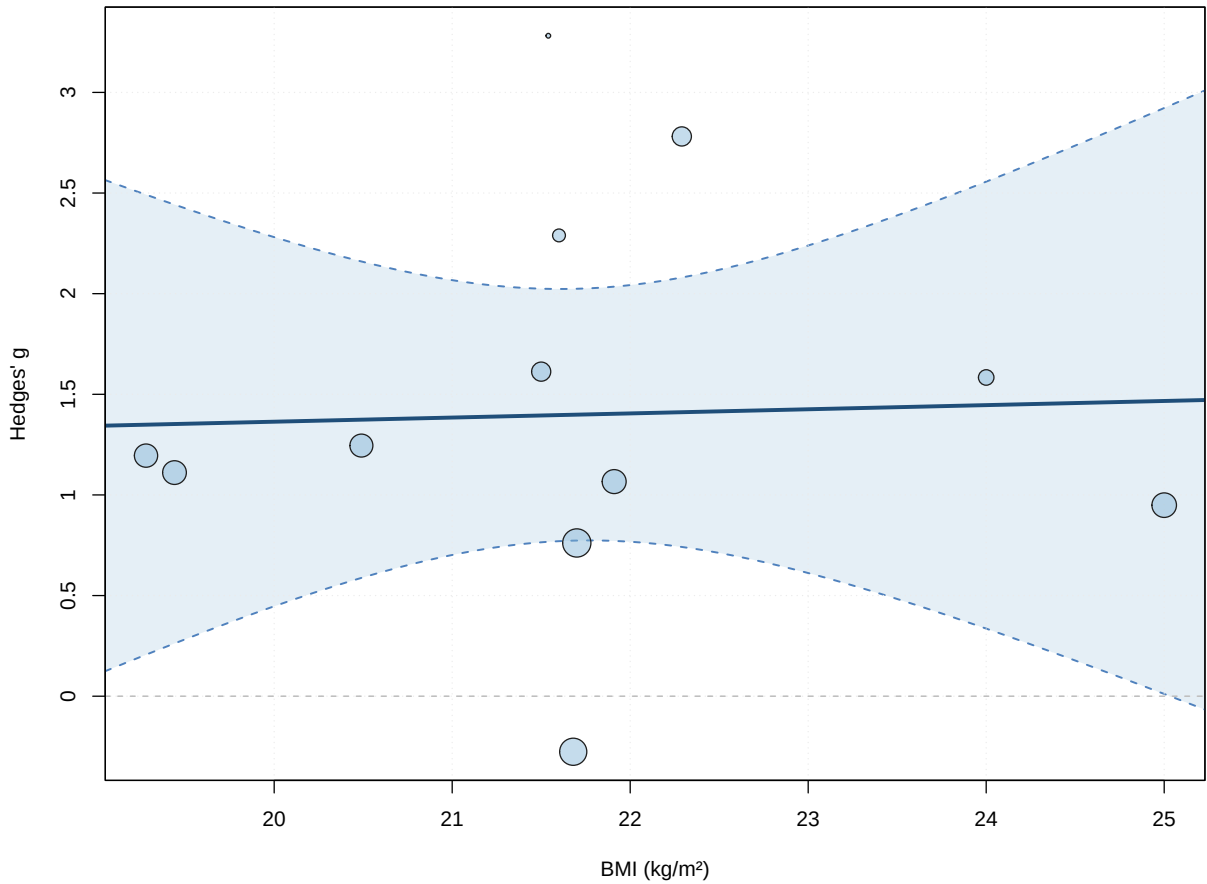
```
      estimate      se    tval  df    pval    ci.lb
intrcpt    0.9518  3.8746  0.2457  10  0.8109  -7.6813
bmi         0.0206  0.1782  0.1156  10  0.9102  -0.3764

      ci.ub
intrcpt    9.5849
bmi         0.4176

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

*i* Note

Not significant



4.7.5.3. Duration

Mixed-Effects Model (k = 15; tau^2 estimator: HE)

|          |          |         |         |         |
|----------|----------|---------|---------|---------|
| logLik   | deviance | AIC     | BIC     | AICc    |
| -13.7530 | 30.8355  | 33.5060 | 35.6301 | 35.6878 |

#### 4. Th1 — Immediate Post-Exercise

```
tau^2 (estimated amount of residual heterogeneity):    0.3425 (SE = 0.1937)
tau (square root of estimated tau^2 value):           0.5853
I^2 (residual heterogeneity / unaccounted variability): 74.35%
H^2 (unaccounted variability / sampling variability):   3.90
R^2 (amount of heterogeneity accounted for):           65.26%
```

Test for Residual Heterogeneity:

QE(df = 13) = 31.7293, p-val = 0.0026

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 13) = 20.8420, p-val = 0.0005

Model Results:

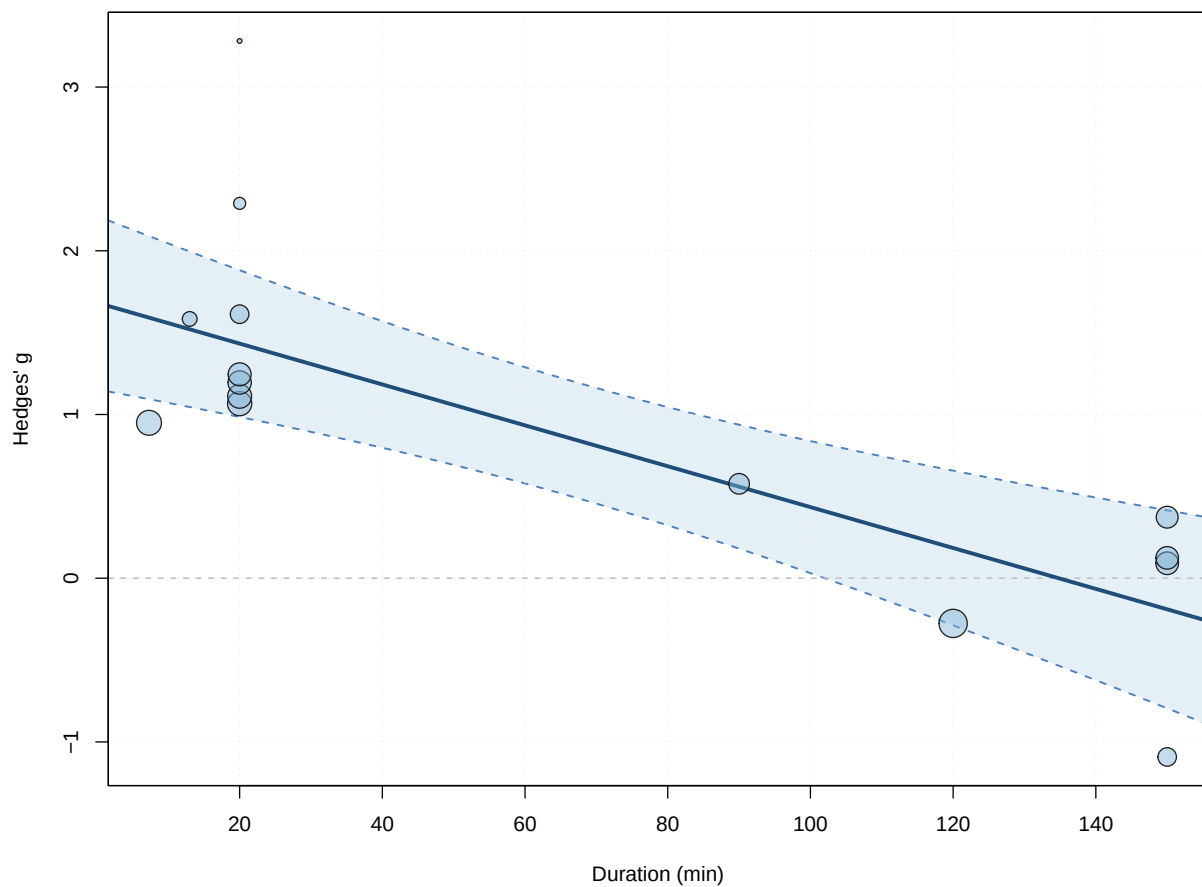
|          | estimate | se     | tval    | df | pval   | ci.lb   |
|----------|----------|--------|---------|----|--------|---------|
| intrcpt  | 1.6827   | 0.2451 | 6.8662  | 13 | <.0001 | 1.1532  |
| duration | -0.0125  | 0.0027 | -4.5653 | 13 | 0.0005 | -0.0184 |
|          | ci.ub    |        |         |    |        |         |
| intrcpt  | 2.2121   | ***    |         |    |        |         |
| duration | -0.0066  | ***    |         |    |        |         |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

! Important

Significant



#### 4.7.5.4. Dose

Mixed-Effects Model ( $k = 14$ ;  $\tau^2$  estimator: HE)

| logLik   | deviance | AIC     | BIC     | AICc    |
|----------|----------|---------|---------|---------|
| -12.5188 | 27.2639  | 31.0376 | 32.9547 | 33.4376 |

$\tau^2$  (estimated amount of residual heterogeneity): 0.3246 (SE = 0.1970)  
 $\tau$  (square root of estimated  $\tau^2$  value): 0.5697  
 $I^2$  (residual heterogeneity / unaccounted variability): 71.72%  
 $H^2$  (unaccounted variability / sampling variability): 3.54  
 $R^2$  (amount of heterogeneity accounted for): 65.62%

Test for Residual Heterogeneity:

QE(df = 12) = 25.4779, p-val = 0.0127

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 12) = 19.2815, p-val = 0.0009

Model Results:

#### 4. Th1 — Immediate Post-Exercise

```

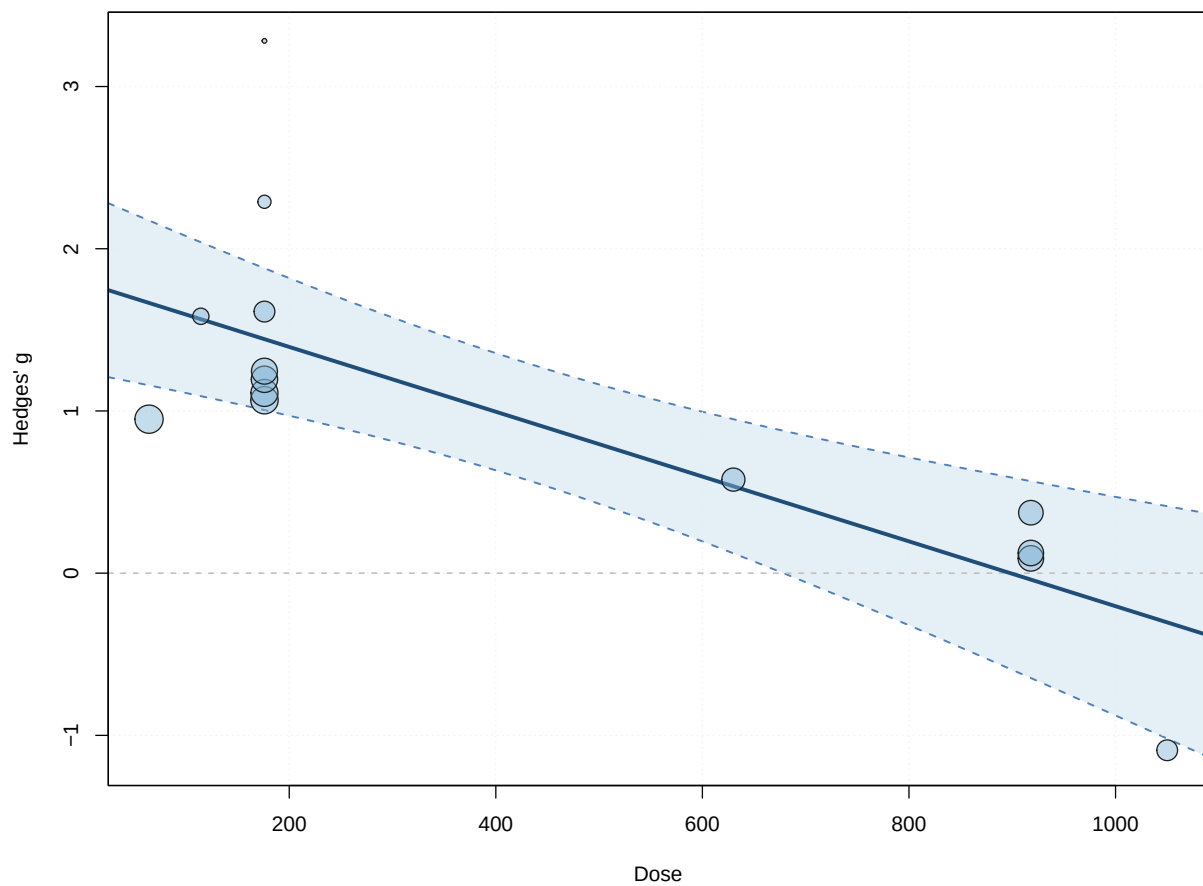
      estimate      se      tval  df      pval      ci.lb
intrcpt      1.7943  0.2547   7.0457  12  <.0001    1.2394
dose        -0.0020  0.0005  -4.3911  12  0.0009   -0.0030
      ci.ub
intrcpt      2.3491  ***
dose        -0.0010  ***

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

#### ! Important

Significant — collinear with duration



#### 4.7.5.5. IFN

Mixed-Effects Model (k = 7; tau<sup>2</sup> estimator: HE)

```

logLik  deviance      AIC      BIC     AICc
-9.1465  20.7348  24.2929  24.1307  32.2929

```

```

tau^2 (estimated amount of residual heterogeneity):    0.9848 (SE = 0.7226)
tau (square root of estimated tau^2 value):           0.9924
I^2 (residual heterogeneity / unaccounted variability): 88.95%
H^2 (unaccounted variability / sampling variability):  9.05
R^2 (amount of heterogeneity accounted for):           0.00%

```

Test for Residual Heterogeneity:

QE(df = 5) = 27.1266, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.0809, p-val = 0.7875

Model Results:

|         | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|---------|----|--------|---------|-------|
| intrcpt | 2.0560   | 1.7626 | 1.1665  | 5  | 0.2960 | -2.4748 |       |
| IFN     | -0.2718  | 0.9556 | -0.2845 | 5  | 0.7875 | -2.7282 |       |
| intrcpt | 6.5869   |        |         |    |        |         |       |
| IFN     | 2.1846   |        |         |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

#### 4.7.5.6. IL-4

Mixed-Effects Model (k = 9; tau^2 estimator: HE)

| logLik   | deviance | AIC     | BIC     | AICc    |
|----------|----------|---------|---------|---------|
| -10.3229 | 23.4488  | 26.6457 | 27.2374 | 31.4457 |

```

tau^2 (estimated amount of residual heterogeneity):    0.6578 (SE = 0.4320)
tau (square root of estimated tau^2 value):           0.8111
I^2 (residual heterogeneity / unaccounted variability): 86.45%
H^2 (unaccounted variability / sampling variability):  7.38
R^2 (amount of heterogeneity accounted for):           0.00%

```

Test for Residual Heterogeneity:

QE(df = 7) = 30.1320, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 7) = 0.8032, p-val = 0.3999

Model Results:

#### 4. Th1 — Immediate Post-Exercise

```

      estimate      se    tval  df    pval    ci.lb
intrcpt    1.3165  0.3308  3.9793   7  0.0053   0.5342
IL_4       0.2506  0.2796  0.8962   7  0.3999  -0.4105
      ci.ub
intrcpt    2.0988  **
IL_4       0.9117

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

#### Note

Not significant

#### 4.7.5.7. IL-6

Mixed-Effects Model (k = 14; tau<sup>2</sup> estimator: HE)

```

      logLik  deviance      AIC      BIC      AICc
-18.0578   39.5434   42.1155   44.0327   44.5155

```

```

tau^2 (estimated amount of residual heterogeneity):    0.8055 (SE = 0.3889)
tau (square root of estimated tau^2 value):           0.8975
I^2 (residual heterogeneity / unaccounted variability): 88.00%
H^2 (unaccounted variability / sampling variability):   8.34
R^2 (amount of heterogeneity accounted for):           0.00%

```

Test for Residual Heterogeneity:

QE(df = 12) = 63.3467, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 12) = 0.1327, p-val = 0.7220

Model Results:

```

      estimate      se    tval  df    pval    ci.lb
intrcpt    1.2063  0.2945  4.0964  12  0.0015   0.5647
IL_6       0.0604  0.1659  0.3642  12  0.7220  -0.3010
      ci.ub
intrcpt    1.8479  **
IL_6       0.4219

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

**i** Note

Not significant

**4.7.5.8. IL\_10**Mixed-Effects Model (k = 11; tau<sup>2</sup> estimator: HE)

| logLik   | deviance | AIC     | BIC     | AICc    |
|----------|----------|---------|---------|---------|
| -12.3204 | 27.0095  | 30.6409 | 31.8345 | 34.0694 |

|                                                                    |                      |
|--------------------------------------------------------------------|----------------------|
| tau <sup>2</sup> (estimated amount of residual heterogeneity):     | 0.5518 (SE = 0.3375) |
| tau (square root of estimated tau <sup>2</sup> value):             | 0.7428               |
| I <sup>2</sup> (residual heterogeneity / unaccounted variability): | 81.00%               |
| H <sup>2</sup> (unaccounted variability / sampling variability):   | 5.26                 |
| R <sup>2</sup> (amount of heterogeneity accounted for):            | 0.00%                |

Test for Residual Heterogeneity:

QE(df = 9) = 35.0975, p-val &lt; .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 9) = 0.2716, p-val = 0.6148

Model Results:

|         | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|---------|----|--------|---------|-------|
| intrcpt | 1.5574   | 0.2433 | 6.3998  | 9  | 0.0001 | 1.0069  |       |
| IL_10   | -0.3495  | 0.6706 | -0.5211 | 9  | 0.6148 | -1.8664 |       |
| intrcpt | 2.1078   | ***    |         |    |        |         |       |
| IL_10   | 1.1675   |        |         |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

**4.7.5.9. IL\_2**Mixed-Effects Model (k = 7; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -8.6034 | 19.6487  | 23.2069 | 23.0446 | 31.2069 |

|                                                                |                      |
|----------------------------------------------------------------|----------------------|
| tau <sup>2</sup> (estimated amount of residual heterogeneity): | 0.8092 (SE = 0.5809) |
|----------------------------------------------------------------|----------------------|

#### 4. Th1 — Immediate Post-Exercise

```
tau (square root of estimated tau^2 value):      0.8996
I^2 (residual heterogeneity / unaccounted variability): 89.58%
H^2 (unaccounted variability / sampling variability): 9.59
R^2 (amount of heterogeneity accounted for):      0.73%
```

Test for Residual Heterogeneity:

```
QE(df = 5) = 32.1011, p-val < .0001
```

Test of Moderators (coefficient 2):

```
F(df1 = 1, df2 = 5) = 0.8054, p-val = 0.4106
```

Model Results:

|         | estimate | se     | tval   | df | pval   | ci.lb   |
|---------|----------|--------|--------|----|--------|---------|
| intrcpt | 0.2553   | 1.4923 | 0.1711 | 5  | 0.8709 | -3.5808 |
| IL_2    | 0.7369   | 0.8211 | 0.8974 | 5  | 0.4106 | -1.3739 |
|         | ci.ub    |        |        |    |        |         |
| intrcpt | 4.0915   |        |        |    |        |         |
| IL_2    | 2.8477   |        |        |    |        |         |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

## 4.8. Multivariate Models

Multivariate meta-regression tests interactions between candidate moderators. Neither model below yields independent predictors.

### 4.8.1. Intensity \* Duration

Mixed-Effects Model (k = 14; tau^2 estimator: HE)

```
tau^2 (estimated amount of residual heterogeneity): 0.3855 (SE = 0.2452)
tau (square root of estimated tau^2 value):         0.6209
I^2 (residual heterogeneity / unaccounted variability): 74.36%
H^2 (unaccounted variability / sampling variability): 3.90
R^2 (amount of heterogeneity accounted for):         59.17%
```

Test for Residual Heterogeneity:

```
QE(df = 10) = 24.8856, p-val = 0.0056
```

Test of Moderators (coefficients 2:4):

```
F(df1 = 3, df2 = 10) = 5.5862, p-val = 0.0164
```

Model Results:

|                            | estimate | se      | tval    | df |
|----------------------------|----------|---------|---------|----|
| intrcpt                    | 0.7778   | 0.9262  | 0.8398  | 10 |
| intensityVigorous          | 0.8233   | 1.9758  | 0.4167  | 10 |
| duration                   | 0.0403   | 0.0508  | 0.7938  | 10 |
| intensityVigorous:duration | -0.0517  | 0.0523  | -0.9887 | 10 |
|                            | pval     | ci.lb   | ci.ub   |    |
| intrcpt                    | 0.4206   | -1.2859 | 2.8415  |    |
| intensityVigorous          | 0.6857   | -3.5792 | 5.2257  |    |
| duration                   | 0.4457   | -0.0728 | 0.1534  |    |
| intensityVigorous:duration | 0.3461   | -0.1681 | 0.0648  |    |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

#### **i** Note

Not significant

### 4.8.2. IFN \* IL\_2

Mixed-Effects Model (k = 7; tau<sup>2</sup> estimator: HE)

|                                                                    |                      |
|--------------------------------------------------------------------|----------------------|
| tau <sup>2</sup> (estimated amount of residual heterogeneity):     | 0.8792 (SE = 0.8102) |
| tau (square root of estimated tau <sup>2</sup> value):             | 0.9376               |
| I <sup>2</sup> (residual heterogeneity / unaccounted variability): | 88.83%               |
| H <sup>2</sup> (unaccounted variability / sampling variability):   | 8.95                 |
| R <sup>2</sup> (amount of heterogeneity accounted for):            | 0.00%                |

Test for Residual Heterogeneity:

QE(df = 3) = 21.9341, p-val < .0001

Test of Moderators (coefficients 2:4):

F(df1 = 3, df2 = 3) = 0.7599, p-val = 0.5866

Model Results:

|          | estimate | se      | tval    | df | pval   | ci.lb    |
|----------|----------|---------|---------|----|--------|----------|
| intrcpt  | 17.1616  | 16.2628 | 1.0553  | 3  | 0.3688 | -34.5937 |
| IFN      | -9.6844  | 9.0286  | -1.0726 | 3  | 0.3620 | -38.4174 |
| IL_2     | -8.3898  | 9.5134  | -0.8819 | 3  | 0.4428 | -38.6658 |
| IFN:IL_2 | 5.1176   | 5.1389  | 0.9959  | 3  | 0.3927 | -11.2366 |
|          | ci.ub    |         |         |    |        |          |
| intrcpt  | 68.9169  |         |         |    |        |          |
| IFN      | 19.0485  |         |         |    |        |          |
| IL_2     | 21.8862  |         |         |    |        |          |

#### 4. Th1 — Immediate Post-Exercise

IFN:IL\_2 21.4719

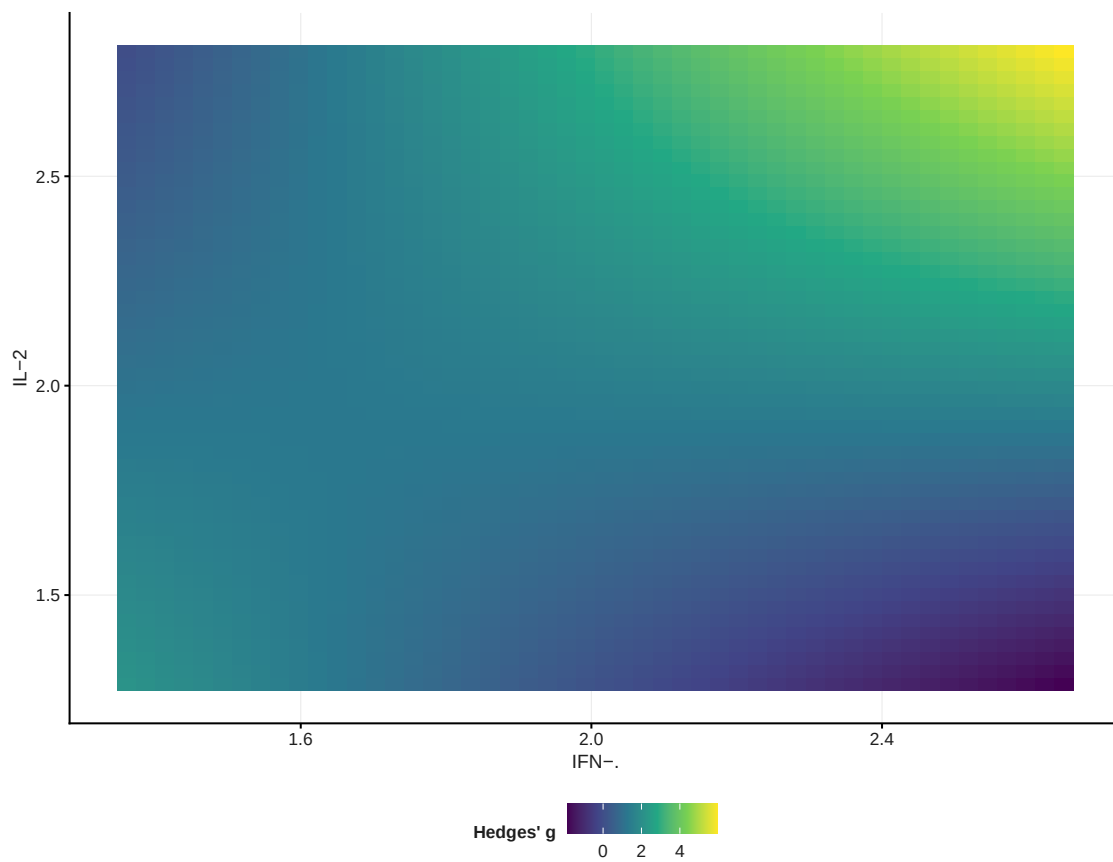
---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

### 4.9. Interaction Surface (IFN- $\times$ IL-2)



## 5. Th1 — ~2 h Post-Exercise

### 5.1. Data Import

The ~2 h window (`effect_size/Th1/pre_post+2h/Th1_pre_post+2h_r_050.csv`) contains few arms and no cytokine/BMI coverage, so it is reported for the main analysis, outlier check, funnel, forest, and asymmetry only — no moderator modelling is possible. No study is excluded a priori in this window.

### 5.2. Primary Meta-Analysis

#### 5.2.1. {meta}

Number of studies:  $k = 5$

|                           | SMD     | 95% CI            | t     |
|---------------------------|---------|-------------------|-------|
| Random effects model (HK) | -0.4137 | [-1.0063; 0.1789] | -1.94 |
| Prediction interval       |         | [-1.5190; 0.6917] |       |
|                           | p-value |                   |       |
| Random effects model (HK) | 0.1247  |                   |       |
| Prediction interval       |         |                   |       |

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.1110$  [0.0000; 1.8375];  $\tau = 0.3332$  [0.0000; 1.3555]  
 $I^2 = 42.5\%$  [0.0%; 78.9%];  $H = 1.32$  [1.00; 2.17]

Test of heterogeneity:

| Q    | d.f. | p-value |
|------|------|---------|
| 6.96 | 4    | 0.1382  |

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model ( $df = 4$ )
- Prediction interval based on t-distribution ( $df = 4$ )

#### 5.2.2. {metafor}

Random-Effects Model ( $k = 5$ ;  $\tau^2$  estimator: HE)

```

tau^2 (estimated amount of total heterogeneity): 0.1110 (SE = 0.1699)
tau (square root of estimated tau^2 value):      0.3332
I^2 (total heterogeneity / total variability):   46.86%
H^2 (total variability / sampling variability):  1.88

```

Test for Heterogeneity:

Q(df = 4) = 6.9560, p-val = 0.1382

Model Results:

```

estimate      se      tval  df    pval    ci.lb    ci.ub
-0.4137  0.2134  -1.9381   4   0.1247  -1.0063   0.1789

```

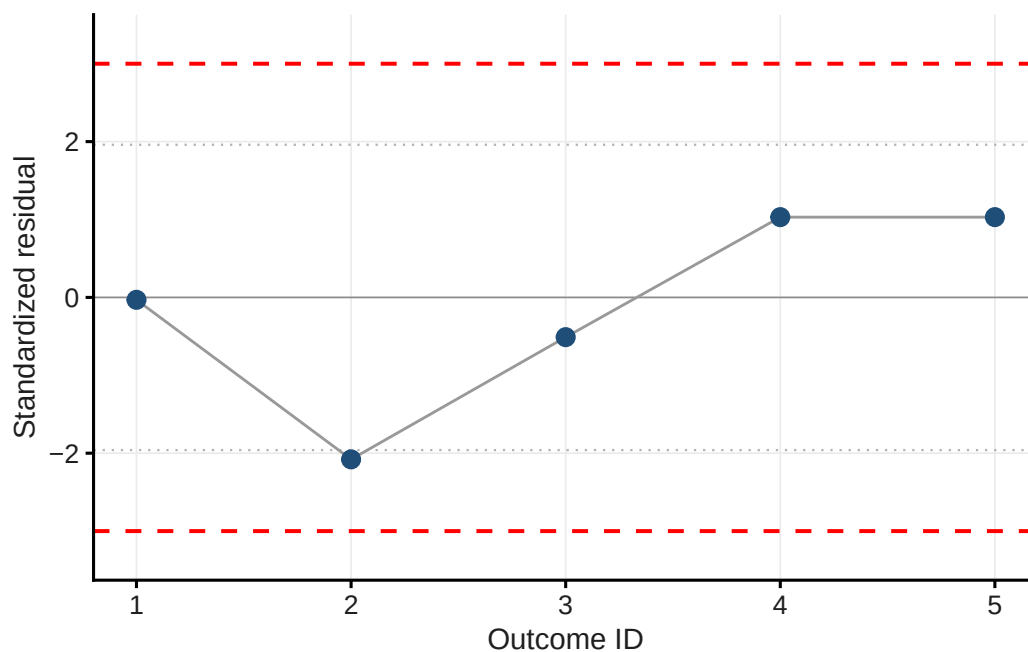
---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

### 5.3. Outlier Detection

**Table 5.1.:** Standardised residuals

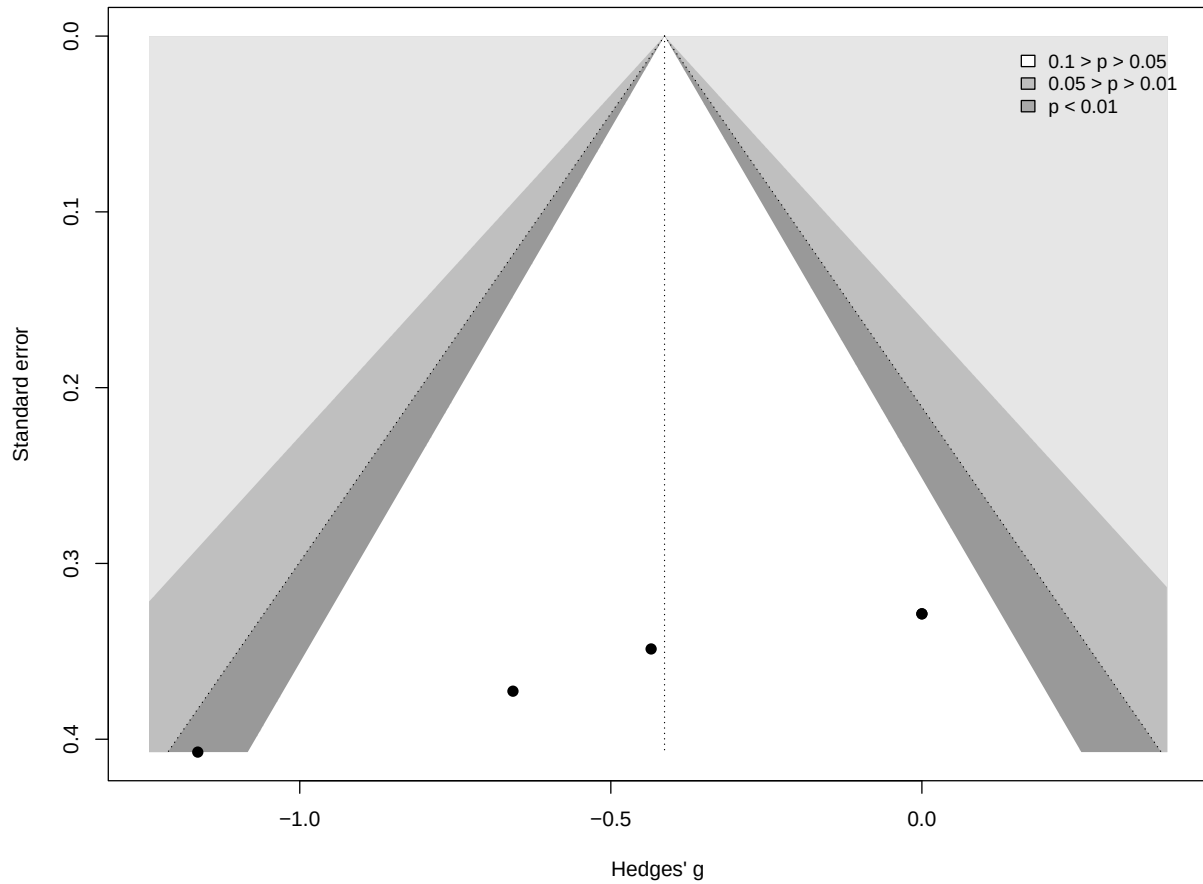
| Short_reference          | g      | std_res |
|--------------------------|--------|---------|
| Ibfelt et al. (2002)     | -0.435 | -0.031  |
| Steensberg et al. (2001) | -1.164 | -2.079  |
| Lancaster et al. (2005)a | -0.657 | -0.510  |
| Lancaster et al. (2005)b | 0.000  | 1.031   |
| Lancaster et al. (2005)c | 0.000  | 1.031   |



#### **i** Note

No outcome has standardized residuals above 3 or below -3.

## 5.4. Funnel Plot & Influential Cases



**Table 5.2.:** DFBETAS influence diagnostics

| Short_reference          | g      | dfbetas     |
|--------------------------|--------|-------------|
| Ibfelt et al. (2002)     | -0.435 | -0.08474579 |
| Steensberg et al. (2001) | -1.164 | -0.87614442 |
| Lancaster et al. (2005)a | -0.657 | -0.38036241 |
| Lancaster et al. (2005)b | 0.000  | 0.73086010  |
| Lancaster et al. (2005)c | 0.000  | 0.73086010  |

One value below  $-1$  and one above  $1$  were identified as influential cases. Neither is an outlier (Kostrzewa-Nowak & Nowak (2020a)a and Tsubakihara et al. (2012)); all outcomes are retained.

## 5.5. Forest Plot

#| fig-width: 8 #| fig-height: 4 #| out-width: 100% forest\_main(m1)

## 5.6. Asymmetry Test

**Note:** the number of studies ( $k = 5$ ) is too small to reliably test for small-study effects (minimum  $k = 10$ ; Borenstein et al., 2009). No Egger's test or trim-and-fill is performed.

## 6. Th1 — 17 h Post-Exercise

### 6.1. Data Import

### 6.2. Primary Meta-Analysis

As in the immediate window, **Kostrzewa-Nowak et al. (2019)** is removed before pooling. Outlier detection then flags **Steensberg et al. (2001)**, which is both an outlier and an influential case and is therefore withdrawn from the downstream analyses (m2\_1).

#### 6.2.1. {meta}

Number of studies: k = 11

|                           | SMD     | 95% CI            | t    |
|---------------------------|---------|-------------------|------|
| Random effects model (HK) | 1.8184  | [ 0.8759; 2.7610] | 4.30 |
| Prediction interval       |         | [-2.1567; 5.7935] |      |
|                           | p-value |                   |      |
| Random effects model (HK) | 0.0016  |                   |      |
| Prediction interval       |         |                   |      |

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 2.8848$  [0.3797; 7.3113];  $\tau = 1.6985$  [0.6162; 2.7039]  
 $I^2 = 83.2\%$  [71.3%; 90.1%];  $H = 2.44$  [1.87; 3.18]

Test of heterogeneity:

| Q     | d.f. | p-value  |
|-------|------|----------|
| 59.37 | 10   | < 0.0001 |

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model (df = 10)
- Prediction interval based on t-distribution (df = 10)

#### 6.2.2. {metafor}

Random-Effects Model (k = 11;  $\tau^2$  estimator: HE)

```
tau^2 (estimated amount of total heterogeneity): 2.8848 (SE = 2.0141)
tau (square root of estimated tau^2 value):      1.6985
I^2 (total heterogeneity / total variability):    95.32%
H^2 (total variability / sampling variability):    21.39
```

Test for Heterogeneity:

```
Q(df = 10) = 59.3676, p-val < .0001
```

Model Results:

```
estimate      se      tval  df      pval   ci.lb   ci.ub
    1.8184  0.4230  4.2986  10   0.0016  0.8759  2.7610  **
```

---

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

### 6.2.3. Adapted Modell {meta}

#### Caution

Re-Fit: Removed Konstrezw-Nowak et al. (2019)

Number of studies: k = 10

```

                        SMD           95% CI      t
Random effects model (HK) 1.6316 [ 0.8788; 2.3845] 4.90
Prediction interval                [-0.8087; 4.0720]
                        p-value
Random effects model (HK) 0.0008
Prediction interval
```

Quantifying heterogeneity (with 95% CIs):

```
tau^2 = 1.0428 [0.3167; 3.7647]; tau = 1.0212 [0.5628; 1.9403]
I^2 = 83.8% [71.7%; 90.7%]; H = 2.48 [1.88; 3.28]
```

Test of heterogeneity:

```
Q d.f.  p-value
55.52   9 < 0.0001
```

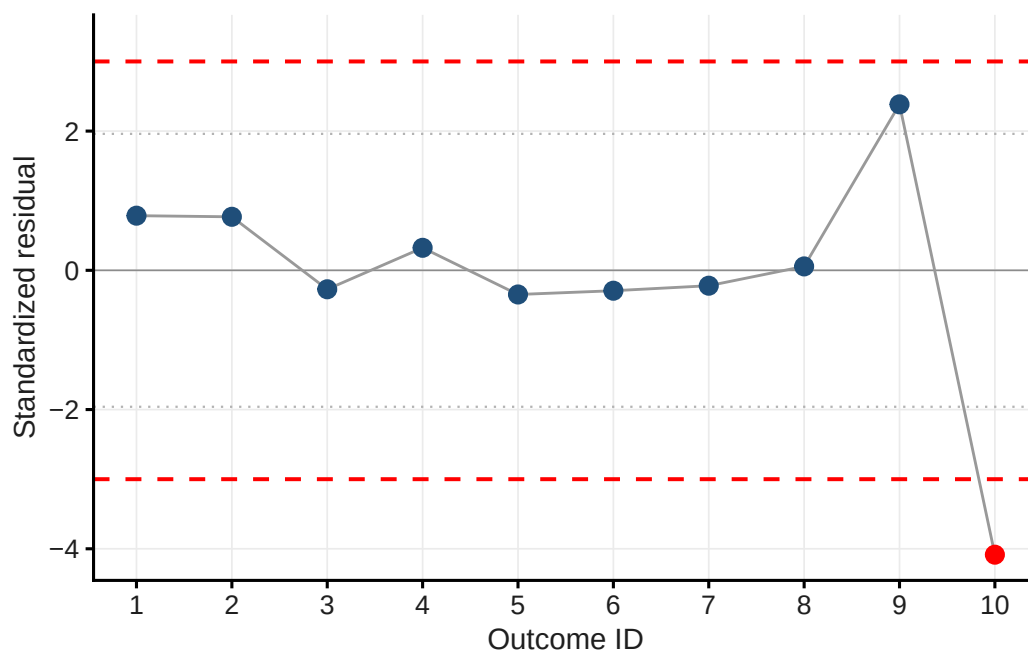
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for tau^2
- Q-Profile method for confidence interval of tau^2 and tau
- Calculation of I^2 based on Q
- Hartung-Knapp adjustment for random effects model (df = 9)
- Prediction interval based on t-distribution (df = 9)

## 6.3. Outlier Detection

**Table 6.1.:** Standardised residuals

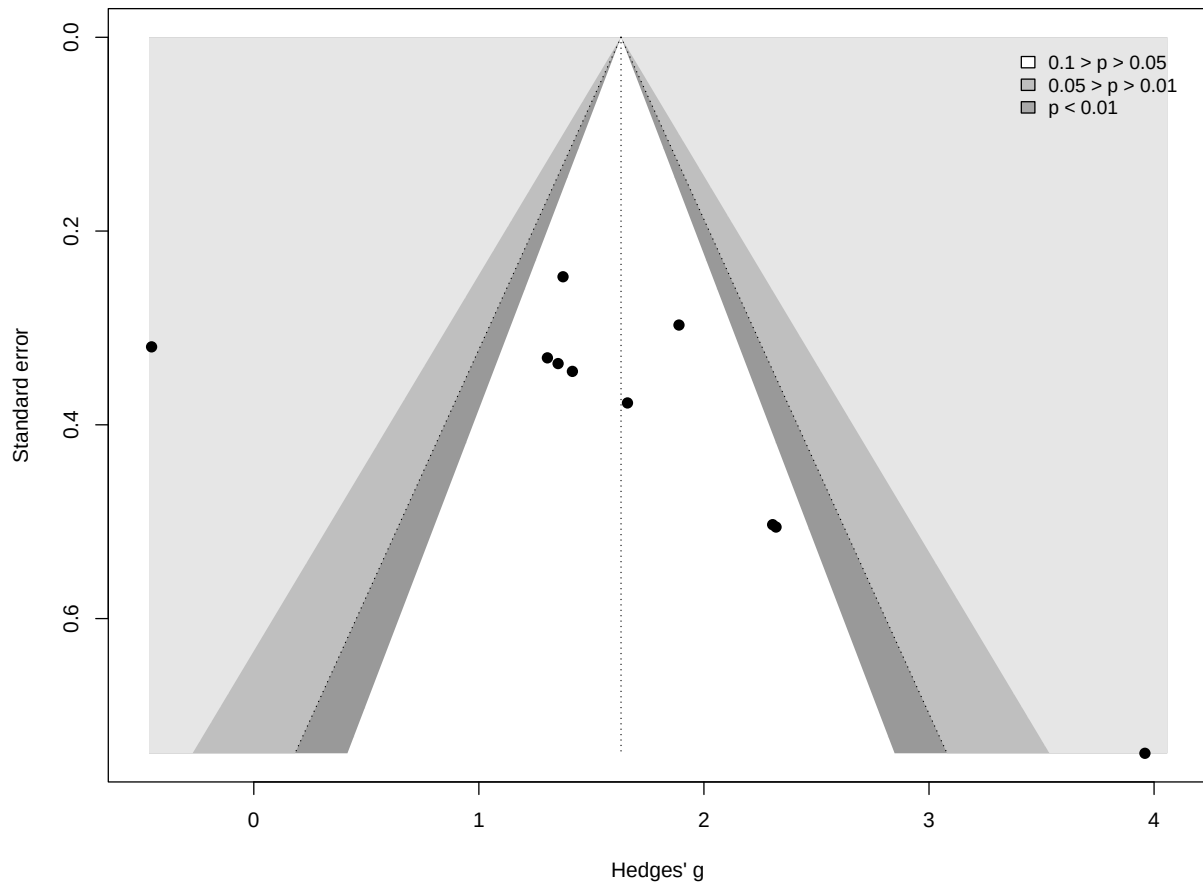
| Short_reference                    | g      | std_res |
|------------------------------------|--------|---------|
| Kostrzewska-Nowak & Nowak (2018)a  | 2.320  | 0.785   |
| Kostrzewska-Nowak & Nowak (2018)b  | 2.305  | 0.769   |
| Kostrzewska-Nowak & Nowak (2020a)a | 1.374  | -0.272  |
| Kostrzewska-Nowak & Nowak (2020a)b | 1.889  | 0.323   |
| Kostrzewska-Nowak & Nowak (2020b)a | 1.305  | -0.347  |
| Kostrzewska-Nowak & Nowak (2020b)b | 1.352  | -0.293  |
| Kostrzewska-Nowak & Nowak (2020b)c | 1.416  | -0.221  |
| Kostrzewska-Nowak & Nowak (2020b)d | 1.660  | 0.056   |
| Kostrzewska-Nowak & Nowak (2020b)e | 3.959  | 2.384   |
| Steensberg et al. (2001)           | -0.454 | -4.085  |



#### **i** Note

One outcome has standardized residuals below -3 (Steensberg et al. (2001)).

## 6.4. Funnel Plot & Influential Cases



**Table 6.2.:** DFBETAS influence diagnostics

| Short_reference                  | g      | dfbetas     |
|----------------------------------|--------|-------------|
| Kostrzewa-Nowak & Nowak (2018)a  | 2.320  | 0.42461320  |
| Kostrzewa-Nowak & Nowak (2018)b  | 2.305  | 0.42202631  |
| Kostrzewa-Nowak & Nowak (2020a)a | 1.374  | -0.04511648 |
| Kostrzewa-Nowak & Nowak (2020a)b | 1.889  | 0.72827130  |
| Kostrzewa-Nowak & Nowak (2020b)a | 1.305  | -0.10227485 |
| Kostrzewa-Nowak & Nowak (2020b)b | 1.352  | -0.04589921 |
| Kostrzewa-Nowak & Nowak (2020b)c | 1.416  | 0.02298307  |
| Kostrzewa-Nowak & Nowak (2020b)d | 1.660  | 0.22848951  |
| Kostrzewa-Nowak & Nowak (2020b)e | 3.959  | 0.53549707  |
| Steensberg et al. (2001)         | -0.454 | -2.29509141 |

### Caution

Steensberg et al. (2001) is  $< -1$  (influential) and also an outlier  $\rightarrow$  withdrawn.

### 6.4.1. Adapted Model

Number of studies:  $k = 9$

|                           | SMD      | 95% CI           | t    |
|---------------------------|----------|------------------|------|
| Random effects model (HK) | 1.8367   | [1.2690; 2.4043] | 7.46 |
| Prediction interval       |          | [0.0315; 3.6418] |      |
|                           | p-value  |                  |      |
| Random effects model (HK) | < 0.0001 |                  |      |
| Prediction interval       |          |                  |      |

Quantifying heterogeneity (with 95% CIs):

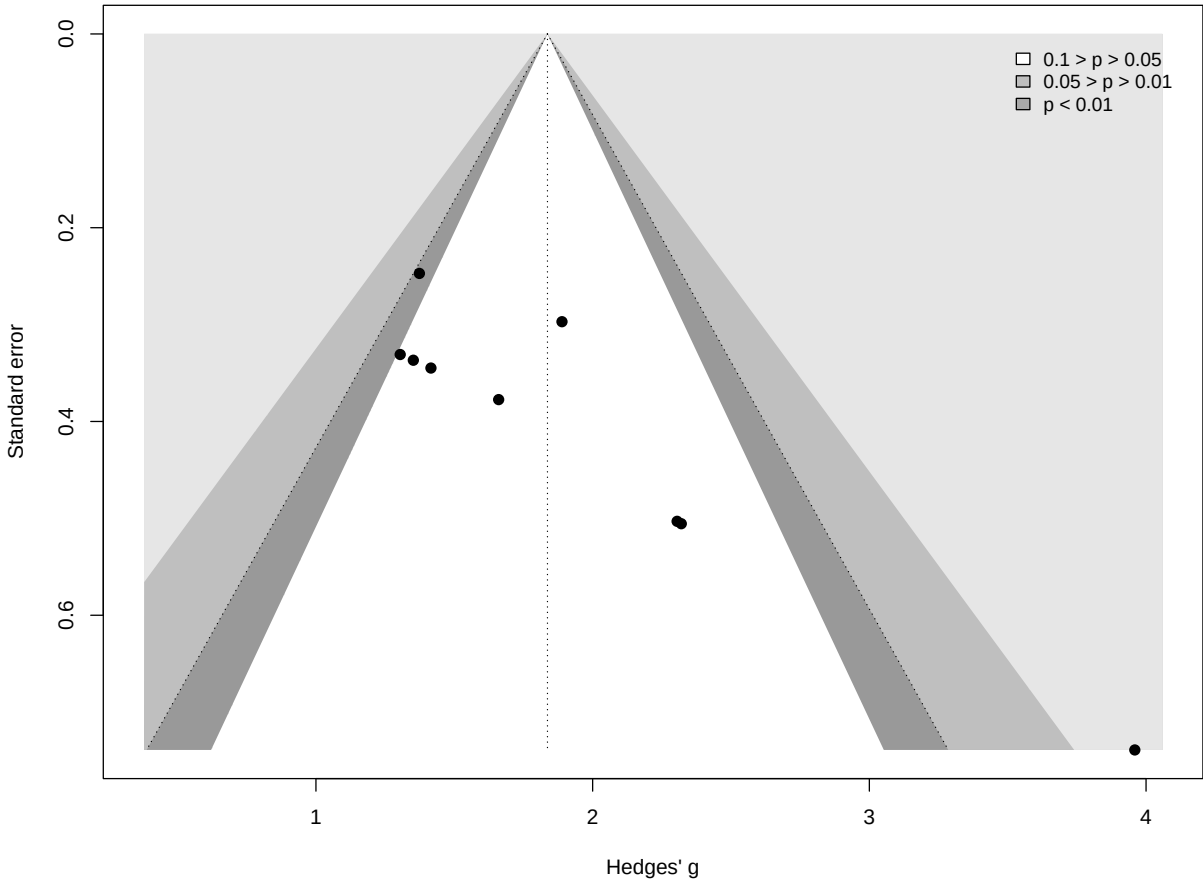
$\tau^2 = 0.5349$  [0.0000; 2.2412];  $\tau = 0.7314$  [0.0000; 1.4971]  
 $I^2 = 54.2\%$  [2.8%; 78.4%];  $H = 1.48$  [1.01; 2.15]

Test of heterogeneity:

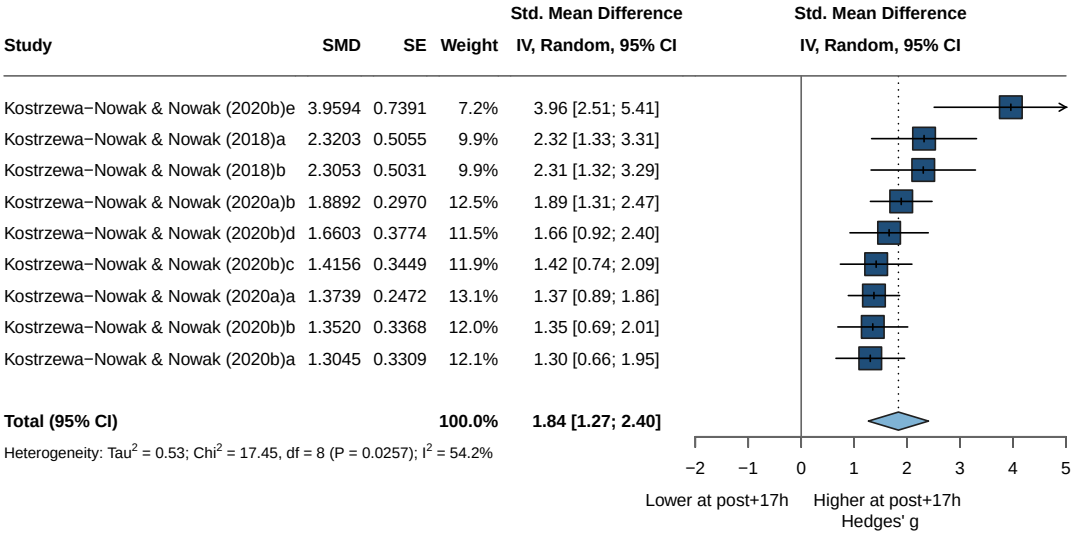
| Q     | d.f. | p-value |
|-------|------|---------|
| 17.45 | 8    | 0.0257  |

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model ( $df = 8$ )
- Prediction interval based on t-distribution ( $df = 8$ )



## 6.5. Forest Plot



## 6.6. Asymmetry Test

### 6.6.1. Egger's regression test

### 6.6.2. Trim and Fill

Estimated number of missing studies on the left side: 0 (SE = 1.9045)

Random-Effects Model (k = 9; tau<sup>2</sup> estimator: HE)

tau<sup>2</sup> (estimated amount of total heterogeneity): 0.5349 (SE = 0.3673)

tau (square root of estimated tau<sup>2</sup> value): 0.7314

I<sup>2</sup> (total heterogeneity / total variability): 80.24%

H<sup>2</sup> (total variability / sampling variability): 5.06

Test for Heterogeneity:

Q(df = 8) = 17.4510, p-val = 0.0257

Model Results:

| estimate | se     | zval   | pval   | ci.lb  | ci.ub  |     |
|----------|--------|--------|--------|--------|--------|-----|
| 1.8367   | 0.2791 | 6.5818 | <.0001 | 1.2897 | 2.3836 | *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

 No missing studies

## 6.7. Moderator Analyses

### 6.7.1. Risk of Bias

#### 6.7.1.1. Model

Number of studies: k = 9

Quantifying heterogeneity (with 95% CIs):

tau<sup>2</sup> = 0.5349 [0.0000; 2.2412]; tau = 0.7314 [0.0000; 1.4971]

I<sup>2</sup> = 54.2% [2.8%; 78.4%]; H = 1.48 [1.01; 2.15]

Test of heterogeneity:

| Q     | d.f. | p-value |
|-------|------|---------|
| 17.45 | 8    | 0.0257  |

Results for subgroups (random effects model (HK)):

| k | SMD | 95% CI | tau <sup>2</sup> | tau |
|---|-----|--------|------------------|-----|
|---|-----|--------|------------------|-----|

```

RoB = Moderate    2 2.3128 [ 2.2169; 2.4086]      0      0
RoB = Serious     2 1.6052 [-1.6511; 4.8615] 0.0581 0.2410
RoB = Low         5 1.8263 [ 0.5408; 3.1117] 1.0891 1.0436
                  Q   I^2
RoB = Moderate    0.00  0.0%
RoB = Serious     1.78 43.8%
RoB = Low         11.76 66.0%

```

Test for subgroup differences (random effects model (HK)):

```

          Q d.f. p-value
Between groups 8.72    2  0.0128

```

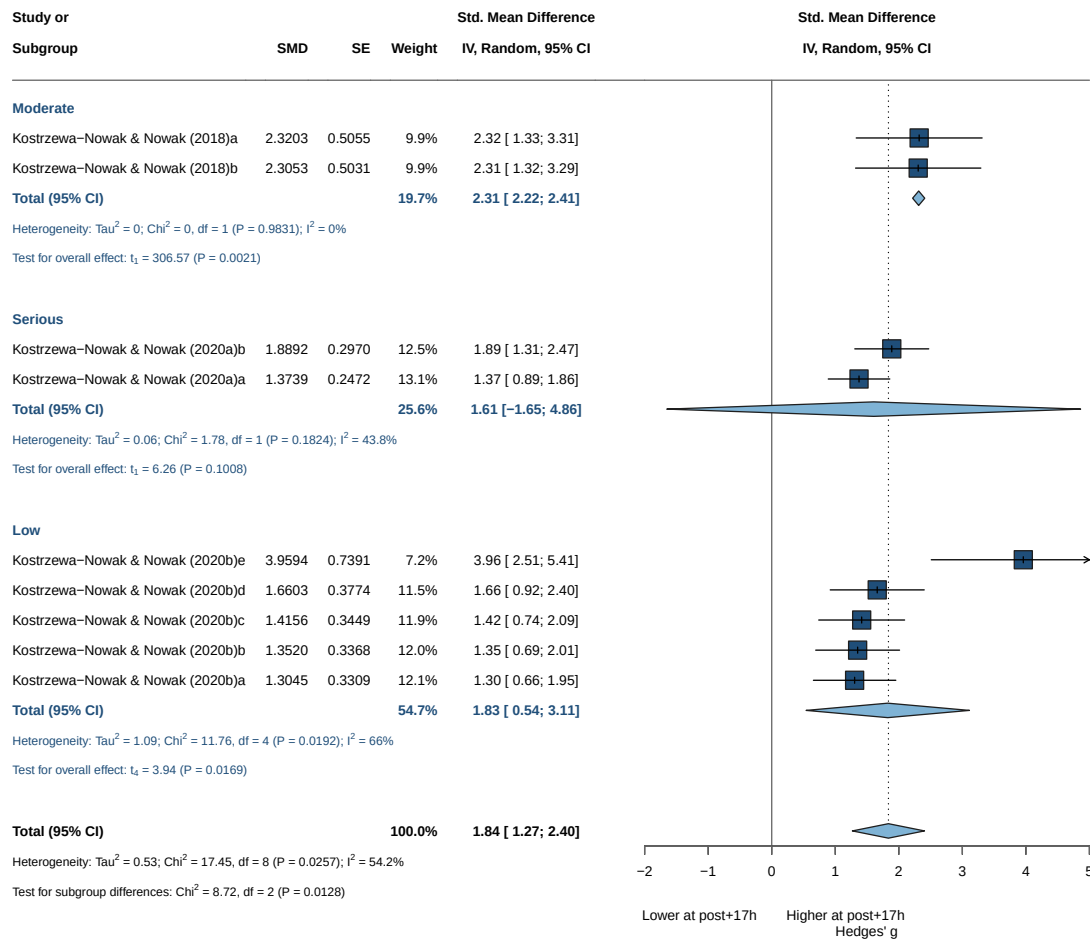
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

#### Note

Not significant results

### 6.7.1.2. Forrest Plot



**Intensity:** all retained arms share the same intensity, so no intensity subgroup analysis is performed.

## 6.7.2. Continuous Moderators (Meta-Regression)

### 6.7.2.1. Age

Mixed-Effects Model ( $k = 9$ ;  $\tau^2$  estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -9.3856 | 19.2634  | 24.7713 | 25.3629 | 29.5713 |

$\tau^2$  (estimated amount of residual heterogeneity): 0.5156 (SE = 0.3831)  
 $\tau$  (square root of estimated  $\tau^2$  value): 0.7181  
 $I^2$  (residual heterogeneity / unaccounted variability): 79.71%  
 $H^2$  (unaccounted variability / sampling variability): 4.93  
 $R^2$  (amount of heterogeneity accounted for): 3.61%

Test for Residual Heterogeneity:

QE( $df = 7$ ) = 15.6877,  $p$ -val = 0.0281

## 6. Th1 — 17 h Post-Exercise

Test of Moderators (coefficient 2):

$F(df1 = 1, df2 = 7) = 1.0164, p\text{-val} = 0.3470$

Model Results:

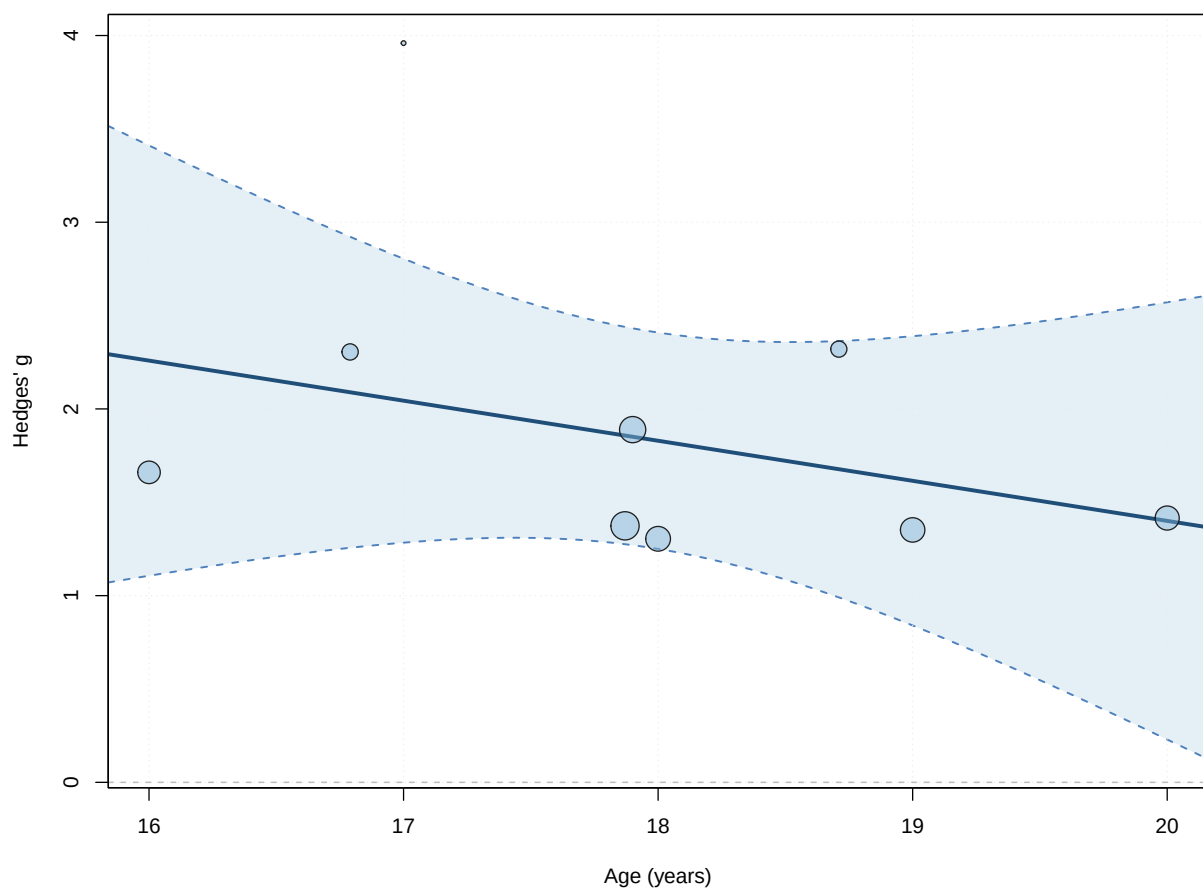
|         | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|---------|----|--------|---------|-------|
| intrcpt | 5.6926   | 3.8350 | 1.4844  | 7  | 0.1813 | -3.3758 |       |
| age     | -0.2146  | 0.2129 | -1.0082 | 7  | 0.3470 | -0.7180 |       |
|         |          |        |         |    |        |         |       |
| intrcpt | 14.7609  |        |         |    |        |         |       |
| age     | 0.2888   |        |         |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant



### 6.7.2.2. BMI

Mixed-Effects Model (k = 9; tau<sup>2</sup> estimator: HE)

|         |          |         |         |         |
|---------|----------|---------|---------|---------|
| logLik  | deviance | AIC     | BIC     | AICc    |
| -9.3661 | 19.2242  | 24.7321 | 25.3238 | 29.5321 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.5147 (SE = 0.3872)  
 tau (square root of estimated tau<sup>2</sup> value): 0.7175  
 I<sup>2</sup> (residual heterogeneity / unaccounted variability): 78.61%  
 H<sup>2</sup> (unaccounted variability / sampling variability): 4.68  
 R<sup>2</sup> (amount of heterogeneity accounted for): 3.77%

Test for Residual Heterogeneity:

QE(df = 7) = 15.9237, p-val = 0.0258

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 7) = 1.0645, p-val = 0.3365

Model Results:

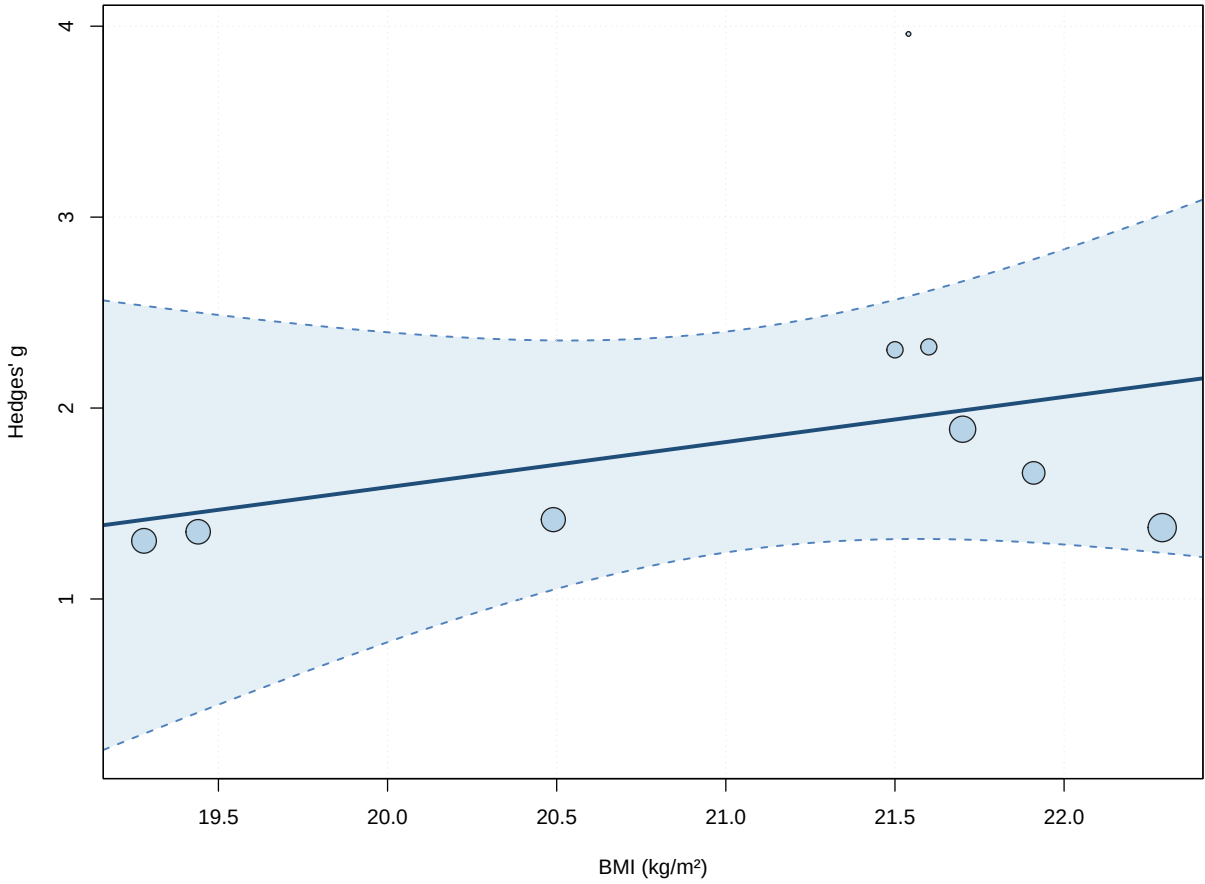
|         | estimate | se     | tval    | df | pval   | ci.lb    | ci.ub |
|---------|----------|--------|---------|----|--------|----------|-------|
| intrcpt | -3.1460  | 4.8331 | -0.6509 | 7  | 0.5359 | -14.5743 |       |
| bmi     | 0.2366   | 0.2293 | 1.0317  | 7  | 0.3365 | -0.3056  |       |
| intrcpt | 8.2824   |        |         |    |        |          |       |
| bmi     | 0.7787   |        |         |    |        |          |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

#### **i** Note

Not significant results



6.7.2.3. Duration

Random-Effects Model (k = 7; tau^2 estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -8.5222 | 15.9889  | 21.0443 | 20.9362 | 24.0443 |

tau^2 (estimated amount of total heterogeneity): 0.6770 (SE = 0.5235)  
tau (square root of estimated tau^2 value): 0.8228  
I^2 (total heterogeneity / total variability): 80.13%  
H^2 (total variability / sampling variability): 5.03

Test for Heterogeneity:  
Q(df = 6) = 15.4832, p-val = 0.0168

Model Results:

| estimate | se     | tval   | df | pval   | ci.lb  | ci.ub  |    |
|----------|--------|--------|----|--------|--------|--------|----|
| 1.9274   | 0.3239 | 5.9501 | 6  | 0.0010 | 1.1348 | 2.7200 | ** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

#### **i** Note

Not significant results

#### 6.7.2.4. Dose

Random-Effects Model (k = 7; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -8.5222 | 15.9889  | 21.0443 | 20.9362 | 24.0443 |

tau<sup>2</sup> (estimated amount of total heterogeneity): 0.6770 (SE = 0.5235)

tau (square root of estimated tau<sup>2</sup> value): 0.8228

I<sup>2</sup> (total heterogeneity / total variability): 80.13%

H<sup>2</sup> (total variability / sampling variability): 5.03

Test for Heterogeneity:

Q(df = 6) = 15.4832, p-val = 0.0168

Model Results:

| estimate | se     | tval   | df | pval   | ci.lb  | ci.ub     |
|----------|--------|--------|----|--------|--------|-----------|
| 1.9274   | 0.3239 | 5.9501 | 6  | 0.0010 | 1.1348 | 2.7200 ** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

#### **i** Note

Not significant results

#### 6.7.2.5. IFN

Mixed-Effects Model (k = 7; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -8.6058 | 18.6411  | 23.2116 | 23.0493 | 31.2116 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.8890 (SE = 0.6816)

tau (square root of estimated tau<sup>2</sup> value): 0.9429

I<sup>2</sup> (residual heterogeneity / unaccounted variability): 87.53%

H<sup>2</sup> (unaccounted variability / sampling variability): 8.02

R<sup>2</sup> (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

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QE(df = 5) = 13.1061, p-val = 0.0224

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.0320, p-val = 0.8651

Model Results:

|         | estimate | se     | tval    | df | pval   | ci.lb   |
|---------|----------|--------|---------|----|--------|---------|
| intrcpt | 1.9455   | 1.1277 | 1.7252  | 5  | 0.1451 | -0.9533 |
| IFN     | -0.0645  | 0.3608 | -0.1788 | 5  | 0.8651 | -0.9920 |
|         | ci.ub    |        |         |    |        |         |
| intrcpt | 4.8442   |        |         |    |        |         |
| IFN     | 0.8630   |        |         |    |        |         |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

### Note

Not significant results

## 6.7.2.6. IL\_4

Mixed-Effects Model (k = 7; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -8.6353 | 18.7000  | 23.2706 | 23.1083 | 31.2706 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.9003 (SE = 0.6946)  
tau (square root of estimated tau<sup>2</sup> value): 0.9488  
I<sup>2</sup> (residual heterogeneity / unaccounted variability): 87.21%  
H<sup>2</sup> (unaccounted variability / sampling variability): 7.82  
R<sup>2</sup> (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 5) = 13.3188, p-val = 0.0206

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.0105, p-val = 0.9225

Model Results:

|         | estimate | se     | tval    | df | pval   | ci.lb   |
|---------|----------|--------|---------|----|--------|---------|
| intrcpt | 1.7842   | 0.4532 | 3.9366  | 5  | 0.0110 | 0.6191  |
| IL_4    | -0.0606  | 0.5927 | -0.1023 | 5  | 0.9225 | -1.5842 |
|         | ci.ub    |        |         |    |        |         |
| intrcpt | 2.9494   | *      |         |    |        |         |
| IL_4    | 1.4629   |        |         |    |        |         |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant results

**6.7.2.7. IL\_6**Mixed-Effects Model (k = 9; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -8.2134 | 16.9190  | 22.4269 | 23.0185 | 27.2269 |

|                                                                    |                      |
|--------------------------------------------------------------------|----------------------|
| tau <sup>2</sup> (estimated amount of residual heterogeneity):     | 0.3862 (SE = 0.3095) |
| tau (square root of estimated tau <sup>2</sup> value):             | 0.6214               |
| I <sup>2</sup> (residual heterogeneity / unaccounted variability): | 75.64%               |
| H <sup>2</sup> (unaccounted variability / sampling variability):   | 4.10                 |
| R <sup>2</sup> (amount of heterogeneity accounted for):            | 27.80%               |

Test for Residual Heterogeneity:

QE(df = 7) = 12.0828, p-val = 0.0979

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 7) = 3.2234, p-val = 0.1157

Model Results:

|         | estimate | se     | tval   | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|--------|----|--------|---------|-------|
| intrcpt | 1.4941   | 0.2761 | 5.4119 | 7  | 0.0010 | 0.8413  |       |
| IL_6    | 0.1709   | 0.0952 | 1.7954 | 7  | 0.1157 | -0.0542 |       |
|         |          |        |        |    |        |         |       |
| intrcpt | 2.1469   | ***    |        |    |        |         |       |
| IL_6    | 0.3960   |        |        |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant results

**6.7.2.8. IL\_10**Mixed-Effects Model (k = 9; tau<sup>2</sup> estimator: HE)

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| logLik   | deviance | AIC     | BIC     | AICc    |
|----------|----------|---------|---------|---------|
| -10.1243 | 20.7407  | 26.2486 | 26.8403 | 31.0486 |

```
tau^2 (estimated amount of residual heterogeneity):    0.6430 (SE = 0.4456)
tau (square root of estimated tau^2 value):           0.8019
I^2 (residual heterogeneity / unaccounted variability): 83.76%
H^2 (unaccounted variability / sampling variability):  6.16
R^2 (amount of heterogeneity accounted for):           0.00%
```

Test for Residual Heterogeneity:

QE(df = 7) = 17.0898, p-val = 0.0168

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 7) = 0.0868, p-val = 0.7768

Model Results:

|         | estimate | se     | tval   | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|--------|----|--------|---------|-------|
| intrcpt | 1.7784   | 0.3578 | 4.9701 | 7  | 0.0016 | 0.9323  |       |
| IL_10   | 0.0714   | 0.2424 | 0.2946 | 7  | 0.7768 | -0.5017 |       |
|         |          |        |        |    |        |         |       |
| intrcpt | 2.6245   |        |        |    |        |         | **    |
| IL_10   | 0.6445   |        |        |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

### Note

Not significant results

## 6.7.2.9. IL\_2

Mixed-Effects Model (k = 7; tau^2 estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -8.6267 | 18.6829  | 23.2535 | 23.0912 | 31.2535 |

```
tau^2 (estimated amount of residual heterogeneity):    0.9041 (SE = 0.6941)
tau (square root of estimated tau^2 value):           0.9509
I^2 (residual heterogeneity / unaccounted variability): 87.75%
H^2 (unaccounted variability / sampling variability):  8.16
R^2 (amount of heterogeneity accounted for):           0.00%
```

Test for Residual Heterogeneity:

QE(df = 5) = 11.8338, p-val = 0.0371

Test of Moderators (coefficient 2):

$F(df1 = 1, df2 = 5) = 0.0501, p\text{-val} = 0.8317$

Model Results:

|         | estimate | se     | tval   | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|--------|----|--------|---------|-------|
| intrcpt | 1.6362   | 0.6316 | 2.5905 | 5  | 0.0488 | 0.0126  |       |
| IL_2    | 0.0662   | 0.2956 | 0.2239 | 5  | 0.8317 | -0.6937 |       |
| intrcpt | 3.2599   |        |        |    |        |         | *     |
| IL_2    | 0.8261   |        |        |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

### **i** Note

Not significant results

## 6.8. Multivariate Models

Mixed-Effects Model (k = 7; tau<sup>2</sup> estimator: HE)

|                                                                    |                      |
|--------------------------------------------------------------------|----------------------|
| tau <sup>2</sup> (estimated amount of residual heterogeneity):     | 1.4592 (SE = 1.3845) |
| tau (square root of estimated tau <sup>2</sup> value):             | 1.2080               |
| I <sup>2</sup> (residual heterogeneity / unaccounted variability): | 90.49%               |
| H <sup>2</sup> (unaccounted variability / sampling variability):   | 10.52                |
| R <sup>2</sup> (amount of heterogeneity accounted for):            | 0.00%                |

Test for Residual Heterogeneity:

$QE(df = 3) = 11.4927, p\text{-val} = 0.0093$

Test of Moderators (coefficients 2:4):

$F(df1 = 3, df2 = 3) = 0.0705, p\text{-val} = 0.9719$

Model Results:

|          | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|----------|----------|--------|---------|----|--------|---------|-------|
| intrcpt  | 1.7103   | 2.6663 | 0.6415  | 3  | 0.5669 | -6.7751 |       |
| IFN      | -0.1372  | 0.8430 | -0.1627 | 3  | 0.8811 | -2.8200 |       |
| IL_2     | 0.5964   | 1.9677 | 0.3031  | 3  | 0.7816 | -5.6658 |       |
| IFN:IL_2 | -0.0966  | 0.4584 | -0.2108 | 3  | 0.8466 | -1.5554 |       |
| intrcpt  | 10.1958  |        |         |    |        |         |       |
| IFN      | 2.5457   |        |         |    |        |         |       |
| IL_2     | 6.8586   |        |         |    |        |         |       |
| IFN:IL_2 | 1.3621   |        |         |    |        |         |       |

---

6. *Th1 — 17 h Post-Exercise*

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

**Part II.**

**Th2**



## 7. Th2 — Immediate Post-Exercise

### 7.1. Data Import

The base analysis uses the pre-computed Hedges'  $g$  file for Th2 at  $r = 0.5$  (`effect_size/Th2/pre_post/Th2_pre_post_r_050`). Sensitivity analyses can substitute the  $r = 0.3 / 0.4 / 0.6 / 0.7$  files in the same folder. The pipeline column names are mapped to the analysis conventions (VE\_g, Sex, Age, BMI) by `load_es()`.

### 7.2. Primary Meta-Analysis

`metagen()` fits a random-effects model on the pre-computed effect sizes ( $g$ ,  $SE_g$ ) using the **Hedges-Olkin (HE)** <sup>2</sup> estimator and the **Knapp-Hartung** small-sample correction (`hakn = TRUE`). **Kostrzewa-Nowak et al. (2019)** is removed before pooling and the model re-fitted; `metafor::rma()` cross-validates.

#### 7.2.1. Full Model {meta}

Number of studies:  $k = 18$

|                           | SMD     | 95% CI            | t    |
|---------------------------|---------|-------------------|------|
| Random effects model (HK) | 0.3214  | [-0.0871; 0.7298] | 1.66 |
| Prediction interval       |         | [-1.9144; 2.5571] |      |
|                           | p-value |                   |      |
| Random effects model (HK) | 0.1152  |                   |      |
| Prediction interval       |         |                   |      |

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 1.0569$  [0.1238; 1.5639];  $\tau = 1.0281$  [0.3518; 1.2506]  
 $I^2 = 71.9\%$  [54.9%; 82.5%];  $H = 1.89$  [1.49; 2.39]

Test of heterogeneity:

| Q     | d.f. | p-value  |
|-------|------|----------|
| 60.51 | 17   | < 0.0001 |

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model ( $df = 17$ )
- Prediction interval based on t-distribution ( $df = 17$ )

### 7.2.2. Full Model {metafor}

Random-Effects Model (k = 18; tau<sup>2</sup> estimator: HE)

tau<sup>2</sup> (estimated amount of total heterogeneity): 1.0569 (SE = 0.4699)  
 tau (square root of estimated tau<sup>2</sup> value): 1.0281  
 I<sup>2</sup> (total heterogeneity / total variability): 93.23%  
 H<sup>2</sup> (total variability / sampling variability): 14.78

Test for Heterogeneity:

Q(df = 17) = 60.5052, p-val < .0001

Model Results:

| estimate | se     | tval   | df | pval   | ci.lb   | ci.ub  |
|----------|--------|--------|----|--------|---------|--------|
| 0.3214   | 0.1936 | 1.6601 | 17 | 0.1152 | -0.0871 | 0.7298 |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

### 7.2.3. Adapted Model {meta}

Number of studies: k = 17

|                           | SMD     | 95% CI            | t    |
|---------------------------|---------|-------------------|------|
| Random effects model (HK) | 0.3791  | [ 0.0865; 0.6718] | 2.75 |
| Prediction interval       |         | [-0.7743; 1.5326] |      |
|                           | p-value |                   |      |
| Random effects model (HK) | 0.0143  |                   |      |
| Prediction interval       |         |                   |      |

Quantifying heterogeneity (with 95% CIs):

tau<sup>2</sup> = 0.2749 [0.0825; 0.7087]; tau = 0.5243 [0.2873; 0.8418]  
 I<sup>2</sup> = 70.2% [51.2%; 81.8%]; H = 1.83 [1.43; 2.35]

Test of heterogeneity:

| Q     | d.f. | p-value  |
|-------|------|----------|
| 53.73 | 16   | < 0.0001 |

Details of meta-analysis methods:

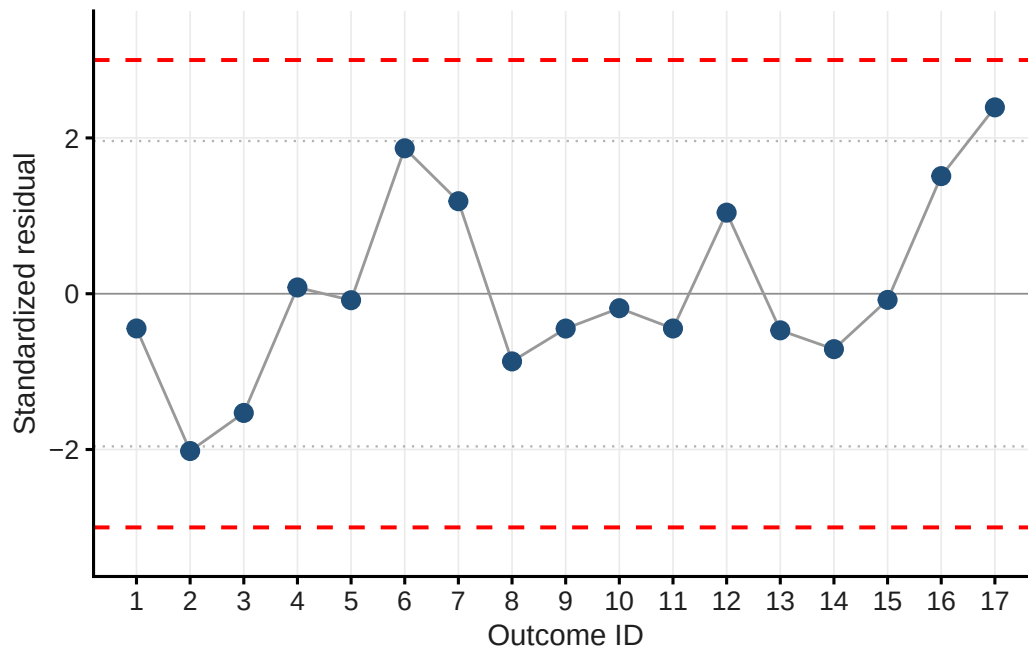
- Inverse variance method
- Hedges estimator for tau<sup>2</sup>
- Q-Profile method for confidence interval of tau<sup>2</sup> and tau
- Calculation of I<sup>2</sup> based on Q
- Hartung-Knapp adjustment for random effects model (df = 16)
- Prediction interval based on t-distribution (df = 16)

## 7.3. Outlier Detection

Leave-one-out standardised residuals are computed via `metaoutliers()`, which is passed the **variance** (`m3$seTE^2`). Studies with  $|z| > 3$  are flagged as outliers;  $|z| > 1.96$  warrants inspection.

**Table 7.1.:** Standardised residuals

| Short_reference                  | g      | std_res |
|----------------------------------|--------|---------|
| Ibfelt et al. (2002)             | 0.137  | -0.446  |
| Kostrzewa-Nowak & Nowak (2018)a  | -0.529 | -2.019  |
| Kostrzewa-Nowak & Nowak (2018)b  | -0.330 | -1.530  |
| Kostrzewa-Nowak & Nowak (2020a)a | 0.410  | 0.080   |
| Kostrzewa-Nowak & Nowak (2020a)b | 0.334  | -0.084  |
| Kostrzewa-Nowak & Nowak (2020b)a | 1.272  | 1.867   |
| Kostrzewa-Nowak & Nowak (2020b)b | 0.941  | 1.187   |
| Kostrzewa-Nowak & Nowak (2020b)c | -0.036 | -0.869  |
| Kostrzewa-Nowak & Nowak (2020b)d | 0.158  | -0.447  |
| Kostrzewa-Nowak & Nowak (2020b)e | 0.282  | -0.186  |
| Steensberg et al. (2001)         | 0.144  | -0.446  |
| Tsubakihara et al. (2012)        | 0.861  | 1.041   |
| Lancaster et al. (2005)a         | 0.124  | -0.470  |
| Lancaster et al. (2005)b         | 0.000  | -0.712  |
| Lancaster et al. (2005)c         | 0.329  | -0.079  |
| Harbaum et al. (2016)a           | 1.096  | 1.510   |
| Harbaum et al. (2016)b           | 1.829  | 2.392   |

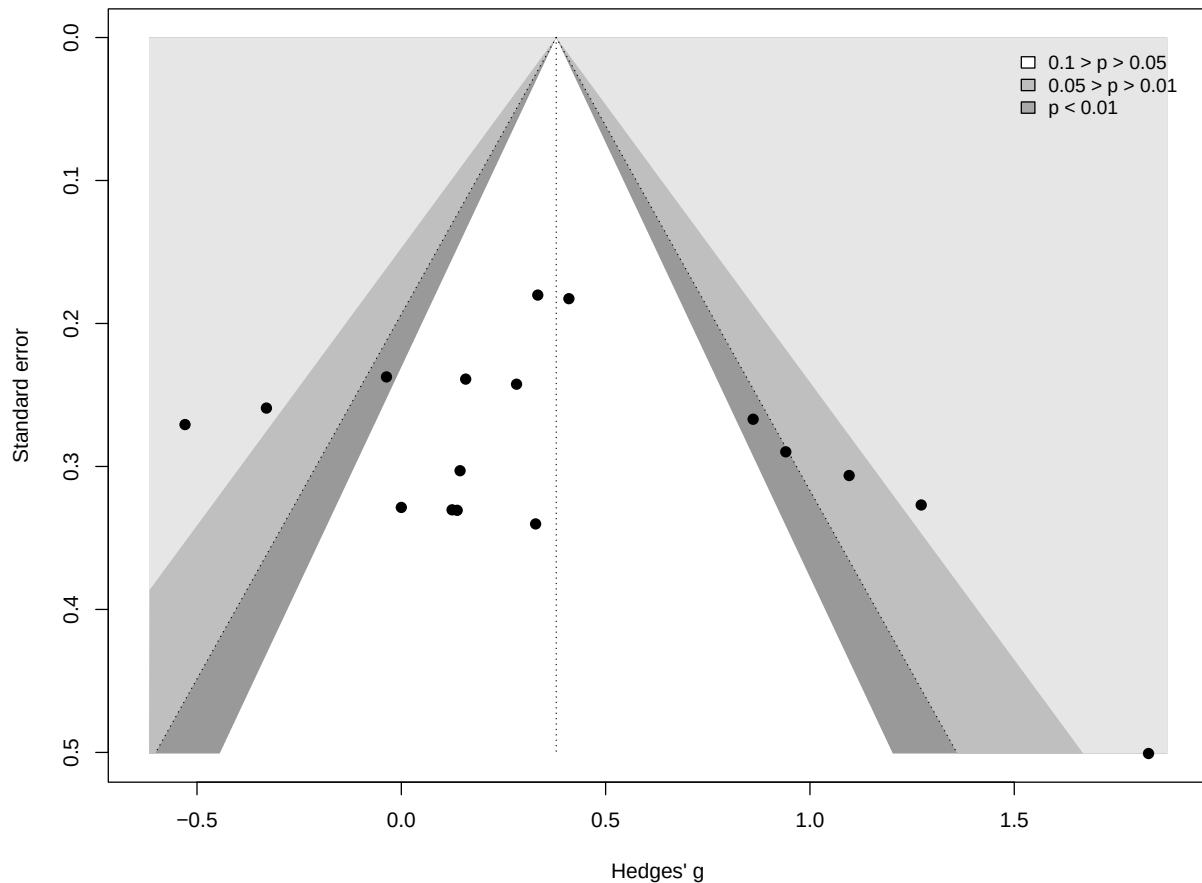


### Caution

One outcome has standardized residuals around -2 (Kostrzewa-Nowak et al. (2019)).

## 7.4. Funnel Plot & Influential Cases

The contour-enhanced funnel plot shows study estimates against their standard errors; shaded regions mark significance zones ( $\alpha = 0.10, 0.05, 0.01$ ). The `dfbetas` diagnostic from `metafor::rma.mv()` identifies **influential cases** — distinct from statistical outliers and not necessarily excluded.



**Table 7.2.:** DFBETAS influence diagnostics

| Short_reference                    | g      | dfbetas       |
|------------------------------------|--------|---------------|
| Ibfelt et al. (2002)               | 0.137  | -0.1178026325 |
| Kostrzewska-Nowak & Nowak (2018)a  | -0.529 | -0.8034118886 |
| Kostrzewska-Nowak & Nowak (2018)b  | -0.330 | -0.6766632755 |
| Kostrzewska-Nowak & Nowak (2020a)a | 0.410  | 0.1818284116  |
| Kostrzewska-Nowak & Nowak (2020a)b | 0.334  | 0.0132850848  |
| Kostrzewska-Nowak & Nowak (2020b)a | 1.272  | 0.5951183620  |
| Kostrzewska-Nowak & Nowak (2020b)b | 0.941  | 0.4971720392  |
| Kostrzewska-Nowak & Nowak (2020b)c | -0.036 | -0.4523407882 |
| Kostrzewska-Nowak & Nowak (2020b)d | 0.158  | -0.2088082167 |
| Kostrzewska-Nowak & Nowak (2020b)e | 0.282  | -0.0547346426 |
| Steensberg et al. (2001)           | 0.144  | -0.1362231954 |
| Tsubakihara et al. (2012)          | 0.861  | 0.5140815471  |
| Lancaster et al. (2005)a           | 0.124  | -0.1258018657 |
| Lancaster et al. (2005)b           | 0.000  | -0.2045859006 |

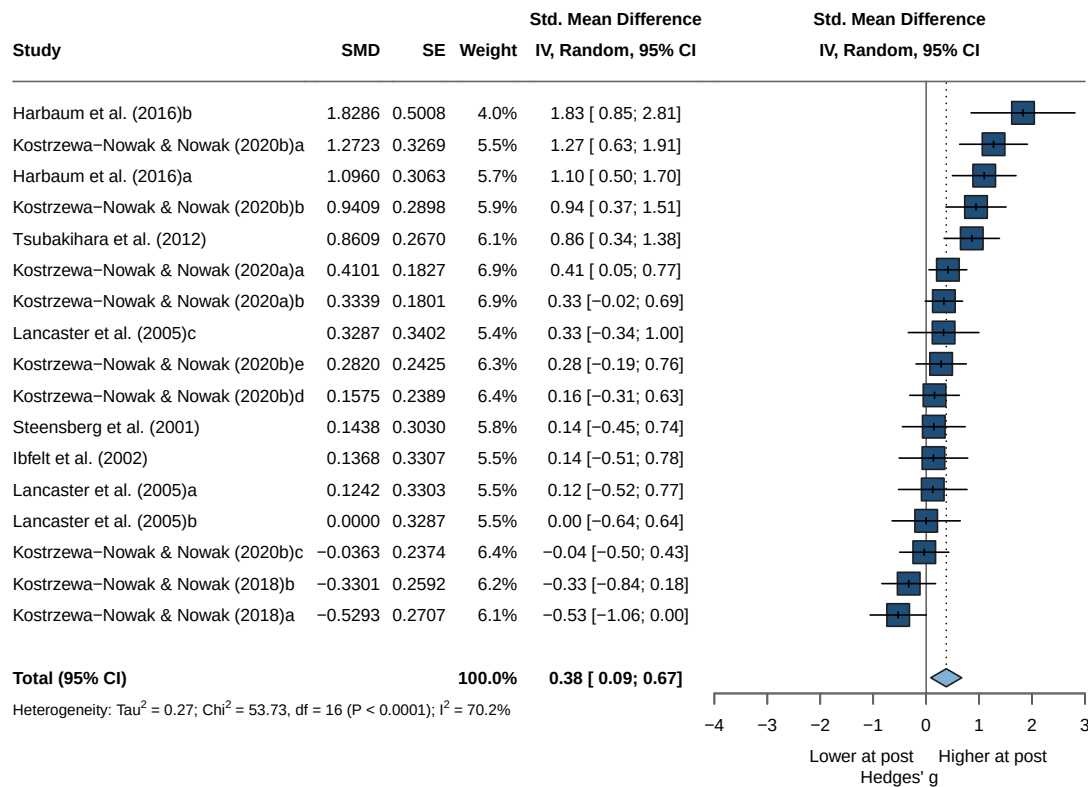
**Table 7.2.:** DFBETAS influence diagnostics

| Short_reference          | g     | dfbetas      |
|--------------------------|-------|--------------|
| Lancaster et al. (2005)c | 0.329 | 0.0003012762 |
| Harbaum et al. (2016)a   | 1.096 | 0.5544242443 |
| Harbaum et al. (2016)b   | 1.829 | 0.3938945044 |

 Tip

No influential case.

## 7.5. Forest Plot



## 7.6. Asymmetry Test

Egger's regression test (`method = "linreg"`) evaluates funnel asymmetry as a proxy for small-study/publication effects (one-tailed,  $p < .10$ ). Where asymmetry is suggested, a trim-and-fill analysis estimates potentially

missing studies.

Linear regression test of funnel plot asymmetry

Test result:  $t = 1.17$ ,  $df = 15$ ,  $p\text{-value} = 0.2595$

Bias estimate: 2.2167 (SE = 1.8916)

Details:

- multiplicative residual heterogeneity variance ( $\tau^2 = 3.2814$ )
- predictor: standard error
- weight: inverse variance
- reference: Egger et al. (1997), BMJ

#### **i** Note

No asymmetry -> no trim-and-fill

## 7.7. Moderator Analyses

### 7.7.1. Risk of Bias

Subgroup analysis by overall RoB rating tests whether study quality moderates the pooled effect; the between-subgroup statistic ( $Q_b$ ) indicates whether subgroup means differ.

#### 7.7.1.1. Model

Number of studies:  $k = 17$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.2749$  [0.0825; 0.7087];  $\tau = 0.5243$  [0.2873; 0.8418]

$I^2 = 70.2\%$  [51.2%; 81.8%];  $H = 1.83$  [1.43; 2.35]

Test of heterogeneity:

| Q     | d.f. | p-value  |
|-------|------|----------|
| 53.73 | 16   | < 0.0001 |

Results for subgroups (random effects model (HK)):

|                | k | SMD     | 95% CI            | $\tau^2$ | $\tau$ |
|----------------|---|---------|-------------------|----------|--------|
| RoB = Serious  | 6 | 0.7080  | [ 0.1099; 1.3062] | 0.2926   | 0.5409 |
| RoB = Moderate | 3 | -0.2584 | [-1.0905; 0.5736] | 0.0421   | 0.2053 |
| RoB = Low      | 8 | 0.3689  | [-0.0172; 0.7550] | 0.1352   | 0.3677 |

|                | Q     | $I^2$ |
|----------------|-------|-------|
| RoB = Serious  | 14.63 | 65.8% |
| RoB = Moderate | 2.84  | 29.5% |
| RoB = Low      | 17.08 | 59.0% |

Test for subgroup differences (random effects model (HK)):

| Q | d.f. | p-value |
|---|------|---------|
|---|------|---------|

Between groups 11.31      2   0.0035

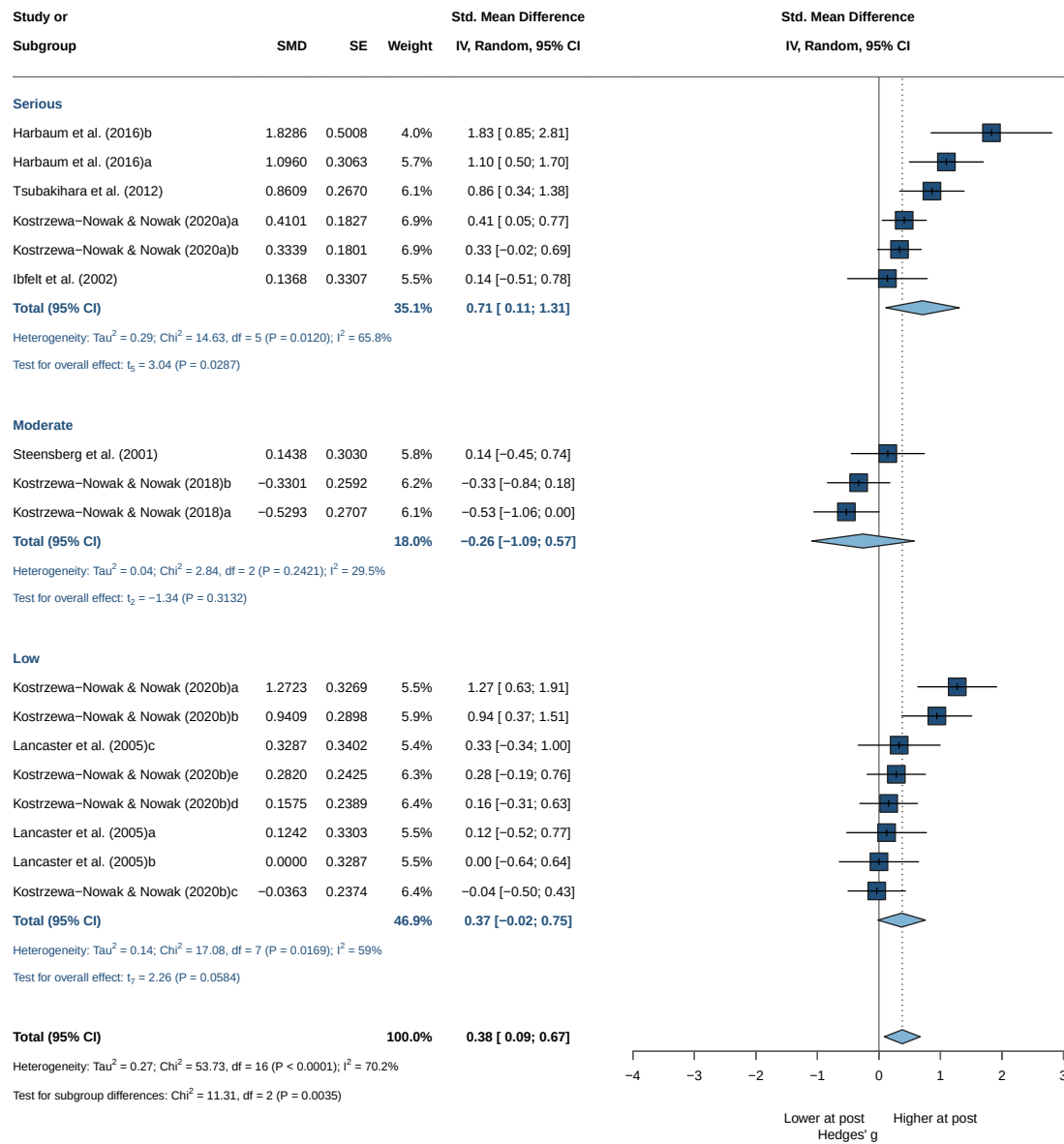
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

! Important

Significant results

## 7.7.1.2. Forrest Plot



## 7.7.1.3. Head-to-Head

**Head-to-head RoB comparisons.** Because the overall RoB subgroup test is significant, the individual rating levels are compared pairwise.

**7.7.1.4. Moderate vs Serious**

Number of studies:  $k = 9$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.5204$  [0.2137; 1.9229];  $\tau = 0.7214$  [0.4622; 1.3867]

$I^2 = 98.0\%$  [97.3%; 98.6%];  $H = 7.11$  [6.08; 8.31]

Test of heterogeneity:

| Q      | d.f. | p-value  |
|--------|------|----------|
| 403.91 | 8    | < 0.0001 |

Results for subgroups (random effects model (HK)):

|                | k | SMD     | 95% CI            | $\tau^2$ | $\tau$ |
|----------------|---|---------|-------------------|----------|--------|
| RoB = Serious  | 6 | 0.7540  | [ 0.1167; 1.3914] | 0.3759   | 0.6131 |
| RoB = Moderate | 3 | -0.2420 | [-1.0969; 0.6129] | 0.1135   | 0.3368 |

|                | Q      | $I^2$ |
|----------------|--------|-------|
| RoB = Serious  | 133.26 | 96.2% |
| RoB = Moderate | 33.35  | 94.0% |

Test for subgroup differences (random effects model (HK)):

|                | Q    | d.f. | p-value |
|----------------|------|------|---------|
| Between groups | 9.83 | 1    | 0.0017  |

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

**!** Important

Significant

**7.7.1.5. Moderate vs Low**

Number of studies:  $k = 11$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.2563$  [0.1206; 0.8046];  $\tau = 0.5063$  [0.3473; 0.8970]

$I^2 = 97.2\%$  [96.1%; 97.9%];  $H = 5.96$  [5.10; 6.97]

Test of heterogeneity:

| Q      | d.f. | p-value  |
|--------|------|----------|
| 355.37 | 10   | < 0.0001 |

Results for subgroups (random effects model (HK)):

|  | k | SMD | 95% CI | $\tau^2$ | $\tau$ |
|--|---|-----|--------|----------|--------|
|--|---|-----|--------|----------|--------|

## 7. Th2 — Immediate Post-Exercise

```
RoB = Moderate  3 -0.2420 [-1.0969; 0.6129] 0.1135 0.3368
RoB = Low       8  0.3817 [-0.0113; 0.7748] 0.2141 0.4627
               Q    I^2
RoB = Moderate 33.35 94.0%
RoB = Low      193.09 96.4%
```

Test for subgroup differences (random effects model (HK)):

```
      Q d.f. p-value
Between groups 5.80    1  0.0161
```

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

**!** Important

Significant

### 7.7.1.6. Low vs Serious

Number of studies:  $k = 14$

Quantifying heterogeneity (with 95% CIs):

```
tau^2 = 0.2999 [0.1367; 0.7776]; tau = 0.5476 [0.3697; 0.8818]
I^2 = 96.3% [95.0%; 97.2%]; H = 5.18 [4.46; 6.03]
```

Test of heterogeneity:

```
      Q d.f. p-value
349.48  13 < 0.0001
```

Results for subgroups (random effects model (HK)):

```
      k    SMD      95% CI  tau^2    tau
RoB = Serious  6 0.7540 [ 0.1167; 1.3914] 0.3759 0.6131
RoB = Low      8 0.3817 [-0.0113; 0.7748] 0.2141 0.4627
               Q    I^2
RoB = Serious 133.26 96.2%
RoB = Low     193.09 96.4%
```

Test for subgroup differences (random effects model (HK)):

```
      Q d.f. p-value
Between groups 1.56    1  0.2123
```

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$

- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

### **i** Note

Not significant

## 7.7.2. Sex

### 7.7.2.1. Model

Number of studies:  $k = 17$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.2749$  [0.0825; 0.7087];  $\tau = 0.5243$  [0.2873; 0.8418]

$I^2 = 70.2\%$  [51.2%; 81.8%];  $H = 1.83$  [1.43; 2.35]

Test of heterogeneity:

| Q     | d.f. | p-value  |
|-------|------|----------|
| 53.73 | 16   | < 0.0001 |

Results for subgroups (random effects model (HK)):

|              | k  | SMD    | 95% CI            | $\tau^2$ | $\tau$ |
|--------------|----|--------|-------------------|----------|--------|
| sex = male   | 14 | 0.2235 | [-0.0321; 0.4792] | 0.1284   | 0.3584 |
| sex = female | 1  | 0.8609 | [ 0.3377; 1.3842] | --       | --     |
| sex = mixed  | 2  | 1.3552 | [-3.0956; 5.8059] | 0.0960   | 0.3099 |

|              | Q     | $I^2$ |
|--------------|-------|-------|
| sex = male   | 32.25 | 59.7% |
| sex = female | 0.00  | --    |
| sex = mixed  | 1.56  | 35.8% |

Test for subgroup differences (random effects model (HK)):

|                | Q     | d.f. | p-value |
|----------------|-------|------|---------|
| Between groups | 12.61 | 2    | 0.0018  |

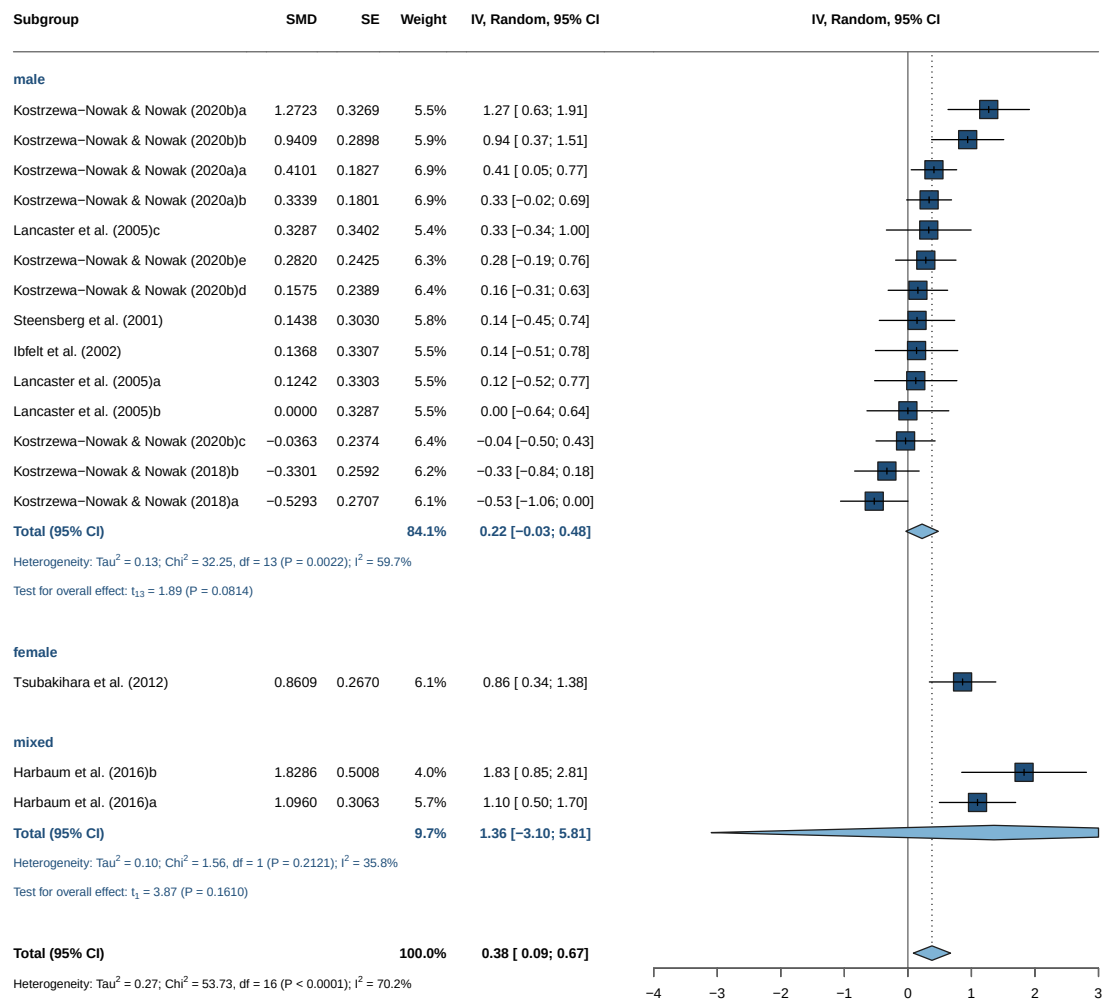
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

### **!** Important

Significant results but 15 vs 1 vs 2

## 7.7.2.2. Forrest Plot



## 7.7.3. Intensity

## 7.7.3.1. Model

Number of studies:  $k = 16$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.2839$  [0.0797; 0.7558];  $\tau = 0.5328$  [0.2822; 0.8694]

$I^2 = 69.7\%$  [49.4%; 81.8%];  $H = 1.82$  [1.41; 2.35]

Test of heterogeneity:

| Q     | d.f. | p-value  |
|-------|------|----------|
| 49.50 | 15   | < 0.0001 |

Results for subgroups (random effects model (HK)):

|                      | k  | SMD    | 95% CI            | $\tau^2$ |
|----------------------|----|--------|-------------------|----------|
| intensity = Vigorous | 5  | 0.1443 | [ 0.0026; 0.2860] | 0        |
| intensity = Maximal  | 11 | 0.4478 | [-0.0115; 0.9070] | 0.4316   |
|                      |    | $\tau$ | Q                 | $I^2$    |
| intensity = Vigorous | 0  | 0.49   | 0.0%              |          |

intensity = Maximal 0.6570 47.65 79.0%

Test for subgroup differences (random effects model (HK)):

Q d.f. p-value  
Between groups 2.04 1 0.1529

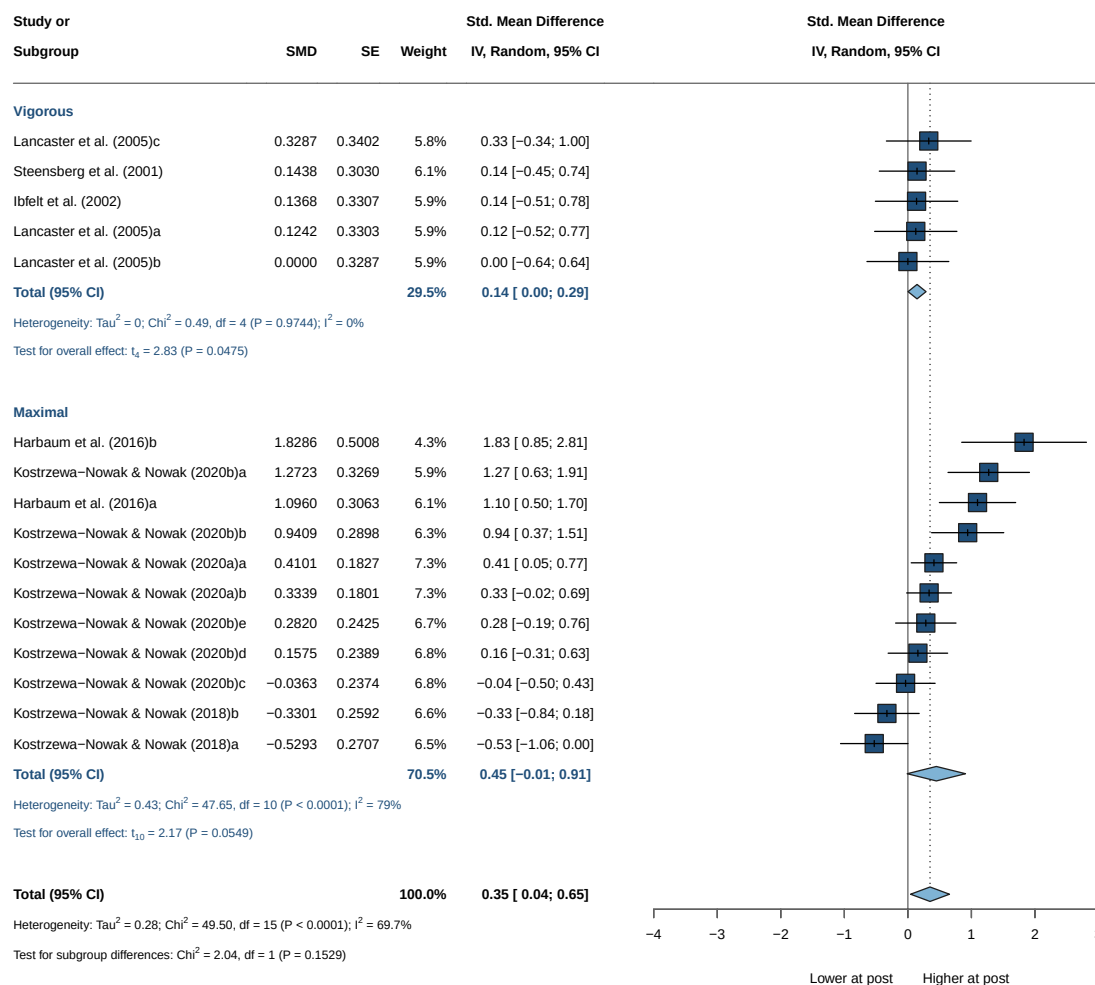
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

### Note

Not significant

#### 7.7.3.2. Forrest Plot



### 7.7.4. Staining

Number of studies:  $k = 17$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.2749$  [0.0825; 0.7087];  $\tau = 0.5243$  [0.2873; 0.8418]

$I^2 = 70.2\%$  [51.2%; 81.8%];  $H = 1.83$  [1.43; 2.35]

Test of heterogeneity:

| Q     | d.f. | p-value  |
|-------|------|----------|
| 53.73 | 16   | < 0.0001 |

Results for subgroups (random effects model (HK)):

|                  | k  | SMD    | 95% CI            | $\tau^2$ | $\tau$ |
|------------------|----|--------|-------------------|----------|--------|
| staining = intra | 14 | 0.2235 | [-0.0321; 0.4792] | 0.1284   | 0.3584 |
| staining = extra | 3  | 1.1539 | [ 0.0415; 2.2663] | 0.1161   | 0.3407 |

|                  | Q     | $I^2$ |
|------------------|-------|-------|
| staining = intra | 32.25 | 59.7% |
| staining = extra | 2.91  | 31.3% |

Test for subgroup differences (random effects model (HK)):

|                | Q     | d.f. | p-value |
|----------------|-------|------|---------|
| Between groups | 10.71 | 1    | 0.0011  |

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

#### ! Important

Significant results

### 7.7.5. Continuous Moderators (Meta-Regression)

Continuous moderators are tested via weighted meta-regression (`metareg()`). A significant slope indicates the moderator explains part of the between-study heterogeneity ( $\tau^2$ ); bubble plots show the fitted line with 95% CI.

#### 7.7.5.1. Age

Mixed-Effects Model ( $k = 16$ ;  $\tau^2$  estimator: HE)

| logLik   | deviance | AIC     | BIC     | AICc    |
|----------|----------|---------|---------|---------|
| -10.1238 | 31.2182  | 26.2475 | 28.5653 | 28.2475 |

```

tau^2 (estimated amount of residual heterogeneity):    0.1586 (SE = 0.0936)
tau (square root of estimated tau^2 value):           0.3982
I^2 (residual heterogeneity / unaccounted variability): 68.97%
H^2 (unaccounted variability / sampling variability):  3.22
R^2 (amount of heterogeneity accounted for):           44.13%

```

Test for Residual Heterogeneity:

QE(df = 14) = 37.4912, p-val = 0.0006

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 14) = 6.4163, p-val = 0.0239

Model Results:

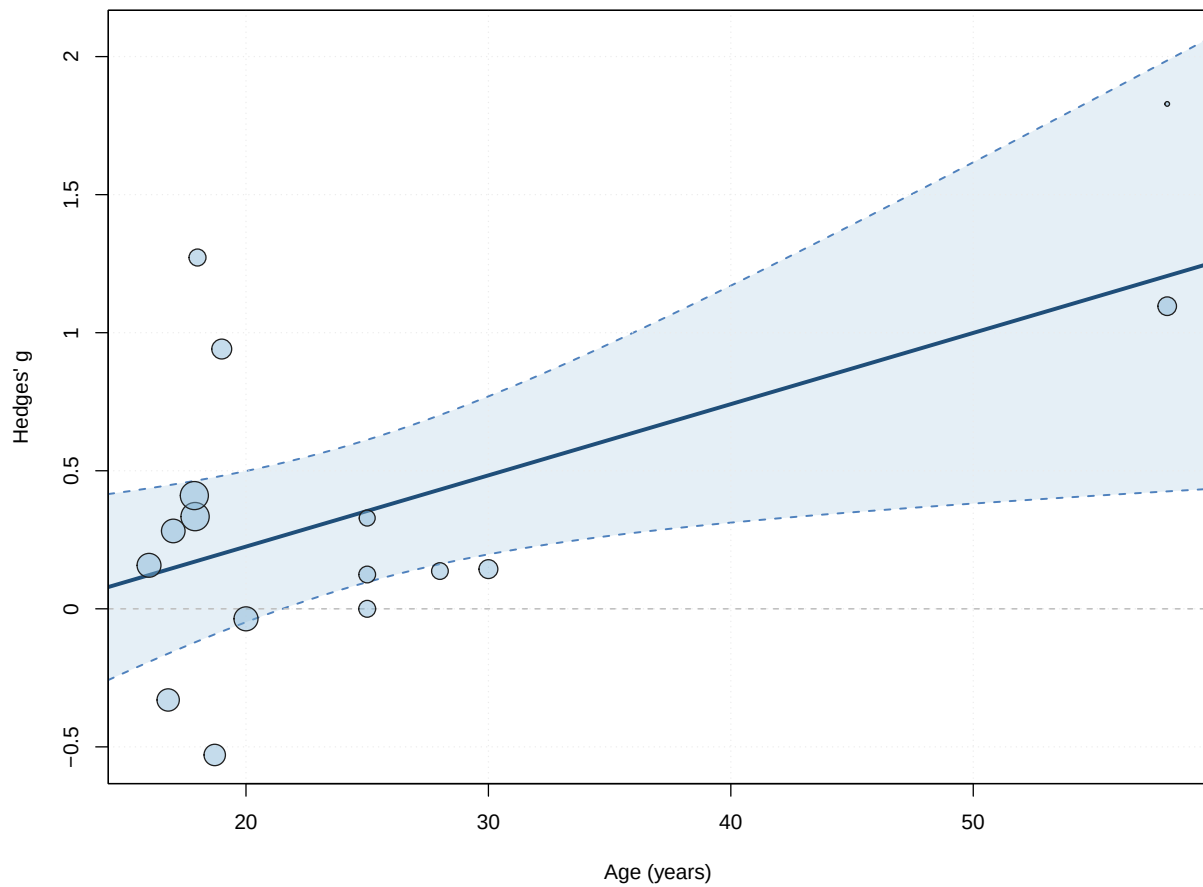
|         | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|---------|----|--------|---------|-------|
| intrcpt | -0.2905  | 0.2747 | -1.0574 | 14 | 0.3082 | -0.8798 |       |
| age     | 0.0258   | 0.0102 | 2.5330  | 14 | 0.0239 | 0.0040  |       |
| intrcpt | 0.2988   |        |         |    |        |         |       |
| age     | 0.0476   |        |         |    |        |         | *     |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

! Important

Significant



#### 7.7.5.2. BMI

Mixed-Effects Model (k = 12; tau<sup>2</sup> estimator: HE)

| logLik   | deviance | AIC     | BIC     | AICc    |
|----------|----------|---------|---------|---------|
| -11.5110 | 32.7869  | 29.0221 | 30.4768 | 32.0221 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.4184 (SE = 0.2231)  
 tau (square root of estimated tau<sup>2</sup> value): 0.6468  
 I<sup>2</sup> (residual heterogeneity / unaccounted variability): 87.04%  
 H<sup>2</sup> (unaccounted variability / sampling variability): 7.71  
 R<sup>2</sup> (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 10) = 50.4537, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 10) = 0.4072, p-val = 0.5377

Model Results:

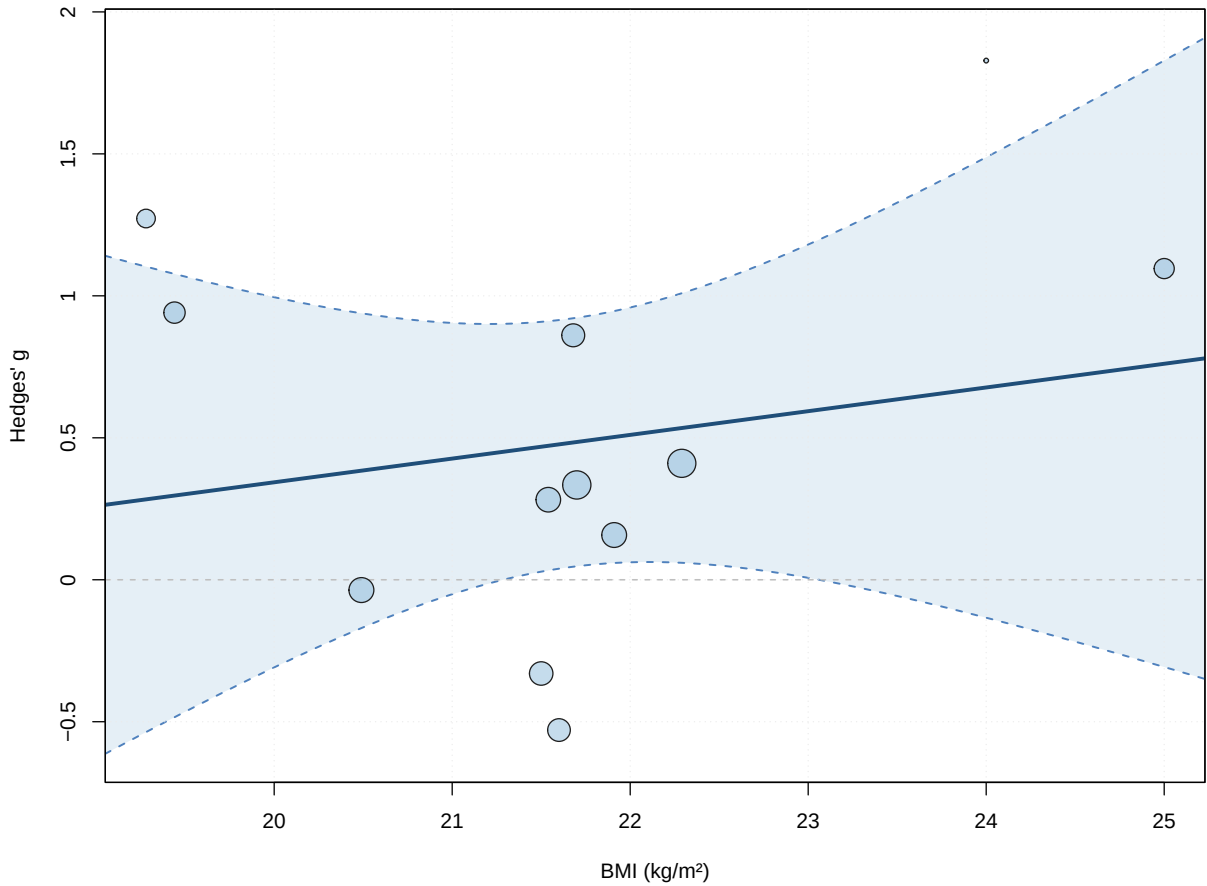
```
      estimate      se      tval  df      pval      ci.lb
intrcpt    -1.3265  2.8404   -0.4670  10   0.6505   -7.6554
bmi         0.0835  0.1308    0.6381  10   0.5377   -0.2080

      ci.ub
intrcpt    5.0023
bmi        0.3750

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

*i* Note

Not significant



7.7.5.3. Duration

Mixed-Effects Model (k = 15; tau^2 estimator: HE)

|          |          |         |         |         |
|----------|----------|---------|---------|---------|
| logLik   | deviance | AIC     | BIC     | AICc    |
| -13.4344 | 35.4902  | 32.8688 | 34.9929 | 35.0506 |

## 7. Th2 — Immediate Post-Exercise

```
tau^2 (estimated amount of residual heterogeneity):    0.3252 (SE = 0.1663)
tau (square root of estimated tau^2 value):           0.5702
I^2 (residual heterogeneity / unaccounted variability): 79.32%
H^2 (unaccounted variability / sampling variability):   4.84
R^2 (amount of heterogeneity accounted for):           0.00%
```

Test for Residual Heterogeneity:  
 QE(df = 13) = 53.3069, p-val < .0001

Test of Moderators (coefficient 2):  
 F(df1 = 1, df2 = 13) = 0.5003, p-val = 0.4919

Model Results:

|          | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|----------|----------|--------|---------|----|--------|---------|-------|
| intrcpt  | 0.5088   | 0.2399 | 2.1212  | 13 | 0.0537 | -0.0094 |       |
| duration | -0.0020  | 0.0028 | -0.7073 | 13 | 0.4919 | -0.0079 |       |
| intrcpt  | 1.0270   |        |         |    |        |         |       |
| duration | 0.0040   |        |         |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

### 7.7.5.4. Dose

Mixed-Effects Model (k = 14; tau^2 estimator: HE)

| logLik   | deviance | AIC     | BIC     | AICc    |
|----------|----------|---------|---------|---------|
| -12.4175 | 32.6531  | 30.8349 | 32.7521 | 33.2349 |

```
tau^2 (estimated amount of residual heterogeneity):    0.3219 (SE = 0.1725)
tau (square root of estimated tau^2 value):           0.5673
I^2 (residual heterogeneity / unaccounted variability): 78.97%
H^2 (unaccounted variability / sampling variability):   4.76
R^2 (amount of heterogeneity accounted for):           3.33%
```

Test for Residual Heterogeneity:  
 QE(df = 12) = 47.3298, p-val < .0001

Test of Moderators (coefficient 2):  
 F(df1 = 1, df2 = 12) = 1.0162, p-val = 0.3333

Model Results:

```

      estimate      se      tval  df      pval      ci.lb
intrcpt    0.5419  0.2552   2.1233  12  0.0552  -0.0142
dose      -0.0005  0.0005  -1.0080  12  0.3333  -0.0015
      ci.ub
intrcpt    1.0980  .
dose       0.0005

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

#### Note

Not significant

#### 7.7.5.5. IFN

Mixed-Effects Model (k = 7; tau<sup>2</sup> estimator: HE)

```

logLik  deviance      AIC      BIC      AICc
-3.8105  14.8706  13.6209  13.4587  21.6209

```

```

tau^2 (estimated amount of residual heterogeneity):    0.1830 (SE = 0.1575)
tau (square root of estimated tau^2 value):           0.4277
I^2 (residual heterogeneity / unaccounted variability): 75.67%
H^2 (unaccounted variability / sampling variability):   4.11
R^2 (amount of heterogeneity accounted for):           0.00%

```

Test for Residual Heterogeneity:

QE(df = 5) = 15.1720, p-val = 0.0097

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.1169, p-val = 0.7463

Model Results:

```

      estimate      se      tval  df      pval      ci.lb
intrcpt    0.7247  0.8225   0.8811   5  0.4186  -1.3896
IFN        -0.1511  0.4421  -0.3419   5  0.7463  -1.2876
      ci.ub
intrcpt    2.8389
IFN         0.9853

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

**i** Note

Not significant

**7.7.5.6. IL\_4**Mixed-Effects Model (k = 9; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -7.3445 | 22.0122  | 20.6889 | 21.2806 | 25.4889 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.3176 (SE = 0.2182)  
 tau (square root of estimated tau<sup>2</sup> value): 0.5636  
 I<sup>2</sup> (residual heterogeneity / unaccounted variability): 82.89%  
 H<sup>2</sup> (unaccounted variability / sampling variability): 5.84  
 R<sup>2</sup> (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 7) = 27.0308, p-val = 0.0003

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 7) = 0.3834, p-val = 0.5554

Model Results:

|         | estimate | se     | tval   | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|--------|----|--------|---------|-------|
| intrcpt | 0.5539   | 0.2387 | 2.3206 | 7  | 0.0534 | -0.0105 |       |
| IL_4    | 0.1255   | 0.2026 | 0.6192 | 7  | 0.5554 | -0.3536 |       |
| intrcpt | 1.1183   |        |        |    |        |         |       |
| IL_4    | 0.6045   |        |        |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

**7.7.5.7. IL\_6**Mixed-Effects Model (k = 14; tau<sup>2</sup> estimator: HE)

| logLik   | deviance | AIC     | BIC     | AICc    |
|----------|----------|---------|---------|---------|
| -11.6641 | 33.3733  | 29.3282 | 31.2454 | 31.7282 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.3033 (SE = 0.1621)

```

tau (square root of estimated tau^2 value):      0.5507
I^2 (residual heterogeneity / unaccounted variability): 81.28%
H^2 (unaccounted variability / sampling variability): 5.34
R^2 (amount of heterogeneity accounted for):      8.75%

```

Test for Residual Heterogeneity:

```
QE(df = 12) = 41.9420, p-val < .0001
```

Test of Moderators (coefficient 2):

```
F(df1 = 1, df2 = 12) = 1.9487, p-val = 0.1880
```

Model Results:

|         | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|---------|----|--------|---------|-------|
| intrcpt | 0.5182   | 0.1867 | 2.7751  | 12 | 0.0168 | 0.1114  |       |
| IL_6    | -0.1428  | 0.1023 | -1.3960 | 12 | 0.1880 | -0.3656 |       |
| intrcpt | 0.9251   | *      |         |    |        |         |       |
| IL_6    | 0.0801   |        |         |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

#### 7.7.5.8. IL\_10

Mixed-Effects Model (k = 11; tau^2 estimator: HE)

| logLik   | deviance | AIC     | BIC     | AICc    |
|----------|----------|---------|---------|---------|
| -10.9326 | 30.8268  | 27.8652 | 29.0589 | 31.2937 |

```

tau^2 (estimated amount of residual heterogeneity): 0.4665 (SE = 0.2620)
tau (square root of estimated tau^2 value):         0.6830
I^2 (residual heterogeneity / unaccounted variability): 87.31%
H^2 (unaccounted variability / sampling variability):  7.88
R^2 (amount of heterogeneity accounted for):         0.00%

```

Test for Residual Heterogeneity:

```
QE(df = 9) = 46.2625, p-val < .0001
```

Test of Moderators (coefficient 2):

```
F(df1 = 1, df2 = 9) = 0.3720, p-val = 0.5570
```

Model Results:

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```

      estimate      se      tval  df      pval      ci.lb
intrcpt    0.4614  0.2143   2.1530   9  0.0597  -0.0234
IL_10     -0.3677  0.6029  -0.6099   9  0.5570  -1.7315
      ci.ub
intrcpt    0.9461  .
IL_10     0.9961

```

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

### Note

Not significant

### 7.7.5.9. IL\_2

Mixed-Effects Model (k = 7; tau<sup>2</sup> estimator: HE)

```

logLik  deviance      AIC      BIC      AICc
-3.9212  15.0921   13.8425   13.6802   21.8425

```

```

tau^2 (estimated amount of residual heterogeneity):    0.1931 (SE = 0.1621)
tau (square root of estimated tau^2 value):           0.4394
I^2 (residual heterogeneity / unaccounted variability): 78.08%
H^2 (unaccounted variability / sampling variability):   4.56
R^2 (amount of heterogeneity accounted for):           0.00%

```

Test for Residual Heterogeneity:

QE(df = 5) = 15.3694, p-val = 0.0089

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.0000, p-val = 0.9986

Model Results:

```

      estimate      se      tval  df      pval      ci.lb
intrcpt    0.4500  0.7121  0.6319   5  0.5552  -1.3805
IL_2       0.0007  0.3851  0.0019   5  0.9986  -0.9892
      ci.ub
intrcpt    2.2806
IL_2       0.9906

```

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

### Note

Not significant

## 7.8. Multivariate Models

### 7.8.1. Intensity \* Duration

Mixed-Effects Model (k = 14; tau<sup>2</sup> estimator: HE)

```
tau^2 (estimated amount of residual heterogeneity):    0.2579 (SE = 0.1595)
tau (square root of estimated tau^2 value):           0.5079
I^2 (residual heterogeneity / unaccounted variability): 75.59%
H^2 (unaccounted variability / sampling variability):  4.10
R^2 (amount of heterogeneity accounted for):           22.54%
```

Test for Residual Heterogeneity:

QE(df = 10) = 34.9688, p-val = 0.0001

Test of Moderators (coefficients 2:4):

F(df1 = 3, df2 = 10) = 1.6514, p-val = 0.2396

Model Results:

|                            | estimate | se     | tval    | df |
|----------------------------|----------|--------|---------|----|
| intrcpt                    | 2.1195   | 0.8475 | 2.5008  | 10 |
| intensityVigorous          | -1.9999  | 1.7640 | -1.1337 | 10 |
| duration                   | -0.0926  | 0.0456 | -2.0281 | 10 |
| intensityVigorous:duration | 0.0928   | 0.0470 | 1.9753  | 10 |

|                            | pval   | ci.lb   | ci.ub    |
|----------------------------|--------|---------|----------|
| intrcpt                    | 0.0314 | 0.2311  | 4.0079 * |
| intensityVigorous          | 0.2834 | -5.9302 | 1.9305   |
| duration                   | 0.0700 | -0.1943 | 0.0091 . |
| intensityVigorous:duration | 0.0765 | -0.0119 | 0.1974 . |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

#### Note

Not significant

### 7.8.2. IL\_10 \* IL\_4

Mixed-Effects Model (k = 9; tau<sup>2</sup> estimator: HE)

```
tau^2 (estimated amount of residual heterogeneity):    0.3863 (SE = 0.3079)
tau (square root of estimated tau^2 value):           0.6215
I^2 (residual heterogeneity / unaccounted variability): 83.42%
H^2 (unaccounted variability / sampling variability):  6.03
R^2 (amount of heterogeneity accounted for):           0.00%
```

Test for Residual Heterogeneity:

QE(df = 5) = 23.7108, p-val = 0.0002

Test of Moderators (coefficients 2:4):

F(df1 = 3, df2 = 5) = 0.3848, p-val = 0.7691

Model Results:

|            | estimate | se     | tval    | df | pval   | ci.lb   |
|------------|----------|--------|---------|----|--------|---------|
| intrcpt    | 0.3992   | 0.3244 | 1.2307  | 5  | 0.2732 | -0.4346 |
| IL_10      | -0.5850  | 0.7088 | -0.8253 | 5  | 0.4468 | -2.4069 |
| IL_4       | 0.1813   | 0.2347 | 0.7725  | 5  | 0.4747 | -0.4220 |
| IL_10:IL_4 | 1.1806   | 1.5451 | 0.7641  | 5  | 0.4793 | -2.7911 |
|            | ci.ub    |        |         |    |        |         |
| intrcpt    | 1.2330   |        |         |    |        |         |
| IL_10      | 1.2370   |        |         |    |        |         |
| IL_4       | 0.7845   |        |         |    |        |         |
| IL_10:IL_4 | 5.1524   |        |         |    |        |         |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

#### Note

Not significant

## 8. Th2 — ~2 h Post-Exercise

### 8.1. Data Import

The ~2 h window (effect\_size/Th2/pre\_post+2h/Th2\_pre\_post+2h\_r\_050.csv) is reported for the main analysis, outlier check, funnel, and forest only; the limited data do not support moderator modelling. No study is excluded a priori.

### 8.2. Primary Meta-Analysis

#### 8.2.1. {meta}

Number of studies: k = 5

|                           | SMD     | 95% CI            | t    |
|---------------------------|---------|-------------------|------|
| Random effects model (HK) | 0.0744  | [-0.0938; 0.2426] | 1.23 |
| Prediction interval       |         | [-0.3288; 0.4775] |      |
|                           | p-value |                   |      |
| Random effects model (HK) | 0.2869  |                   |      |
| Prediction interval       |         |                   |      |

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0$  [0.0000; 0.0495];  $\tau = 0$  [0.0000; 0.2226]  
 $I^2 = 0.0\%$  [0.0%; 79.2%];  $H = 1.00$  [1.00; 2.19]

Test of heterogeneity:

| Q    | d.f. | p-value |
|------|------|---------|
| 0.70 | 4    | 0.9518  |

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model (df = 4)
- Prediction interval based on t-distribution (df = 4)

#### 8.2.2. {metafor}

Random-Effects Model (k = 5;  $\tau^2$  estimator: HE)

```

tau^2 (estimated amount of total heterogeneity): 0 (SE = 0.0752)
tau (square root of estimated tau^2 value):      0
I^2 (total heterogeneity / total variability):    0.00%
H^2 (total variability / sampling variability):    1.00

```

Test for Heterogeneity:

Q(df = 4) = 0.6963, p-val = 0.9518

Model Results:

```

estimate      se      tval  df      pval      ci.lb      ci.ub
    0.0744  0.0606  1.2277   4  0.2869  -0.0938  0.2426

```

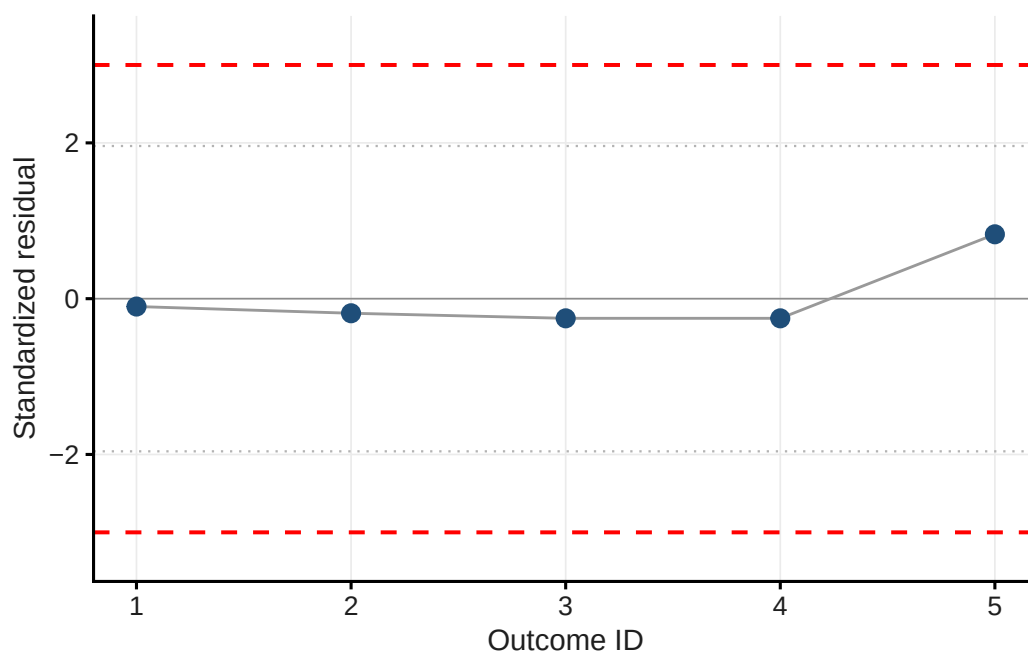
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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

### 8.3. Outlier Detection

**Table 8.1.:** Standardised residuals

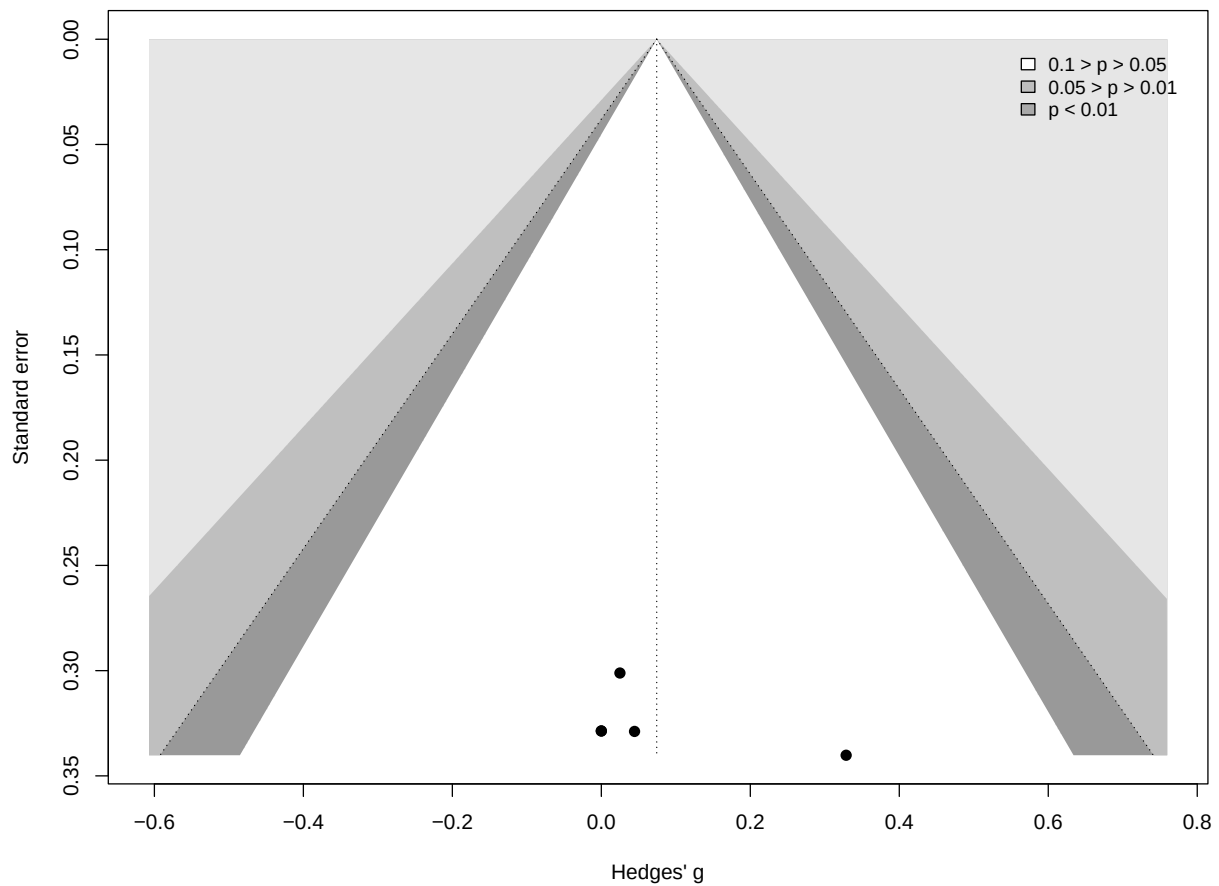
| Short_reference          | g     | std_res |
|--------------------------|-------|---------|
| Ibfelt et al. (2002)     | 0.045 | -0.101  |
| Steensberg et al. (2001) | 0.025 | -0.187  |
| Lancaster et al. (2005)a | 0.000 | -0.252  |
| Lancaster et al. (2005)b | 0.000 | -0.252  |
| Lancaster et al. (2005)c | 0.329 | 0.827   |



**i** Note

One outcome has standardized residuals around -2 (Kostrzewa-Nowak et al. (2019)).

## 8.4. Funnel Plot & Influential Cases



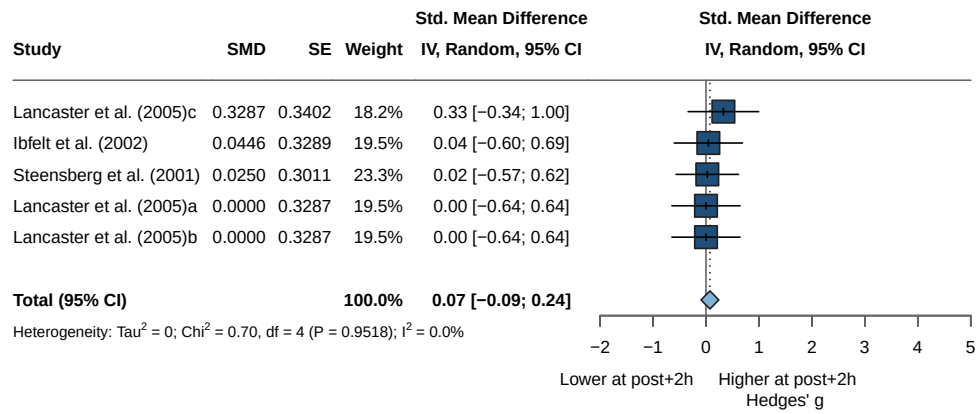
**Table 8.2.:** DFBETAS influence diagnostics

| Short_reference          | g     | dfbetas     |
|--------------------------|-------|-------------|
| Ibfelt et al. (2002)     | 0.045 | -0.04966493 |
| Steensberg et al. (2001) | 0.025 | -0.10307622 |
| Lancaster et al. (2005)a | 0.000 | -0.12421976 |
| Lancaster et al. (2005)b | 0.000 | -0.12421976 |
| Lancaster et al. (2005)c | 0.329 | 0.39010883  |

**i** Note

No influential case.

## 8.5. Forest Plot



## 8.6. Asymmetry Test

**Note:** the number of studies in this window is too small to reliably test for small-study effects (minimum  $k = 10$ ). No Egger's test or trim-and-fill is performed.

## 9. Th2 — 17 h Post-Exercise

### 9.1. Data Import

### 9.2. Primary Meta-Analysis

Kostrzewa-Nowak et al. (2019) is removed before pooling and the model re-fitted.

#### 9.2.1. {meta}

Number of studies: k = 11

|                           | SMD     | 95% CI            | t    |
|---------------------------|---------|-------------------|------|
| Random effects model (HK) | 0.2468  | [-0.4359; 0.9295] | 0.81 |
| Prediction interval       |         | [-2.8310; 3.3247] |      |
|                           | p-value |                   |      |
| Random effects model (HK) | 0.4392  |                   |      |
| Prediction interval       |         |                   |      |

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 1.7347$  [0.0804; 4.1148];  $\tau = 1.3171$  [0.2835; 2.0285]  
 $I^2 = 72.2\%$  [48.9%; 84.9%];  $H = 1.90$  [1.40; 2.57]

Test of heterogeneity:

| Q     | d.f. | p-value  |
|-------|------|----------|
| 35.94 | 10   | < 0.0001 |

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model (df = 10)
- Prediction interval based on t-distribution (df = 10)

#### 9.2.2. {metafor}

Random-Effects Model (k = 11;  $\tau^2$  estimator: HE)

$\tau^2$  (estimated amount of total heterogeneity): 1.7347 (SE = 0.9828)  
 $\tau$  (square root of estimated  $\tau^2$  value): 1.3171

$I^2$  (total heterogeneity / total variability): 96.28%  
 $H^2$  (total variability / sampling variability): 26.85

Test for Heterogeneity:  
 $Q(df = 10) = 35.9359$ ,  $p\text{-val} < .0001$

Model Results:

| estimate | se     | tval   | df | pval   | ci.lb   | ci.ub  |
|----------|--------|--------|----|--------|---------|--------|
| 0.2468   | 0.3064 | 0.8056 | 10 | 0.4392 | -0.4359 | 0.9295 |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

### 9.2.3. Adapted Model {meta}

Number of studies:  $k = 10$

|                           | SMD     | 95% CI            | t    |
|---------------------------|---------|-------------------|------|
| Random effects model (HK) | 0.4009  | [ 0.0582; 0.7436] | 2.65 |
| Prediction interval       |         | [-0.6056; 1.4073] |      |
|                           | p-value |                   |      |
| Random effects model (HK) | 0.0266  |                   |      |
| Prediction interval       |         |                   |      |

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.1740$  [0.0376; 0.7263];  $\tau = 0.4171$  [0.1939; 0.8522]  
 $I^2 = 69.1\%$  [40.3%; 83.9%];  $H = 1.80$  [1.29; 2.50]

Test of heterogeneity:

| Q     | d.f. | p-value |
|-------|------|---------|
| 29.08 | 9    | 0.0006  |

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model ( $df = 9$ )
- Prediction interval based on t-distribution ( $df = 9$ )

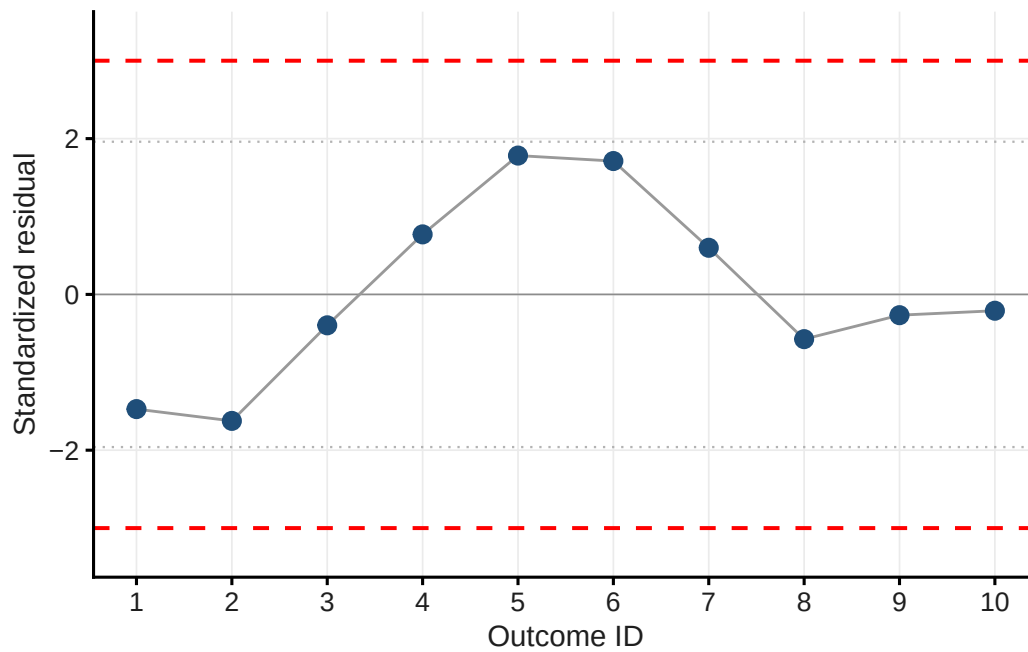
## 9.3. Outlier Detection

**Table 9.1.:** Standardised residuals

| Short_reference                 | g      | std_res |
|---------------------------------|--------|---------|
| Kostrzewa-Nowak & Nowak (2018)a | -0.199 | -1.472  |
| Kostrzewa-Nowak & Nowak (2018)b | -0.250 | -1.624  |

**Table 9.1.:** Standardised residuals

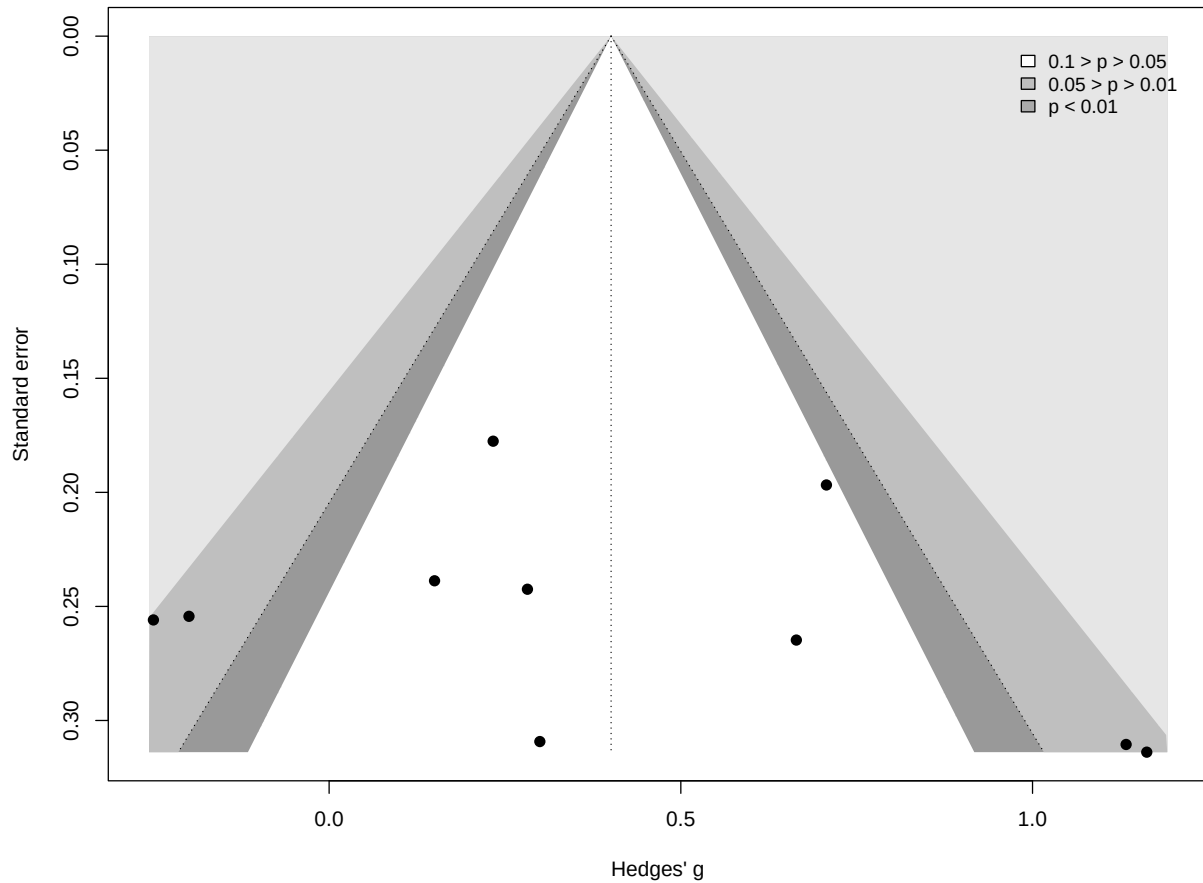
| Short_reference                    | g     | std_res |
|------------------------------------|-------|---------|
| Kostrzewska-Nowak & Nowak (2020a)a | 0.233 | -0.397  |
| Kostrzewska-Nowak & Nowak (2020a)b | 0.707 | 0.771   |
| Kostrzewska-Nowak & Nowak (2020b)a | 1.162 | 1.782   |
| Kostrzewska-Nowak & Nowak (2020b)b | 1.133 | 1.712   |
| Kostrzewska-Nowak & Nowak (2020b)c | 0.664 | 0.599   |
| Kostrzewska-Nowak & Nowak (2020b)d | 0.150 | -0.574  |
| Kostrzewska-Nowak & Nowak (2020b)e | 0.282 | -0.267  |
| Steensberg et al. (2001)           | 0.300 | -0.210  |



**i** Note

No outlier.

## 9.4. Funnel Plot & Influential Cases



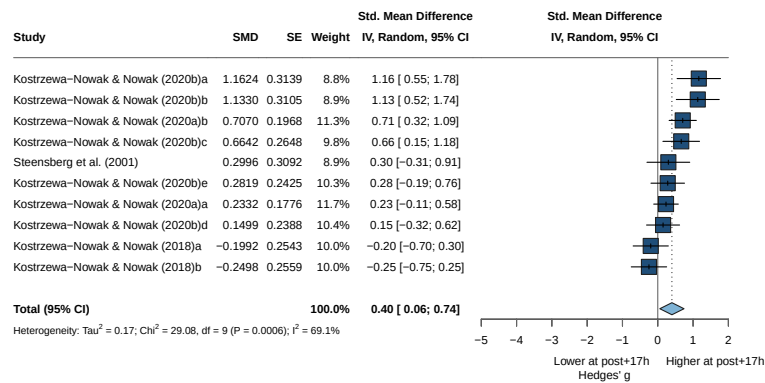
**Table 9.2.:** DFBETAS influence diagnostics

| Short_reference                    | g      | dfbetas    |
|------------------------------------|--------|------------|
| Kostrzewska-Nowak & Nowak (2018)a  | -0.199 | -0.7487610 |
| Kostrzewska-Nowak & Nowak (2018)b  | -0.250 | -0.8040583 |
| Kostrzewska-Nowak & Nowak (2020a)a | 0.233  | -0.4154425 |
| Kostrzewska-Nowak & Nowak (2020a)b | 0.707  | 0.7911110  |
| Kostrzewska-Nowak & Nowak (2020b)a | 1.162  | 0.6594062  |
| Kostrzewska-Nowak & Nowak (2020b)b | 1.133  | 0.6497221  |
| Kostrzewska-Nowak & Nowak (2020b)c | 0.664  | 0.3526656  |
| Kostrzewska-Nowak & Nowak (2020b)d | 0.150  | -0.3340006 |
| Kostrzewska-Nowak & Nowak (2020b)e | 0.282  | -0.1299465 |
| Steensberg et al. (2001)           | 0.300  | -0.0613988 |

### Note

No influential case.

## 9.5. Forest Plot



## 9.6. Asymmetry Test

Linear regression test of funnel plot asymmetry

Test result:  $t = 0.76$ ,  $df = 8$ ,  $p\text{-value} = 0.4679$

Bias estimate: 2.3883 (SE = 3.1343)

Details:

- multiplicative residual heterogeneity variance ( $\tau^2 = 3.3889$ )
- predictor: standard error
- weight: inverse variance
- reference: Egger et al. (1997), BMJ

### **i** Note

No asymmetry -> no trim-and-fill

## 9.7. Moderator Analyses

### 9.7.1. Risk of Bias

#### 9.7.1.1. Model

Number of studies:  $k = 10$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.1740$  [0.0376; 0.7263];  $\tau = 0.4171$  [0.1939; 0.8522]

$I^2 = 69.1\%$  [40.3%; 83.9%];  $H = 1.80$  [1.29; 2.50]

Test of heterogeneity:

| Q     | d.f. | p-value |
|-------|------|---------|
| 29.08 | 9    | 0.0006  |

Results for subgroups (random effects model (HK)):

|                | k | SMD     | 95% CI            | $\tau^2$ | $\tau$ |
|----------------|---|---------|-------------------|----------|--------|
| RoB = Moderate | 3 | -0.0845 | [-0.7928; 0.6239] | 0.0169   | 0.1301 |
| RoB = Serious  | 2 | 0.4625  | [-2.5460; 3.4710] | 0.0771   | 0.2777 |
| RoB = Low      | 5 | 0.6449  | [ 0.0667; 1.2230] | 0.1432   | 0.3785 |

|                | Q     | $I^2$ |
|----------------|-------|-------|
| RoB = Moderate | 2.16  | 7.5%  |
| RoB = Serious  | 3.20  | 68.7% |
| RoB = Low      | 11.47 | 65.1% |

Test for subgroup differences (random effects model (HK)):

|                | Q    | d.f. | p-value |
|----------------|------|------|---------|
| Between groups | 8.52 | 2    | 0.0141  |

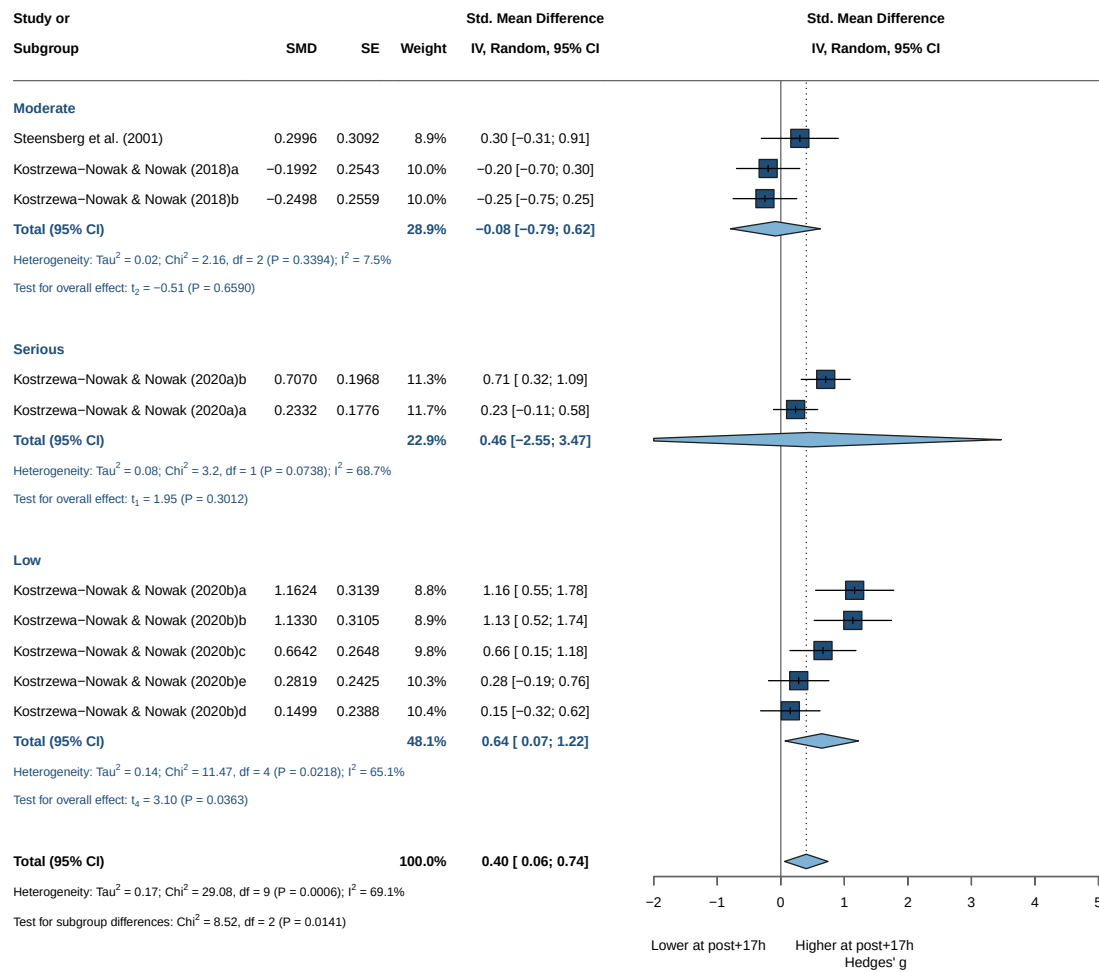
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

#### **i** Note

Not significant results

## 9.7.1.2. Forrest Plot



**Sex:** all retained arms are male, so no sex subgroup analysis is performed.

## 9.7.2. Intensity

Number of studies:  $k = 10$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.1740$  [0.0376; 0.7263];  $\tau = 0.4171$  [0.1939; 0.8522]

$I^2 = 69.1\%$  [40.3%; 83.9%];  $H = 1.80$  [1.29; 2.50]

Test of heterogeneity:

| Q     | d.f. | p-value |
|-------|------|---------|
| 29.08 | 9    | 0.0006  |

Results for subgroups (random effects model (HK)):

|                      | k | SMD    | 95% CI            | $\tau^2$ |
|----------------------|---|--------|-------------------|----------|
| intensity = Maximal  | 9 | 0.4129 | [0.0242; 0.8016]  | 0.2053   |
| intensity = Vigorous | 1 | 0.2996 | [-0.3065; 0.9057] | --       |
|                      |   | $\tau$ | Q                 | $I^2$    |
| intensity = Maximal  |   | 0.4531 | 29.02             | 72.4%    |
| intensity = Vigorous |   | --     | 0.00              | --       |

Test for subgroup differences (random effects model (HK)):

|                | Q    | d.f. | p-value |
|----------------|------|------|---------|
| Between groups | 0.10 | 1    | 0.7477  |

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

#### **i** Note

Not significant but 10 vs 1

### 9.7.3. Continuous Moderators (Meta-Regression)

#### 9.7.3.1. Age

Mixed-Effects Model (k = 10;  $\tau^2$  estimator: HE)

|  | logLik  | deviance | AIC     | BIC     | AICc    |
|--|---------|----------|---------|---------|---------|
|  | -6.5029 | 22.1591  | 19.0057 | 19.9135 | 23.0057 |

|                                                           |                      |
|-----------------------------------------------------------|----------------------|
| $\tau^2$ (estimated amount of residual heterogeneity):    | 0.2063 (SE = 0.1360) |
| $\tau$ (square root of estimated $\tau^2$ value):         | 0.4542               |
| $I^2$ (residual heterogeneity / unaccounted variability): | 77.93%               |
| $H^2$ (unaccounted variability / sampling variability):   | 4.53                 |
| $R^2$ (amount of heterogeneity accounted for):            | 0.00%                |

Test for Residual Heterogeneity:

QE(df = 8) = 28.7965, p-val = 0.0003

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 8) = 0.0326, p-val = 0.8613

Model Results:

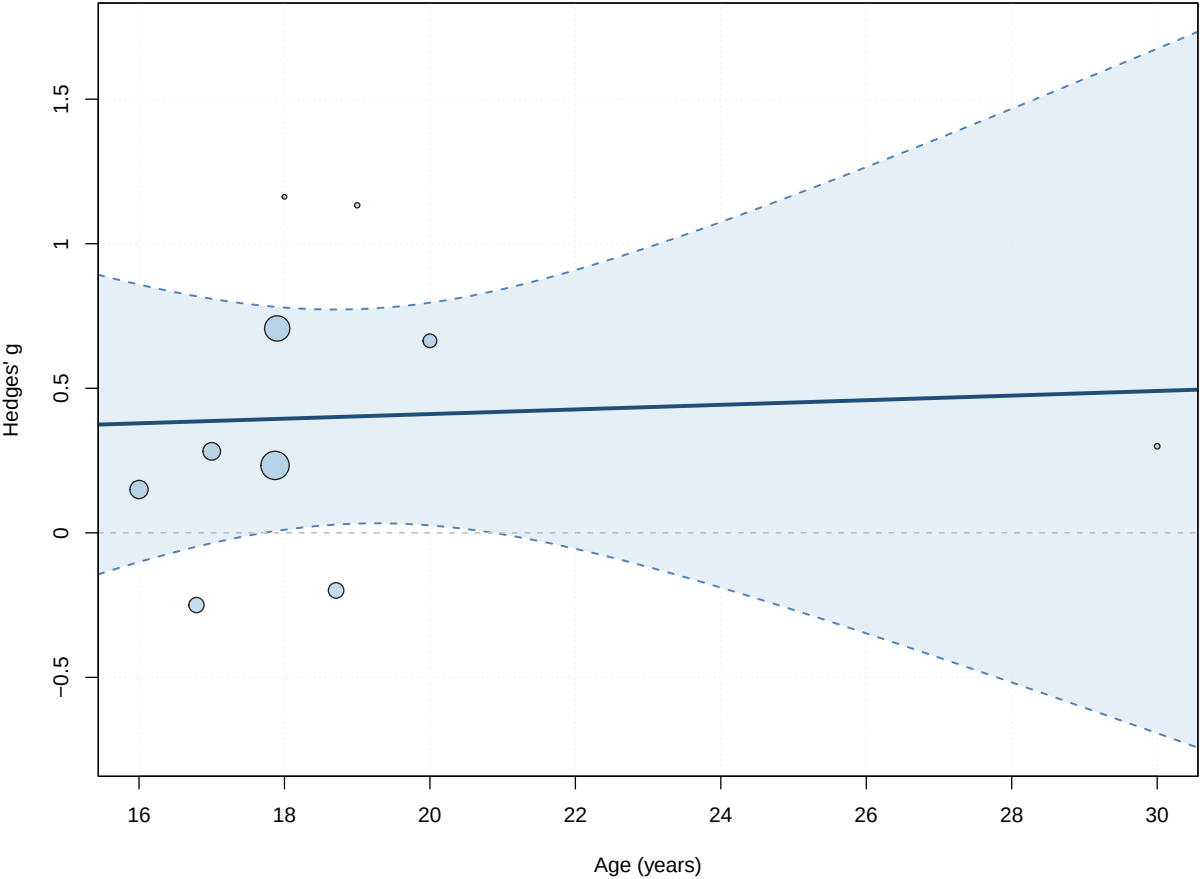
|         | estimate | se     | tval   | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|--------|----|--------|---------|-------|
| intrcpt | 0.2511   | 0.8554 | 0.2936 | 8  | 0.7766 | -1.7214 |       |
| age     | 0.0080   | 0.0443 | 0.1805 | 8  | 0.8613 | -0.0941 |       |
| intrcpt | 2.2236   |        |        |    |        |         |       |
| age     | 0.1100   |        |        |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

*i* Note

Not significant



9.7.3.2. BMI

Mixed-Effects Model (k = 9; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -2.3155 | 13.2751  | 10.6311 | 11.2227 | 15.4311 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.0517 (SE = 0.0618)  
tau (square root of estimated tau<sup>2</sup> value): 0.2273  
I<sup>2</sup> (residual heterogeneity / unaccounted variability): 47.38%  
H<sup>2</sup> (unaccounted variability / sampling variability): 1.90  
R<sup>2</sup> (amount of heterogeneity accounted for): 74.84%

Test for Residual Heterogeneity:  
QE(df = 7) = 15.1650, p-val = 0.0339

Test of Moderators (coefficient 2):

## 9. Th2 — 17 h Post-Exercise

$F(df1 = 1, df2 = 7) = 8.4875, p\text{-val} = 0.0226$

Model Results:

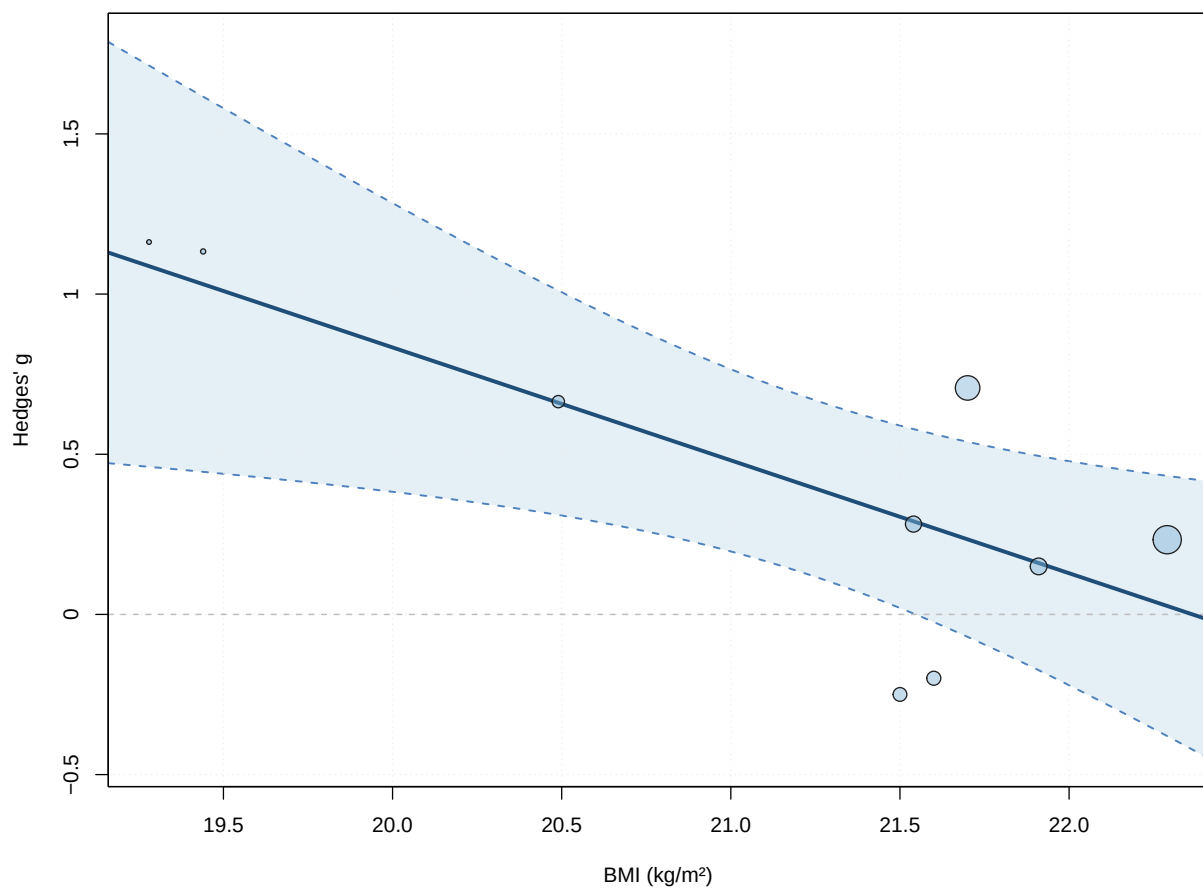
|         | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|---------|----|--------|---------|-------|
| intrcpt | 7.8813   | 2.5727 | 3.0635  | 7  | 0.0182 | 1.7980  |       |
| bmi     | -0.3524  | 0.1210 | -2.9133 | 7  | 0.0226 | -0.6384 |       |
| intrcpt | 13.9647  | *      |         |    |        |         |       |
| bmi     | -0.0664  | *      |         |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

! Important

Significant



### 9.7.3.3. Duration

Mixed-Effects Model ( $k = 8$ ;  $\tau^2$  estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -5.9984 | 18.1177  | 17.9969 | 18.2352 | 23.9969 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.2675 (SE = 0.1968)  
 tau (square root of estimated tau<sup>2</sup> value): 0.5172  
 I<sup>2</sup> (residual heterogeneity / unaccounted variability): 79.17%  
 H<sup>2</sup> (unaccounted variability / sampling variability): 4.80  
 R<sup>2</sup> (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 6) = 25.3804, p-val = 0.0003

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 6) = 0.0249, p-val = 0.8798

Model Results:

|          | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|----------|----------|--------|---------|----|--------|---------|-------|
| intrcpt  | 0.4150   | 0.2673 | 1.5525  | 6  | 0.1715 | -0.2391 |       |
| duration | -0.0008  | 0.0049 | -0.1578 | 6  | 0.8798 | -0.0127 |       |
| intrcpt  | 1.0691   |        |         |    |        |         |       |
| duration | 0.0112   |        |         |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

#### 9.7.3.4. Dose

Mixed-Effects Model (k = 8; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -5.9984 | 18.1177  | 17.9969 | 18.2352 | 23.9969 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.2675 (SE = 0.1968)  
 tau (square root of estimated tau<sup>2</sup> value): 0.5172  
 I<sup>2</sup> (residual heterogeneity / unaccounted variability): 79.17%  
 H<sup>2</sup> (unaccounted variability / sampling variability): 4.80  
 R<sup>2</sup> (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 6) = 25.3804, p-val = 0.0003

Test of Moderators (coefficient 2):

$F(df1 = 1, df2 = 6) = 0.0249, p\text{-val} = 0.8798$

Model Results:

|         | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|---------|----|--------|---------|-------|
| intrcpt | 0.4198   | 0.2876 | 1.4593  | 6  | 0.1948 | -0.2841 |       |
| dose    | -0.0001  | 0.0007 | -0.1578 | 6  | 0.8798 | -0.0019 |       |
|         |          |        |         |    |        |         |       |
| intrcpt | 1.1236   |        |         |    |        |         |       |
| dose    | 0.0017   |        |         |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

#### **i** Note

Not significant

### 9.7.3.5. IFN

Mixed-Effects Model ( $k = 7$ ;  $\tau^2$  estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -2.4686 | 11.7933  | 10.9373 | 10.7750 | 18.9373 |

$\tau^2$  (estimated amount of residual heterogeneity): 0.1048 (SE = 0.1101)  
 $\tau$  (square root of estimated  $\tau^2$  value): 0.3238  
 $I^2$  (residual heterogeneity / unaccounted variability): 62.72%  
 $H^2$  (unaccounted variability / sampling variability): 2.68  
 $R^2$  (amount of heterogeneity accounted for): 5.44%

Test for Residual Heterogeneity:

$QE(df = 5) = 10.3245, p\text{-val} = 0.0665$

Test of Moderators (coefficient 2):

$F(df1 = 1, df2 = 5) = 1.4551, p\text{-val} = 0.2817$

Model Results:

|         | estimate | se     | tval   | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|--------|----|--------|---------|-------|
| intrcpt | 0.0498   | 0.4627 | 0.1077 | 5  | 0.9184 | -1.1396 |       |
| IFN     | 0.1801   | 0.1493 | 1.2063 | 5  | 0.2817 | -0.2037 |       |
|         |          |        |        |    |        |         |       |
| intrcpt | 1.2392   |        |        |    |        |         |       |
| IFN     | 0.5640   |        |        |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

**9.7.3.6. IL\_4**Mixed-Effects Model (k = 7; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC    | AICc    |
|---------|----------|---------|--------|---------|
| -2.0026 | 10.8611  | 10.0052 | 9.8429 | 18.0052 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.0779 (SE = 0.0935)  
 tau (square root of estimated tau<sup>2</sup> value): 0.2790  
 I<sup>2</sup> (residual heterogeneity / unaccounted variability): 55.08%  
 H<sup>2</sup> (unaccounted variability / sampling variability): 2.23  
 R<sup>2</sup> (amount of heterogeneity accounted for): 29.77%

Test for Residual Heterogeneity:

QE(df = 5) = 10.3727, p-val = 0.0653

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 2.1405, p-val = 0.2033

Model Results:

|         | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|---------|----|--------|---------|-------|
| intrcpt | 0.7575   | 0.1881 | 4.0262  | 5  | 0.0101 | 0.2739  |       |
| IL_4    | -0.3349  | 0.2289 | -1.4630 | 5  | 0.2033 | -0.9234 |       |
| intrcpt | 1.2411   | *      |         |    |        |         |       |
| IL_4    | 0.2535   |        |         |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

**9.7.3.7. IL\_6**Mixed-Effects Model (k = 9; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -4.7669 | 18.1779  | 15.5338 | 16.1255 | 20.3338 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.1548 (SE = 0.1178)

## 9. Th2 — 17 h Post-Exercise

```
tau (square root of estimated tau^2 value):      0.3934
I^2 (residual heterogeneity / unaccounted variability): 72.75%
H^2 (unaccounted variability / sampling variability): 3.67
R^2 (amount of heterogeneity accounted for):      24.62%
```

Test for Residual Heterogeneity:

```
QE(df = 7) = 21.5167, p-val = 0.0031
```

Test of Moderators (coefficient 2):

```
F(df1 = 1, df2 = 7) = 2.7235, p-val = 0.1429
```

Model Results:

|         | estimate | se     | tval    | df | pval   | ci.lb   |
|---------|----------|--------|---------|----|--------|---------|
| intrcpt | 0.6354   | 0.2046 | 3.1055  | 7  | 0.0172 | 0.1516  |
| IL_6    | -0.1053  | 0.0638 | -1.6503 | 7  | 0.1429 | -0.2562 |
|         | ci.ub    |        |         |    |        |         |
| intrcpt | 1.1193   | *      |         |    |        |         |
| IL_6    | 0.0456   |        |         |    |        |         |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

! Important

Significant

### 9.7.3.8. IL\_10

Mixed-Effects Model (k = 9; tau^2 estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -5.8893 | 20.4226  | 17.7785 | 18.3702 | 22.5785 |

```
tau^2 (estimated amount of residual heterogeneity): 0.2170 (SE = 0.1510)
tau (square root of estimated tau^2 value):         0.4658
I^2 (residual heterogeneity / unaccounted variability): 79.03%
H^2 (unaccounted variability / sampling variability): 4.77
R^2 (amount of heterogeneity accounted for):         0.00%
```

Test for Residual Heterogeneity:

```
QE(df = 7) = 26.1288, p-val = 0.0005
```

Test of Moderators (coefficient 2):

```
F(df1 = 1, df2 = 7) = 0.6854, p-val = 0.4351
```

Model Results:

|         | estimate | se     | tval    | df | pval   | ci.lb   |
|---------|----------|--------|---------|----|--------|---------|
| intrcpt | 0.5383   | 0.2288 | 2.3529  | 7  | 0.0509 | -0.0027 |
| IL_10   | -0.1244  | 0.1503 | -0.8279 | 7  | 0.4351 | -0.4798 |
|         | ci.ub    |        |         |    |        |         |
| intrcpt | 1.0793   | .      |         |    |        |         |
| IL_10   | 0.2309   |        |         |    |        |         |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

**9.7.3.9. IL\_2**Mixed-Effects Model (k = 7; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -3.2017 | 13.2593  | 12.4033 | 12.2410 | 20.4033 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.1367 (SE = 0.1309)tau (square root of estimated tau<sup>2</sup> value): 0.3697I<sup>2</sup> (residual heterogeneity / unaccounted variability): 68.35%H<sup>2</sup> (unaccounted variability / sampling variability): 3.16R<sup>2</sup> (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 5) = 13.5802, p-val = 0.0185

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.1962, p-val = 0.6763

Model Results:

|         | estimate | se     | tval   | df | pval   | ci.lb   |
|---------|----------|--------|--------|----|--------|---------|
| intrcpt | 0.4754   | 0.2972 | 1.5994 | 5  | 0.1706 | -0.2887 |
| IL_2    | 0.0608   | 0.1372 | 0.4430 | 5  | 0.6763 | -0.2919 |
|         | ci.ub    |        |        |    |        |         |
| intrcpt | 1.2395   |        |        |    |        |         |
| IL_2    | 0.4134   |        |        |    |        |         |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

## 9.8. Multivariate Models

Mixed-Effects Model (k = 8; tau<sup>2</sup> estimator: HE)

```
tau^2 (estimated amount of residual heterogeneity):    0.2675 (SE = 0.1968)
tau (square root of estimated tau^2 value):           0.5172
I^2 (residual heterogeneity / unaccounted variability): 79.17%
H^2 (unaccounted variability / sampling variability):   4.80
R^2 (amount of heterogeneity accounted for):           0.00%
```

Test for Residual Heterogeneity:

QE(df = 6) = 25.3804, p-val = 0.0003

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 6) = 0.0249, p-val = 0.8798

Model Results:

|                   | estimate | se     | tval    | df | pval   |
|-------------------|----------|--------|---------|----|--------|
| intrcpt           | 0.3996   | 0.2176 | 1.8360  | 6  | 0.1160 |
| intensityVigorous | -0.1000  | 0.6338 | -0.1578 | 6  | 0.8798 |
|                   | ci.lb    | ci.ub  |         |    |        |
| intrcpt           | -0.1330  | 0.9322 |         |    |        |
| intensityVigorous | -1.6510  | 1.4509 |         |    |        |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

### **i** Note

Not significant

**Part III.**

**Th1/Th2 Ratio**



## 10. Th1/Th2 Ratio - Immediate Post-Exercise

### 10.1. Data Import

The base analysis uses the pre-computed Hedges'  $g$  file for the Th1/Th2 ratio at  $r = 0.5$  (`effect_size/Ratio/pre_post/Ratio_0.5`). Sensitivity analyses can substitute the  $r = 0.3 / 0.4 / 0.6 / 0.7$  files in the same folder. The pipeline column names are mapped to the analysis conventions (`VE_g`, `Sex`, `Age`, `BMI`) by `load_es()`.

### 10.2. Primary Meta-Analysis

`metagen()` fits a random-effects model on the pre-computed effect sizes ( $g$ ,  $SE_g$ ) using the **Hedges-Olkin (HE)** <sup>2</sup> estimator and the **Knapp-Hartung** small-sample correction (`hakn = TRUE`). **Kostrzewa-Nowak et al. (2019)** is removed before pooling and the model re-fitted; `metafor::rma()` cross-validates.

#### 10.2.1. Full Model {meta}

Number of studies:  $k = 18$

|                           | SMD     | 95% CI            | t    |
|---------------------------|---------|-------------------|------|
| Random effects model (HK) | 0.3544  | [-0.0597; 0.7685] | 1.81 |
| Prediction interval       |         | [-1.7206; 2.4294] |      |
|                           | p-value |                   |      |
| Random effects model (HK) | 0.0887  |                   |      |
| Prediction interval       |         |                   |      |

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.9096$  [0.2029; 1.5996];  $\tau = 0.9537$  [0.4504; 1.2647]  
 $I^2 = 80.5\%$  [70.0%; 87.3%];  $H = 2.26$  [1.83; 2.81]

Test of heterogeneity:

| Q     | d.f. | p-value  |
|-------|------|----------|
| 87.11 | 17   | < 0.0001 |

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model ( $df = 17$ )
- Prediction interval based on t-distribution ( $df = 17$ )

### 10.2.2. Full Model {metafor}

Random-Effects Model (k = 18; tau<sup>2</sup> estimator: HE)

tau<sup>2</sup> (estimated amount of total heterogeneity): 0.9096 (SE = 0.4257)  
 tau (square root of estimated tau<sup>2</sup> value): 0.9537  
 I<sup>2</sup> (total heterogeneity / total variability): 91.77%  
 H<sup>2</sup> (total variability / sampling variability): 12.15

Test for Heterogeneity:

Q(df = 17) = 87.1136, p-val < .0001

Model Results:

| estimate | se     | tval   | df | pval   | ci.lb   | ci.ub  |
|----------|--------|--------|----|--------|---------|--------|
| 0.3544   | 0.1963 | 1.8055 | 17 | 0.0887 | -0.0597 | 0.7685 |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

### 10.2.3. Adapted Model {meta}

Number of studies: k = 17

|                           | SMD     | 95% CI            | t    |
|---------------------------|---------|-------------------|------|
| Random effects model (HK) | 0.3005  | [-0.0558; 0.6568] | 1.79 |
| Prediction interval       |         | [-1.1139; 1.7149] |      |
|                           | p-value |                   |      |
| Random effects model (HK) | 0.0928  |                   |      |
| Prediction interval       |         |                   |      |

Quantifying heterogeneity (with 95% CIs):

tau<sup>2</sup> = 0.4156 [0.1703; 1.0546]; tau = 0.6447 [0.4126; 1.0270]  
 I<sup>2</sup> = 80.6% [69.8%; 87.5%]; H = 2.27 [1.82; 2.83]

Test of heterogeneity:

| Q     | d.f. | p-value  |
|-------|------|----------|
| 82.37 | 16   | < 0.0001 |

Details of meta-analysis methods:

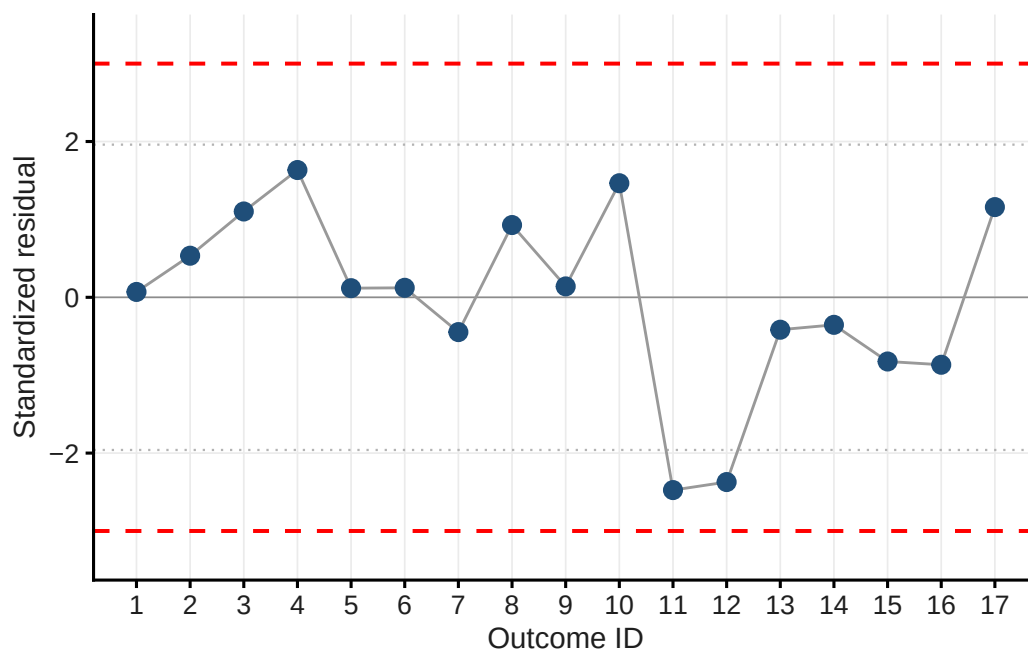
- Inverse variance method
- Hedges estimator for tau<sup>2</sup>
- Q-Profile method for confidence interval of tau<sup>2</sup> and tau
- Calculation of I<sup>2</sup> based on Q
- Hartung-Knapp adjustment for random effects model (df = 16)
- Prediction interval based on t-distribution (df = 16)

## 10.3. Outlier Detection

Leave-one-out standardised residuals are computed via `metaoutliers()`, which is passed the **variance** (`m5$seTE^2`). Studies with  $|z| > 3$  are flagged as outliers;  $|z| > 1.96$  warrants inspection.

**Table 10.1.:** Standardised residuals

| Short_reference                    | g      | std_res |
|------------------------------------|--------|---------|
| Ibfelt et al. (2002)               | 0.347  | 0.069   |
| Kostrzewska-Nowak & Nowak (2018)a  | 0.637  | 0.534   |
| Kostrzewska-Nowak & Nowak (2018)b  | 0.993  | 1.102   |
| Kostrzewska-Nowak & Nowak (2020a)a | 1.209  | 1.634   |
| Kostrzewska-Nowak & Nowak (2020a)b | 0.374  | 0.117   |
| Kostrzewska-Nowak & Nowak (2020b)a | 0.378  | 0.122   |
| Kostrzewska-Nowak & Nowak (2020b)b | 0.028  | -0.447  |
| Kostrzewska-Nowak & Nowak (2020b)c | 0.875  | 0.928   |
| Kostrzewska-Nowak & Nowak (2020b)d | 0.389  | 0.140   |
| Kostrzewska-Nowak & Nowak (2020b)e | 1.206  | 1.465   |
| Steensberg et al. (2001)           | -1.309 | -2.475  |
| Tsubakihara et al. (2012)          | -0.959 | -2.372  |
| Lancaster et al. (2005)a           | 0.033  | -0.416  |
| Lancaster et al. (2005)b           | 0.073  | -0.353  |
| Lancaster et al. (2005)c           | -0.226 | -0.824  |
| Harbaum et al. (2016)a             | -0.215 | -0.866  |
| Harbaum et al. (2016)b             | 1.065  | 1.158   |

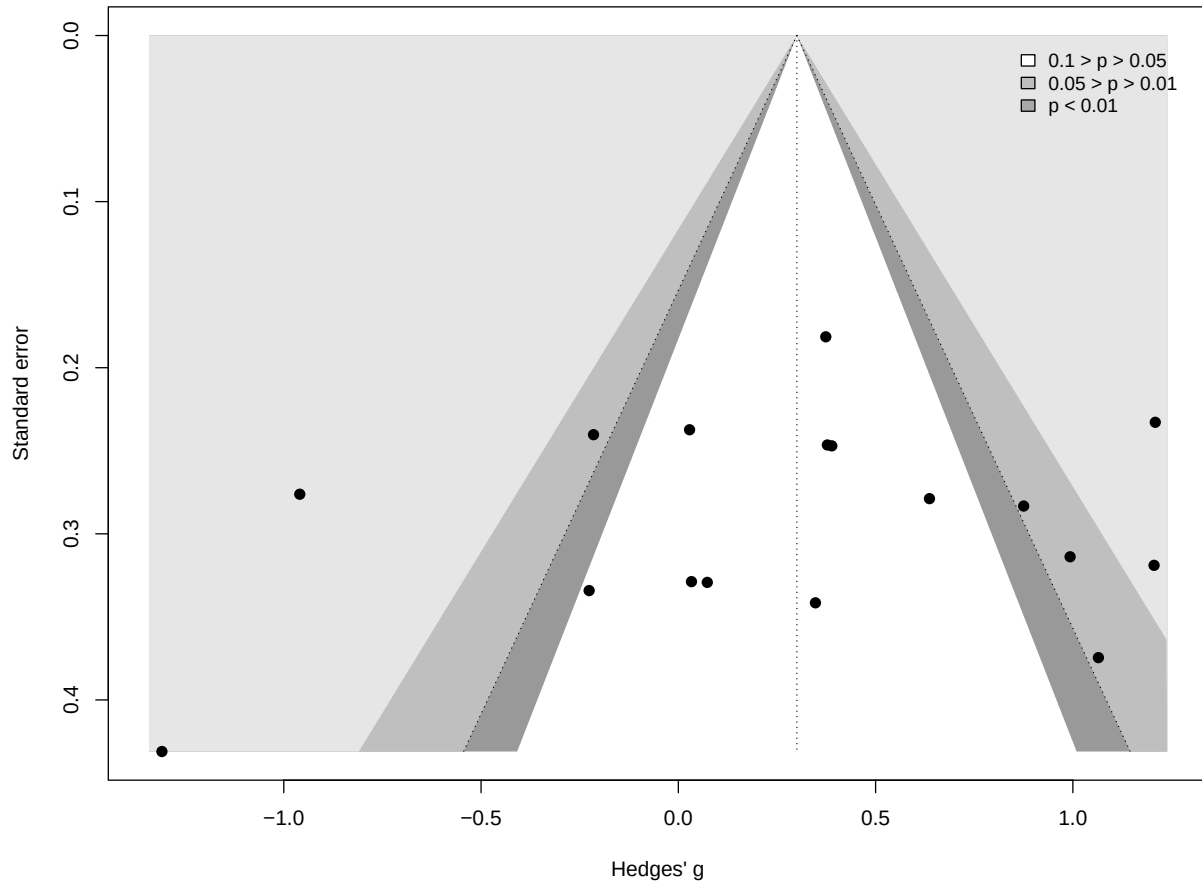


**i** Note

No outlier.

## 10.4. Funnel Plot & Influential Cases

The contour-enhanced funnel plot shows study estimates against their standard errors; shaded regions mark significance zones ( $\alpha = 0.10, 0.05, 0.01$ ). The `dfbetas` diagnostic from `metafor::rma.mv()` identifies **influential cases** - distinct from statistical outliers and not necessarily excluded.



**Table 10.2.:** DFBETAS influence diagnostics

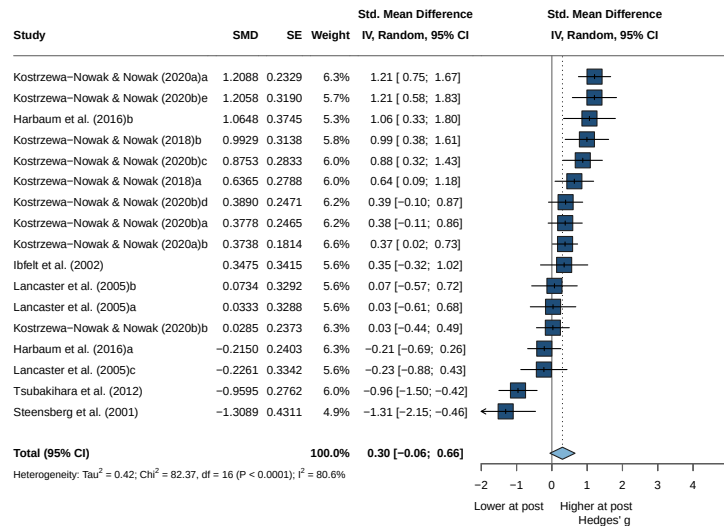
| Short_reference                    | g      | dfbetas     |
|------------------------------------|--------|-------------|
| Ibfelt et al. (2002)               | 0.347  | 0.01004021  |
| Kostrzewska-Nowak & Nowak (2018)a  | 0.637  | 0.27911537  |
| Kostrzewska-Nowak & Nowak (2018)b  | 0.993  | 0.47102525  |
| Kostrzewska-Nowak & Nowak (2020a)a | 1.209  | 1.18025273  |
| Kostrzewska-Nowak & Nowak (2020a)b | 0.374  | 0.10134265  |
| Kostrzewska-Nowak & Nowak (2020b)a | 0.378  | 0.05596990  |
| Kostrzewska-Nowak & Nowak (2020b)b | 0.028  | -0.38958693 |
| Kostrzewska-Nowak & Nowak (2020b)c | 0.875  | 0.48056733  |
| Kostrzewska-Nowak & Nowak (2020b)d | 0.389  | 0.06893600  |
| Kostrzewska-Nowak & Nowak (2020b)e | 1.206  | 0.60156479  |
| Steensberg et al. (2001)           | -1.309 | -0.60449272 |
| Tsubakihara et al. (2012)          | -0.959 | -1.20154886 |
| Lancaster et al. (2005)a           | 0.033  | -0.19186890 |
| Lancaster et al. (2005)b           | 0.073  | -0.16552634 |

**Table 10.2.:** DFBETAS influence diagnostics

| Short_reference          | g      | dfbetas     |
|--------------------------|--------|-------------|
| Lancaster et al. (2005)c | -0.226 | -0.34724912 |
| Harbaum et al. (2016)a   | -0.215 | -0.68475128 |
| Harbaum et al. (2016)b   | 1.065  | 0.36143534  |

One value was above 1 and one below  $-1$ , so both were identified as influential cases. Neither is an outlier; all outcomes are retained and the sources of heterogeneity are examined in the moderator analysis.

## 10.5. Forest Plot



## 10.6. Asymmetry Test

Egger's regression test (`method = "linreg"`) evaluates funnel asymmetry as a proxy for small-study/publication effects (one-tailed,  $p < .10$ ). Where asymmetry is suggested, a trim-and-fill analysis estimates potentially missing studies.

Linear regression test of funnel plot asymmetry

Test result:  $t = -0.53$ ,  $df = 15$ ,  $p\text{-value} = 0.6026$

Bias estimate:  $-1.4500$  (SE = 2.7260)

Details:

## 10. Th1/Th2 Ratio - Immediate Post-Exercise

- multiplicative residual heterogeneity variance ( $\tau^2 = 5.3895$ )
- predictor: standard error
- weight: inverse variance
- reference: Egger et al. (1997), BMJ

### Note

No asymmetry -> no trim-and-fill

## 10.7. Moderator Analyses

### 10.7.1. Risk of Bias

Subgroup analysis by overall RoB rating tests whether study quality moderates the pooled effect; the between-subgroup statistic ( $Q_b$ ) indicates whether subgroup means differ.

#### 10.7.1.1. Model

Number of studies:  $k = 17$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.4156$  [0.1703; 1.0546];  $\tau = 0.6447$  [0.4126; 1.0270]

$I^2 = 80.6\%$  [69.8%; 87.5%];  $H = 2.27$  [1.82; 2.83]

Test of heterogeneity:

Q d.f. p-value

82.37 16 < 0.0001

Results for subgroups (random effects model (HK)):

|                | k | SMD    | 95% CI            | $\tau^2$ | $\tau$ |
|----------------|---|--------|-------------------|----------|--------|
| RoB = Serious  | 6 | 0.2952 | [-0.5524; 1.1427] | 0.5743   | 0.7578 |
| RoB = Moderate | 3 | 0.1355 | [-2.9084; 3.1793] | 1.4143   | 1.1892 |
| RoB = Low      | 8 | 0.3483 | [-0.0432; 0.7398] | 0.1457   | 0.3818 |

Q  $I^2$

RoB = Serious 44.98 88.9%

RoB = Moderate 20.00 90.0%

RoB = Low 17.11 59.1%

Test for subgroup differences (random effects model (HK)):

Q d.f. p-value

Between groups 0.10 2 0.9517

Details of meta-analysis methods:

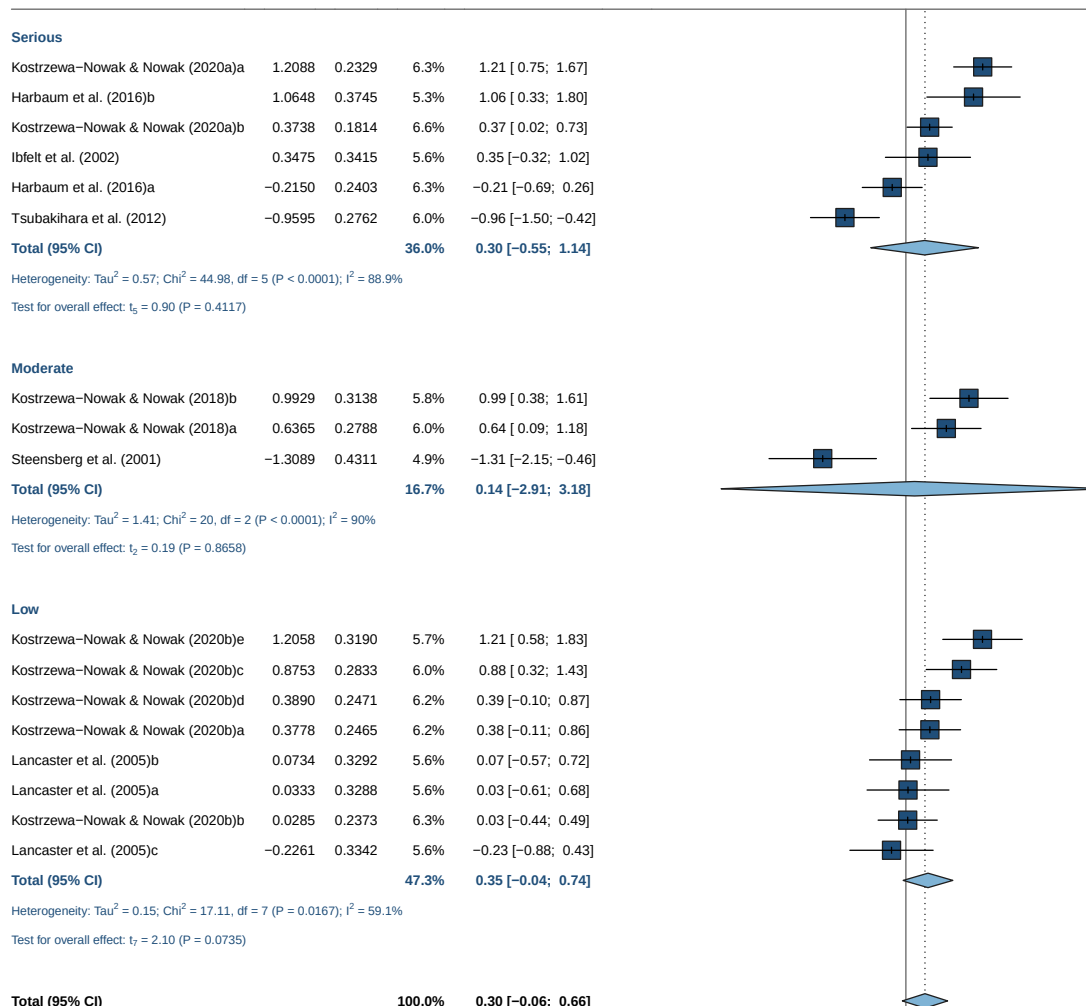
- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model

- Prediction interval based on t-distribution

### **i** Note

Not significant results

## 10.7.1.2. Forrest Plot



## 10.7.2. Sex

### 10.7.2.1. Model

Number of studies:  $k = 17$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.4156$  [0.1703; 1.0546];  $\tau = 0.6447$  [0.4126; 1.0270]

$I^2 = 80.6\%$  [69.8%; 87.5%];  $H = 2.27$  [1.82; 2.83]

Test of heterogeneity:

Q d.f. p-value

82.37 16 < 0.0001

## 10. Th1/Th2 Ratio - Immediate Post-Exercise

Results for subgroups (random effects model (HK)):

|              | k  | SMD     | 95% CI             | tau <sup>2</sup> | tau    |
|--------------|----|---------|--------------------|------------------|--------|
| sex = male   | 14 | 0.3858  | [ 0.0247; 0.7469]  | 0.3417           | 0.5846 |
| sex = female | 1  | -0.9595 | [-1.5008; -0.4182] | --               | --     |
| sex = mixed  | 2  | 0.3927  | [-7.7274; 8.5127]  | 0.7199           | 0.8484 |

|              | Q     | I <sup>2</sup> |
|--------------|-------|----------------|
| sex = male   | 49.15 | 73.5%          |
| sex = female | 0.00  | --             |
| sex = mixed  | 8.27  | 87.9%          |

Test for subgroup differences (random effects model (HK)):

|                | Q     | d.f. | p-value |
|----------------|-------|------|---------|
| Between groups | 17.68 | 2    | 0.0001  |

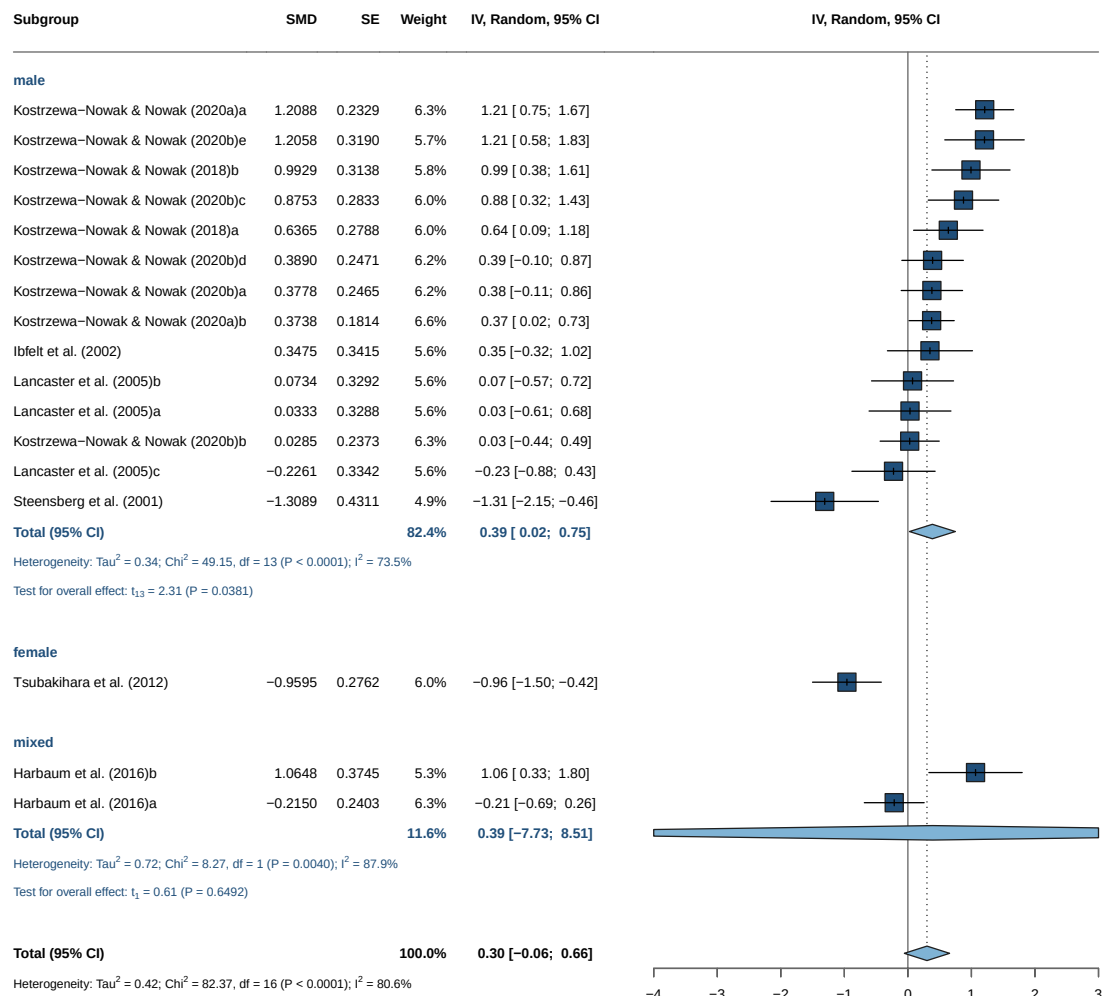
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for tau<sup>2</sup>
- Q-Profile method for confidence interval of tau<sup>2</sup> and tau
- Calculation of I<sup>2</sup> based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

### ! Important

Significant results but 15 vs 1 vs 2

### 10.7.2.2. Forrest Plot



### 10.7.3. Intensity

#### 10.7.3.1. Model

Number of studies:  $k = 16$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.3382$  [0.1128; 0.8975];  $\tau = 0.5815$  [0.3358; 0.9474]

$I^2 = 74.7\%$  [58.6%; 84.5%];  $H = 1.99$  [1.56; 2.54]

Test of heterogeneity:

| Q     | d.f. | p-value  |
|-------|------|----------|
| 59.18 | 15   | < 0.0001 |

Results for subgroups (random effects model (HK)):

|                      | k  | SMD     | 95% CI            | $\tau^2$ |
|----------------------|----|---------|-------------------|----------|
| intensity = Vigorous | 5  | -0.1814 | [-0.9439; 0.5811] | 0.2883   |
| intensity = Maximal  | 11 | 0.5997  | [ 0.2761; 0.9232] | 0.1564   |
|                      |    | $\tau$  | Q                 | $I^2$    |
| intensity = Vigorous |    | 0.5370  | 10.14             | 60.6%    |

## 10. Th1/Th2 Ratio - Immediate Post-Exercise

intensity = Maximal 0.3955 34.33 70.9%

Test for subgroup differences (random effects model (HK)):

Q d.f. p-value  
Between groups 6.32 1 0.0119

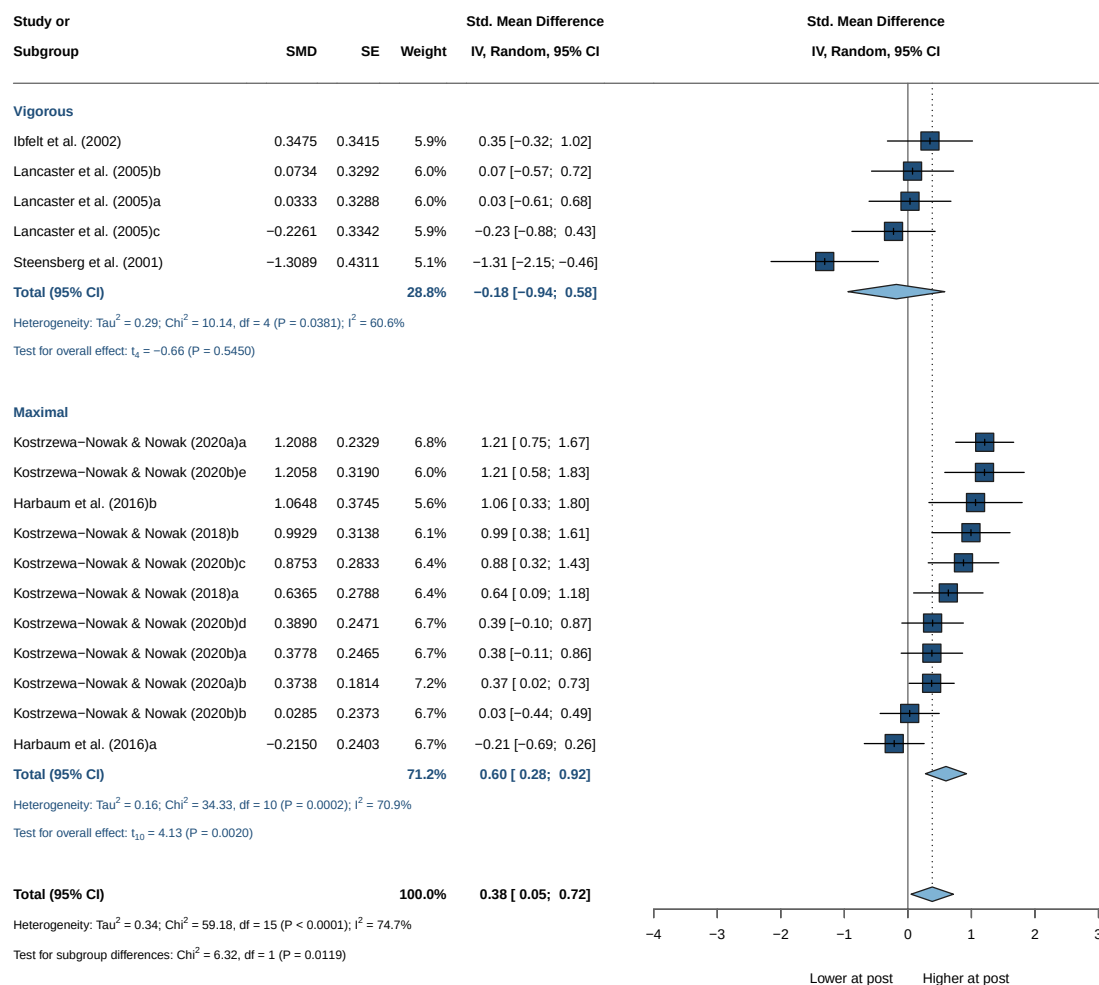
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

**! Important**

Significant

### 10.7.3.2. Forrest Plot



### 10.7.4. Staining

Number of studies:  $k = 17$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.4156$  [0.1703; 1.0546];  $\tau = 0.6447$  [0.4126; 1.0270]

$I^2 = 80.6\%$  [69.8%; 87.5%];  $H = 2.27$  [1.82; 2.83]

Test of heterogeneity:

Q d.f. p-value

82.37 16 < 0.0001

Results for subgroups (random effects model (HK)):

|                  | k  | SMD     | 95% CI            | $\tau^2$ |
|------------------|----|---------|-------------------|----------|
| staining = intra | 14 | 0.3858  | [ 0.0247; 0.7469] | 0.3417   |
| staining = extra | 3  | -0.0594 | [-2.5758; 2.4570] | 0.9569   |

|                  | $\tau$ | Q     | $I^2$ |
|------------------|--------|-------|-------|
| staining = intra | 0.5846 | 49.15 | 73.5% |
| staining = extra | 0.9782 | 18.93 | 89.4% |

Test for subgroup differences (random effects model (HK)):

Q d.f. p-value

Between groups 0.54 1 0.4643

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

#### Note

Not significant results

### 10.7.5. Continuous Moderators (Meta-Regression)

Continuous moderators are tested via weighted meta-regression (`metareg()`). A significant slope indicates the moderator explains part of the between-study heterogeneity ( $\tau^2$ ); bubble plots show the fitted line with 95% CI.

#### 10.7.5.1. Age

Mixed-Effects Model ( $k = 16$ ;  $\tau^2$  estimator: HE)

| logLik   | deviance | AIC     | BIC     | AICc    |
|----------|----------|---------|---------|---------|
| -14.5497 | 39.4622  | 35.0995 | 37.4173 | 37.0995 |

#### 10. Th1/Th2 Ratio - Immediate Post-Exercise

```
tau^2 (estimated amount of residual heterogeneity):    0.3523 (SE = 0.1679)
tau (square root of estimated tau^2 value):           0.5936
I^2 (residual heterogeneity / unaccounted variability): 81.90%
H^2 (unaccounted variability / sampling variability):   5.52
R^2 (amount of heterogeneity accounted for):           0.00%
```

Test for Residual Heterogeneity:

QE(df = 14) = 53.0007, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 14) = 0.7114, p-val = 0.4131

Model Results:

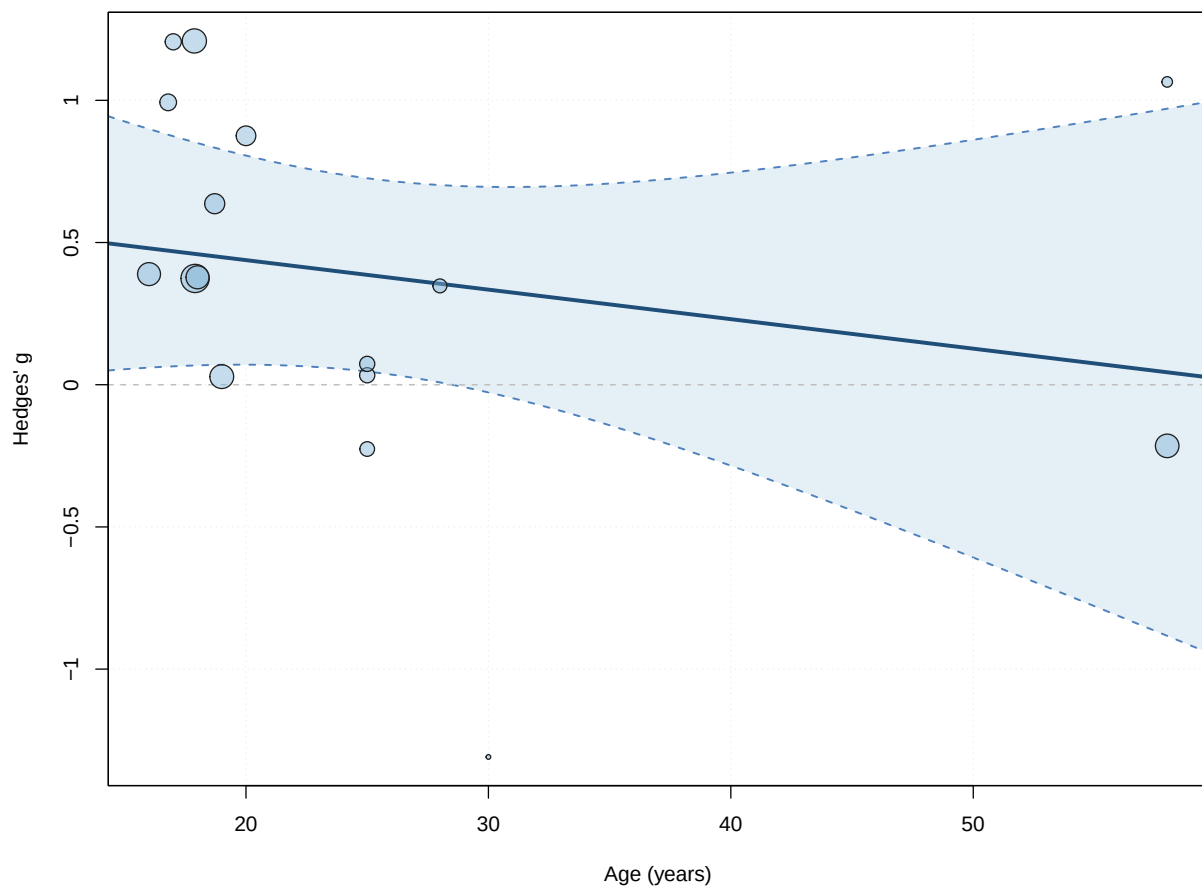
|         | estimate | se     | tval    | df | pval   | ci.lb   |
|---------|----------|--------|---------|----|--------|---------|
| intrcpt | 0.6457   | 0.3495 | 1.8473  | 14 | 0.0859 | -0.1040 |
| age     | -0.0104  | 0.0123 | -0.8435 | 14 | 0.4131 | -0.0368 |
|         | ci.ub    |        |         |    |        |         |
| intrcpt | 1.3953   | .      |         |    |        |         |
| age     | 0.0160   |        |         |    |        |         |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant



### 10.7.5.2. BMI

Mixed-Effects Model ( $k = 12$ ;  $\tau^2$  estimator: HE)

| logLik   | deviance | AIC     | BIC     | AICc    |
|----------|----------|---------|---------|---------|
| -11.3566 | 32.5305  | 28.7132 | 30.1679 | 31.7132 |

$\tau^2$  (estimated amount of residual heterogeneity): 0.3880 (SE = 0.2073)  
 $\tau$  (square root of estimated  $\tau^2$  value): 0.6229  
 $I^2$  (residual heterogeneity / unaccounted variability): 85.24%  
 $H^2$  (unaccounted variability / sampling variability): 6.77  
 $R^2$  (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 10) = 61.5799, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 10) = 0.0011, p-val = 0.9744

Model Results:

### 10. Th1/Th2 Ratio - Immediate Post-Exercise

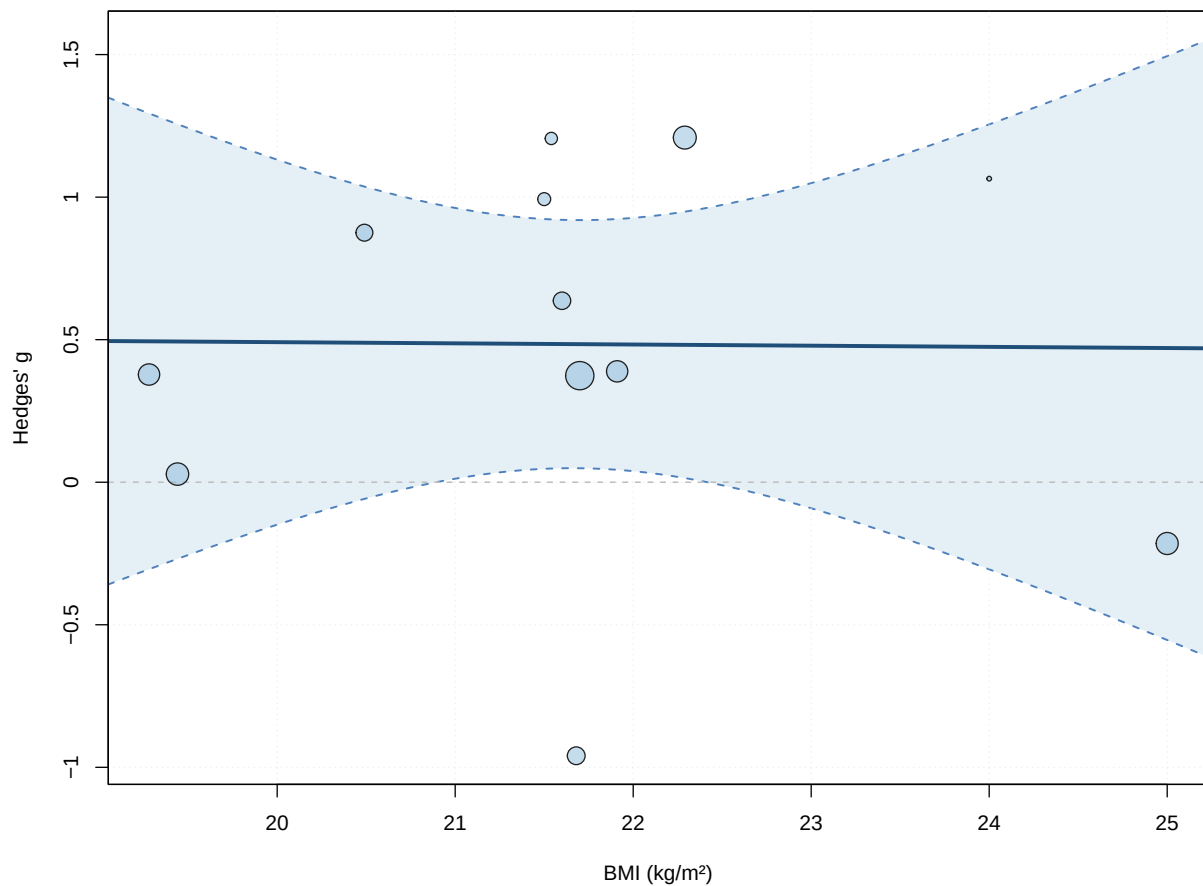
|         | estimate | se     | tval    | df | pval   | ci.lb   |
|---------|----------|--------|---------|----|--------|---------|
| intrcpt | 0.5737   | 2.7249 | 0.2106  | 10 | 0.8375 | -5.4977 |
| bmi     | -0.0041  | 0.1254 | -0.0329 | 10 | 0.9744 | -0.2834 |
|         | ci.ub    |        |         |    |        |         |
| intrcpt | 6.6452   |        |         |    |        |         |
| bmi     | 0.2752   |        |         |    |        |         |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

#### Note

Not significant



### 10.7.5.3. Duration

Mixed-Effects Model (k = 15; tau<sup>2</sup> estimator: HE)

| logLik   | deviance | AIC     | BIC     | AICc    |
|----------|----------|---------|---------|---------|
| -11.2968 | 31.0389  | 28.5936 | 30.7177 | 30.7754 |

```

tau^2 (estimated amount of residual heterogeneity):    0.2107 (SE = 0.1206)
tau (square root of estimated tau^2 value):           0.4590
I^2 (residual heterogeneity / unaccounted variability): 71.01%
H^2 (unaccounted variability / sampling variability):   3.45
R^2 (amount of heterogeneity accounted for):           49.32%

```

Test for Residual Heterogeneity:

QE(df = 13) = 43.6946, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 13) = 9.0460, p-val = 0.0101

Model Results:

|          | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|----------|----------|--------|---------|----|--------|---------|-------|
| intrcpt  | 0.6802   | 0.2053 | 3.3142  | 13 | 0.0056 | 0.2368  |       |
| duration | -0.0073  | 0.0024 | -3.0077 | 13 | 0.0101 | -0.0126 |       |
|          |          |        |         |    |        |         |       |
| intrcpt  | 1.1237   | **     |         |    |        |         |       |
| duration | -0.0021  | *      |         |    |        |         |       |

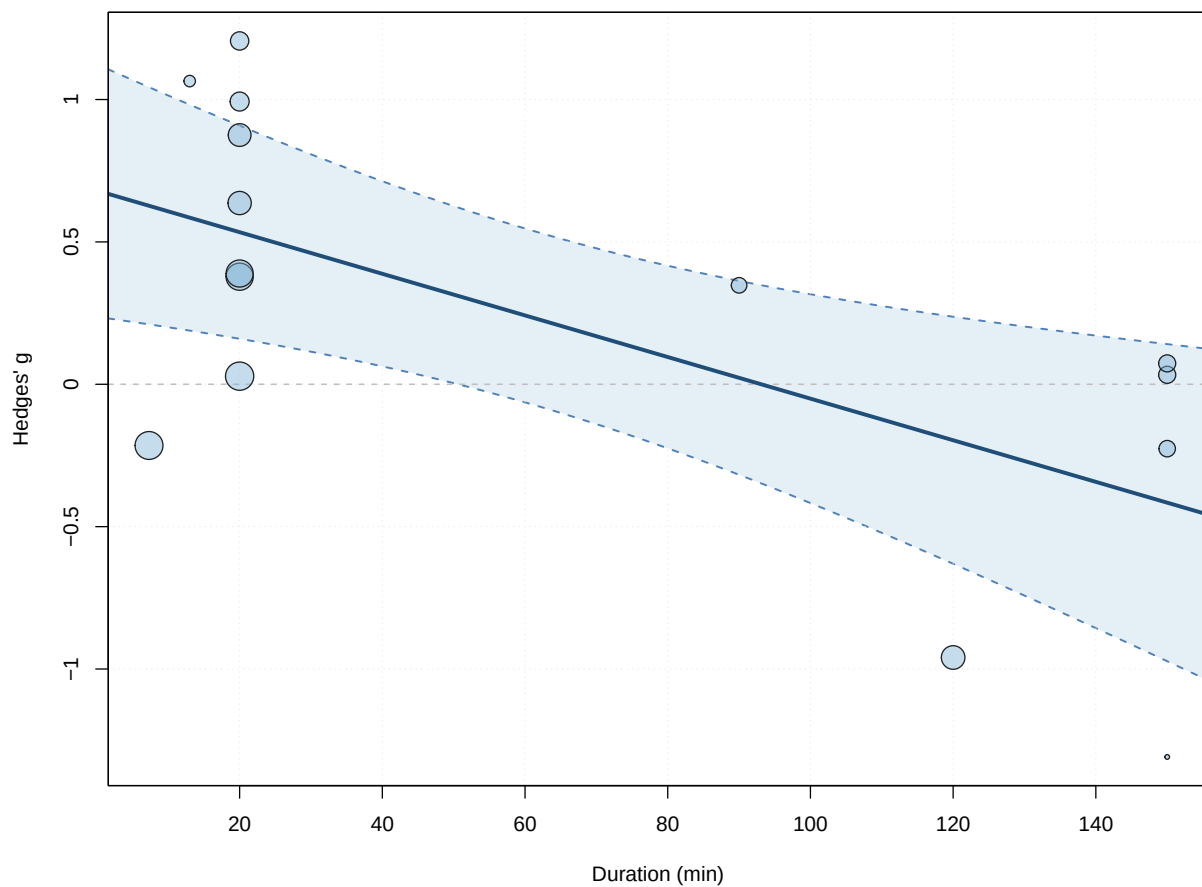
---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

! Important

Significant

## 10. Th1/Th2 Ratio - Immediate Post-Exercise



### 10.7.5.4. Dose

Mixed-Effects Model (k = 14; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -9.3620 | 26.4336  | 24.7240 | 26.6412 | 27.1240 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.1656 (SE = 0.1075)  
tau (square root of estimated tau<sup>2</sup> value): 0.4070  
I<sup>2</sup> (residual heterogeneity / unaccounted variability): 65.73%  
H<sup>2</sup> (unaccounted variability / sampling variability): 2.92  
R<sup>2</sup> (amount of heterogeneity accounted for): 51.12%

Test for Residual Heterogeneity:

QE(df = 12) = 34.1013, p-val = 0.0007

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 12) = 7.2942, p-val = 0.0193

Model Results:

|         | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|---------|----|--------|---------|-------|
| intrcpt | 0.7315   | 0.2022 | 3.6180  | 12 | 0.0035 | 0.2910  |       |
| dose    | -0.0010  | 0.0004 | -2.7008 | 12 | 0.0193 | -0.0019 |       |
|         |          |        |         |    |        |         |       |
| intrcpt | 1.1721   | **     |         |    |        |         |       |
| dose    | -0.0002  | *      |         |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

**10.7.5.5. IFN**Mixed-Effects Model (k = 7; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -4.0987 | 14.9426  | 14.1973 | 14.0350 | 22.1973 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.1868 (SE = 0.1615)tau (square root of estimated tau<sup>2</sup> value): 0.4322I<sup>2</sup> (residual heterogeneity / unaccounted variability): 74.03%H<sup>2</sup> (unaccounted variability / sampling variability): 3.85R<sup>2</sup> (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 5) = 19.9174, p-val = 0.0013

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.0002, p-val = 0.9881

Model Results:

|         | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|---------|----|--------|---------|-------|
| intrcpt | 0.6319   | 0.8621 | 0.7330  | 5  | 0.4965 | -1.5842 |       |
| IFN     | -0.0073  | 0.4649 | -0.0156 | 5  | 0.9881 | -1.2024 |       |
|         |          |        |         |    |        |         |       |
| intrcpt | 2.8480   |        |         |    |        |         |       |
| IFN     | 1.1878   |        |         |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

#### 10.7.5.6. IL\_4

Mixed-Effects Model (k = 9; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -6.2496 | 20.3845  | 18.4992 | 19.0909 | 23.2992 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.2223 (SE = 0.1585)  
 tau (square root of estimated tau<sup>2</sup> value): 0.4715  
 I<sup>2</sup> (residual heterogeneity / unaccounted variability): 77.77%  
 H<sup>2</sup> (unaccounted variability / sampling variability): 4.50  
 R<sup>2</sup> (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 7) = 30.1193, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 7) = 0.3341, p-val = 0.5814

Model Results:

|         | estimate | se     | tval   | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|--------|----|--------|---------|-------|
| intrcpt | 0.4967   | 0.2145 | 2.3160 | 7  | 0.0537 | -0.0104 |       |
| IL_4    | 0.1043   | 0.1804 | 0.5780 | 7  | 0.5814 | -0.3224 |       |
| intrcpt | 1.0038   |        |        |    |        |         |       |
| IL_4    | 0.5310   |        |        |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

#### 10.7.5.7. IL\_6

Mixed-Effects Model (k = 14; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -8.9389 | 28.0846  | 23.8778 | 25.7949 | 26.2778 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.1675 (SE = 0.1022)  
 tau (square root of estimated tau<sup>2</sup> value): 0.4092  
 I<sup>2</sup> (residual heterogeneity / unaccounted variability): 70.18%  
 H<sup>2</sup> (unaccounted variability / sampling variability): 3.35  
 R<sup>2</sup> (amount of heterogeneity accounted for): 7.58%

Test for Residual Heterogeneity:

QE(df = 12) = 36.6447, p-val = 0.0003

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 12) = 1.8665, p-val = 0.1969

Model Results:

|         | estimate | se     | tval   | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|--------|----|--------|---------|-------|
| intrcpt | 0.3688   | 0.1537 | 2.4001 | 12 | 0.0335 | 0.0340  |       |
| IL_6    | 0.1207   | 0.0884 | 1.3662 | 12 | 0.1969 | -0.0718 |       |
| intrcpt | 0.7036   |        |        |    |        |         | *     |
| IL_6    | 0.3132   |        |        |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

#### Note

Not significant

### 10.7.5.8. IL\_10

Mixed-Effects Model (k = 11; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -7.0987 | 23.2792  | 20.1975 | 21.3911 | 23.6260 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.1760 (SE = 0.1201)

tau (square root of estimated tau<sup>2</sup> value): 0.4195

I<sup>2</sup> (residual heterogeneity / unaccounted variability): 71.18%

H<sup>2</sup> (unaccounted variability / sampling variability): 3.47

R<sup>2</sup> (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 9) = 34.2151, p-val < .0001

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 9) = 0.0965, p-val = 0.7631

Model Results:

|         | estimate | se     | tval   | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|--------|----|--------|---------|-------|
| intrcpt | 0.5975   | 0.1530 | 3.9061 | 9  | 0.0036 | 0.2515  |       |
| IL_10   | 0.1301   | 0.4189 | 0.3107 | 9  | 0.7631 | -0.8174 |       |

#### 10. Th1/Th2 Ratio - Immediate Post-Exercise

```
intrcpt 0.9435 **
IL_10    1.0777
```

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

##### Note

Not significant

#### 10.7.5.9. IL\_2

Mixed-Effects Model (k = 7; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -3.8027 | 14.3505  | 13.6053 | 13.4430 | 21.6053 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.1669 (SE = 0.1434)  
tau (square root of estimated tau<sup>2</sup> value): 0.4086  
I<sup>2</sup> (residual heterogeneity / unaccounted variability): 74.80%  
H<sup>2</sup> (unaccounted variability / sampling variability): 3.97  
R<sup>2</sup> (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 5) = 19.7007, p-val = 0.0014

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.4392, p-val = 0.5368

Model Results:

|         | estimate | se     | tval   | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|--------|----|--------|---------|-------|
| intrcpt | 0.1414   | 0.7407 | 0.1909 | 5  | 0.8561 | -1.7626 |       |
| IL_2    | 0.2684   | 0.4049 | 0.6627 | 5  | 0.5368 | -0.7725 |       |
| intrcpt | 2.0453   |        |        |    |        |         |       |
| IL_2    | 1.3093   |        |        |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

##### Note

Not significant

## 10.8. Multivariate Models

Multivariate meta-regression tests interactions between candidate moderators. None of the models below yields independent predictors.

### 10.8.1. Intensity \* Duration

Mixed-Effects Model (k = 14; tau<sup>2</sup> estimator: HE)

```
tau^2 (estimated amount of residual heterogeneity):    0.1901 (SE = 0.1290)
tau (square root of estimated tau^2 value):           0.4360
I^2 (residual heterogeneity / unaccounted variability): 68.47%
H^2 (unaccounted variability / sampling variability):   3.17
R^2 (amount of heterogeneity accounted for):           43.88%
```

Test for Residual Heterogeneity:

QE(df = 10) = 26.5082, p-val = 0.0031

Test of Moderators (coefficients 2:4):

F(df1 = 3, df2 = 10) = 2.9355, p-val = 0.0857

Model Results:

|                            | estimate | se     | tval    | df |
|----------------------------|----------|--------|---------|----|
| intrcpt                    | -0.2459  | 0.6993 | -0.3516 | 10 |
| intensityVigorous          | 1.5730   | 1.5618 | 1.0071  | 10 |
| duration                   | 0.0452   | 0.0381 | 1.1868  | 10 |
| intensityVigorous:duration | -0.0561  | 0.0394 | -1.4245 | 10 |

|                            | pval   | ci.lb   | ci.ub  |
|----------------------------|--------|---------|--------|
| intrcpt                    | 0.7324 | -1.8040 | 1.3122 |
| intensityVigorous          | 0.3376 | -1.9070 | 5.0529 |
| duration                   | 0.2627 | -0.0397 | 0.1300 |
| intensityVigorous:duration | 0.1848 | -0.1438 | 0.0316 |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

### 10.8.2. IL\_10 \* IL\_4

Mixed-Effects Model (k = 9; tau<sup>2</sup> estimator: HE)

```
tau^2 (estimated amount of residual heterogeneity):    0.3297 (SE = 0.2594)
tau (square root of estimated tau^2 value):           0.5742
```

### 10. Th1/Th2 Ratio - Immediate Post-Exercise

I<sup>2</sup> (residual heterogeneity / unaccounted variability): 81.69%  
H<sup>2</sup> (unaccounted variability / sampling variability): 5.46  
R<sup>2</sup> (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 5) = 30.1034, p-val < .0001

Test of Moderators (coefficients 2:4):

F(df1 = 3, df2 = 5) = 0.0923, p-val = 0.9611

Model Results:

|            | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub  |
|------------|----------|--------|---------|----|--------|---------|--------|
| intrcpt    | 0.5401   | 0.3177 | 1.6997  | 5  | 0.1499 | -0.2767 | 1.3568 |
| IL_10      | 0.0951   | 0.6813 | 0.1395  | 5  | 0.8945 | -1.6563 | 1.8464 |
| IL_4       | 0.0953   | 0.2254 | 0.4229  | 5  | 0.6899 | -0.4842 | 0.6748 |
| IL_10:IL_4 | -0.2817  | 1.4854 | -0.1897 | 5  | 0.8570 | -4.1000 | 3.5365 |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

#### Note

Not significant

### 10.8.3. IL\_2 \* IFN

Mixed-Effects Model (k = 7; tau<sup>2</sup> estimator: HE)

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.2707 (SE = 0.2724)  
tau (square root of estimated tau<sup>2</sup> value): 0.5203  
I<sup>2</sup> (residual heterogeneity / unaccounted variability): 81.40%  
H<sup>2</sup> (unaccounted variability / sampling variability): 5.38  
R<sup>2</sup> (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 3) = 17.0604, p-val = 0.0007

Test of Moderators (coefficients 2:4):

F(df1 = 3, df2 = 3) = 0.2470, p-val = 0.8596

Model Results:

|          | estimate | se     | tval    | df | pval   | ci.lb    |
|----------|----------|--------|---------|----|--------|----------|
| intrcpt  | 5.7989   | 9.5568 | 0.6068  | 3  | 0.5868 | -24.6150 |
| IL_2     | -2.8869  | 5.5986 | -0.5156 | 3  | 0.6417 | -20.7043 |
| IFN      | -3.1991  | 5.2880 | -0.6050 | 3  | 0.5879 | -20.0280 |
| IL_2:IFN | 1.7440   | 3.0132 | 0.5788  | 3  | 0.6033 | -7.8454  |

|          | ci.ub   |
|----------|---------|
| intrcpt  | 36.2129 |
| IL_2     | 14.9304 |
| IFN      | 13.6298 |
| IL_2:IFN | 11.3335 |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

### **i** Note

Not significant



# 11. Th1/Th2 Ratio - ~2 h Post-Exercise

## 11.1. Data Import

The ~2 h window (effect\_size/Ratio/pre\_post+2h/Ratio\_pre\_post+2h\_r\_050.csv) is reported for the main analysis, outlier check, funnel, and forest only; the limited data do not support moderator modelling. No study is excluded a priori.

## 11.2. Primary Meta-Analysis

### 11.2.1. {meta}

Number of studies: k = 5

|                           | SMD     | 95% CI            | t     |
|---------------------------|---------|-------------------|-------|
| Random effects model (HK) | -0.5071 | [-1.1495; 0.1353] | -2.19 |
| Prediction interval       |         | [-1.8150; 0.8007] |       |
|                           | p-value |                   |       |
| Random effects model (HK) | 0.0935  |                   |       |
| Prediction interval       |         |                   |       |

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.1626$  [0.0000; 2.3056];  $\tau = 0.4033$  [0.0000; 1.5184]  
 $I^2 = 44.6\%$  [0.0%; 79.7%];  $H = 1.34$  [1.00; 2.22]

Test of heterogeneity:

| Q    | d.f. | p-value |
|------|------|---------|
| 7.22 | 4    | 0.1245  |

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model (df = 4)
- Prediction interval based on t-distribution (df = 4)

### 11.2.2. {metafor}

Random-Effects Model (k = 5;  $\tau^2$  estimator: HE)

### 11. Th1/Th2 Ratio - ~2 h Post-Exercise

```
tau^2 (estimated amount of total heterogeneity): 0.1626 (SE = 0.2133)
tau (square root of estimated tau^2 value):      0.4033
I^2 (total heterogeneity / total variability):    55.30%
H^2 (total variability / sampling variability):    2.24
```

Test for Heterogeneity:

Q(df = 4) = 7.2248, p-val = 0.1245

Model Results:

```
estimate      se      tval  df    pval    ci.lb    ci.ub
-0.5071  0.2314  -2.1916   4   0.0935  -1.1495   0.1353  .
```

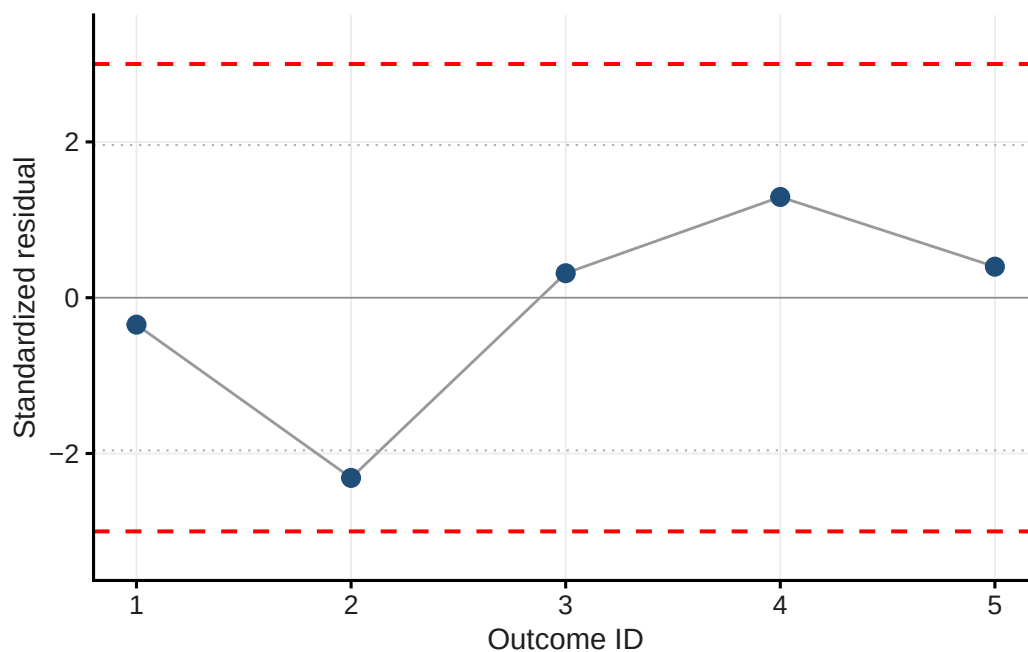
---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

## 11.3. Outlier Detection

**Table 11.1.:** Standardised residuals

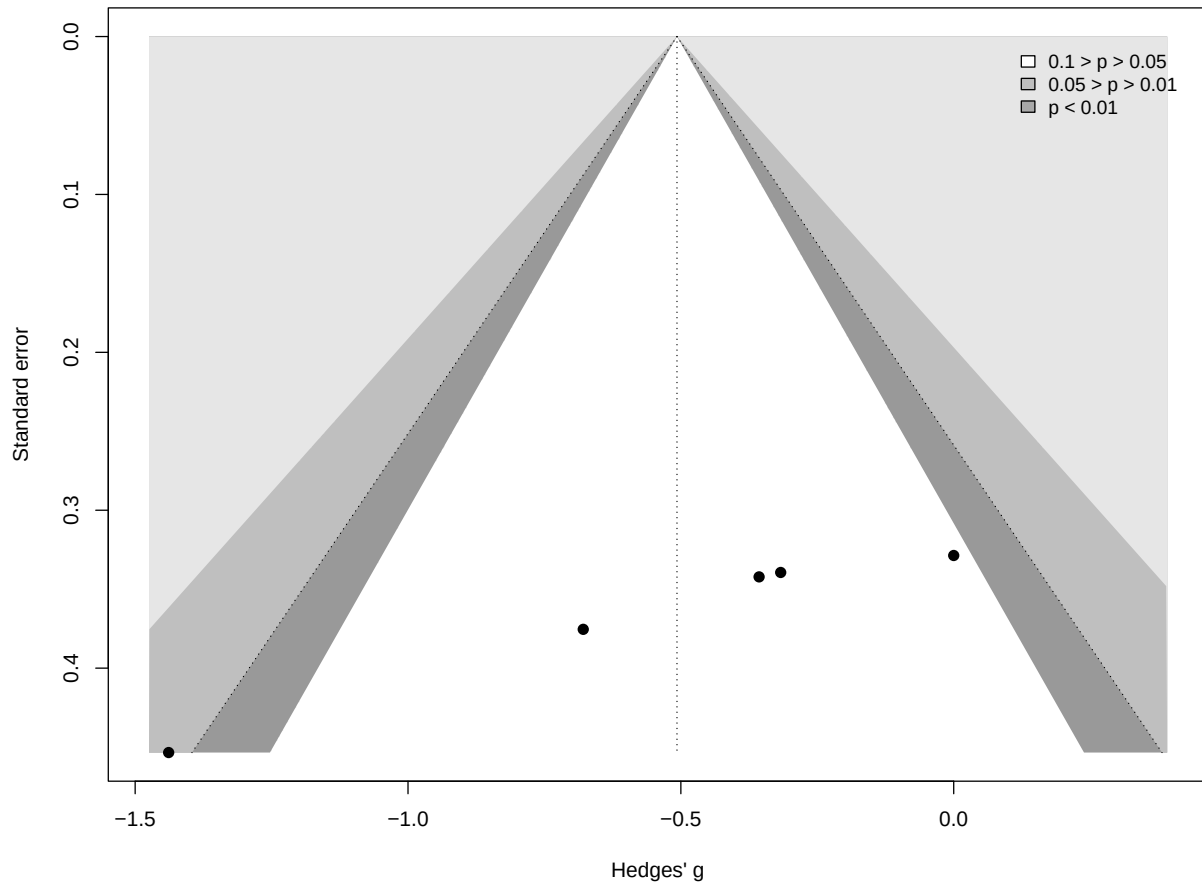
| Short_reference          | g      | std_res |
|--------------------------|--------|---------|
| Ibfelt et al. (2002)     | -0.679 | -0.346  |
| Steensberg et al. (2001) | -1.439 | -2.313  |
| Lancaster et al. (2005)a | -0.357 | 0.314   |
| Lancaster et al. (2005)b | 0.000  | 1.293   |
| Lancaster et al. (2005)c | -0.317 | 0.398   |



**i** Note

No outlier.

## 11.4. Funnel Plot & Influential Cases

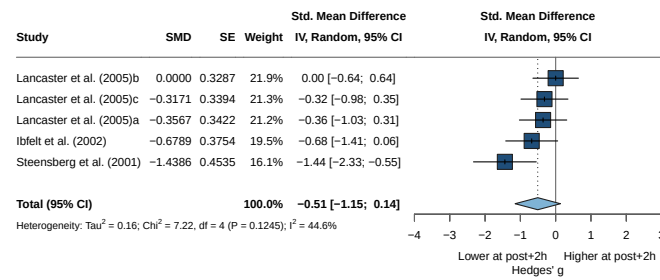


**Table 11.2.:** DFBETAS influence diagnostics

| Short_reference          | g      | dfbetas    |
|--------------------------|--------|------------|
| Ibfelt et al. (2002)     | -0.679 | -0.3097587 |
| Steensberg et al. (2001) | -1.439 | -0.8804841 |
| Lancaster et al. (2005)a | -0.357 | 0.1798076  |
| Lancaster et al. (2005)b | 0.000  | 0.9012816  |
| Lancaster et al. (2005)c | -0.317 | 0.2552408  |

One value above 1 and one below  $-1$  were identified as influential cases; neither is an outlier, so all outcomes are retained.

## 11.5. Forest Plot



## 11.6. Asymmetry Test

**Note:** the number of studies in this window is too small to reliably test for small-study effects (minimum  $k = 10$ ). No Egger's test or trim-and-fill is performed.

## 12. Th1/Th2 Ratio - 17 h Post-Exercise

### 12.1. Data Import

### 12.2. Primary Meta-Analysis

Kostrzewa-Nowak et al. (2019) is removed before pooling and the model re-fitted.

#### 12.2.1. {meta}

Number of studies:  $k = 11$

|                           | SMD     | 95% CI            | t    |
|---------------------------|---------|-------------------|------|
| Random effects model (HK) | 0.6433  | [-0.0265; 1.3131] | 2.14 |
| Prediction interval       |         | [-2.4114; 3.6980] |      |
|                           | p-value |                   |      |
| Random effects model (HK) | 0.0580  |                   |      |
| Prediction interval       |         |                   |      |

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 1.7065$  [0.1105; 3.8540];  $\tau = 1.3063$  [0.3324; 1.9632]  
 $I^2 = 73.5\%$  [51.6%; 85.5%];  $H = 1.94$  [1.44; 2.62]

Test of heterogeneity:

| Q     | d.f. | p-value  |
|-------|------|----------|
| 37.71 | 10   | < 0.0001 |

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model (df = 10)
- Prediction interval based on t-distribution (df = 10)

#### 12.2.2. {metafor}

Random-Effects Model ( $k = 11$ ;  $\tau^2$  estimator: HE)

$\tau^2$  (estimated amount of total heterogeneity): 1.7065 (SE = 1.0895)  
 $\tau$  (square root of estimated  $\tau^2$  value): 1.3063

## 12. Th1/Th2 Ratio - 17 h Post-Exercise

I<sup>2</sup> (total heterogeneity / total variability): 95.92%  
H<sup>2</sup> (total variability / sampling variability): 24.49

Test for Heterogeneity:  
Q(df = 10) = 37.7068, p-val < .0001

Model Results:

| estimate | se     | tval   | df | pval   | ci.lb   | ci.ub  |
|----------|--------|--------|----|--------|---------|--------|
| 0.6433   | 0.3006 | 2.1401 | 10 | 0.0580 | -0.0265 | 1.3131 |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

### 12.2.3. Adapted Model {meta}

Number of studies: k = 10

|                           | SMD     | 95% CI            | t    |
|---------------------------|---------|-------------------|------|
| Random effects model (HK) | 0.5129  | [ 0.0830; 0.9427] | 2.70 |
| Prediction interval       |         | [-0.8512; 1.8770] |      |
|                           | p-value |                   |      |
| Random effects model (HK) | 0.0244  |                   |      |
| Prediction interval       |         |                   |      |

Quantifying heterogeneity (with 95% CIs):  
tau<sup>2</sup> = 0.3238 [0.0746; 1.2164]; tau = 0.5690 [0.2731; 1.1029]  
I<sup>2</sup> = 72.7% [48.5%; 85.6%]; H = 1.92 [1.39; 2.63]

Test of heterogeneity:  
Q d.f. p-value  
33.01 9 0.0001

Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for tau<sup>2</sup>
- Q-Profile method for confidence interval of tau<sup>2</sup> and tau
- Calculation of I<sup>2</sup> based on Q
- Hartung-Knapp adjustment for random effects model (df = 9)
- Prediction interval based on t-distribution (df = 9)

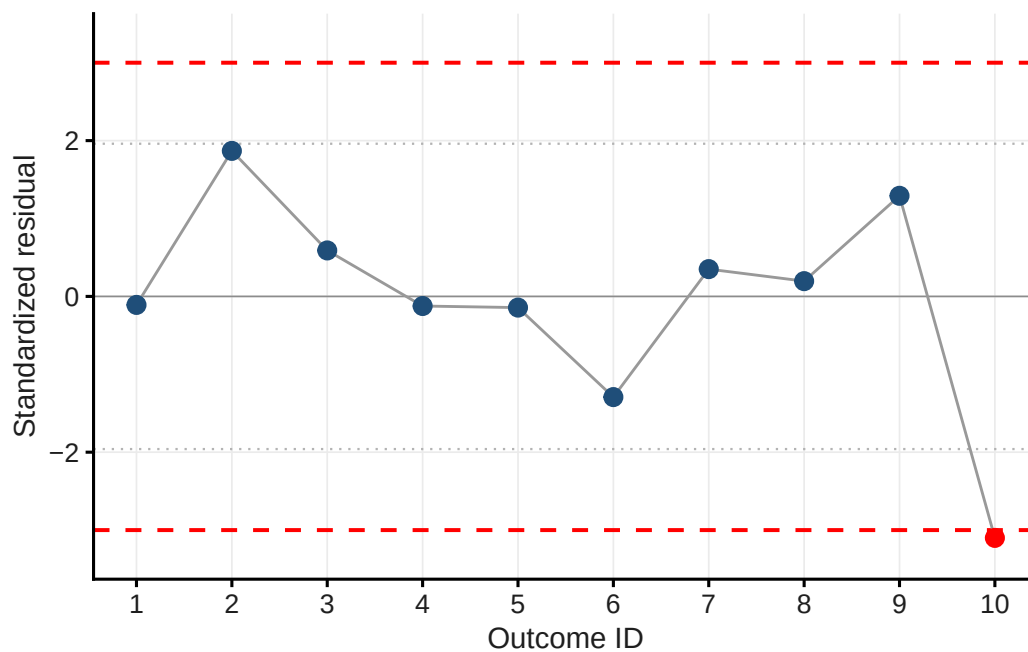
## 12.3. Outlier Detection

Table 12.1.: Standardised residuals

| Short_reference                 | g     | std_res |
|---------------------------------|-------|---------|
| Kostrzewa-Nowak & Nowak (2018)a | 0.458 | -0.109  |
| Kostrzewa-Nowak & Nowak (2018)b | 1.463 | 1.868   |

**Table 12.1.:** Standardised residuals

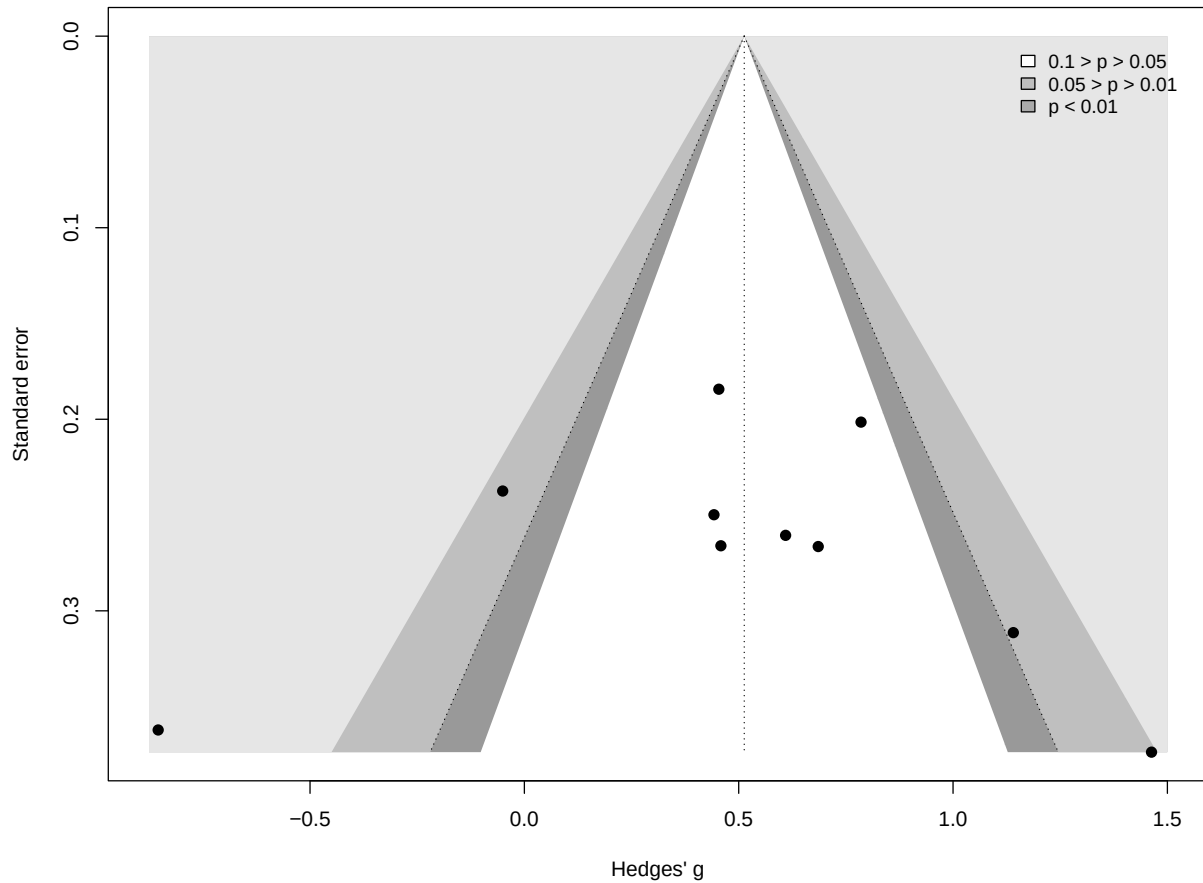
| Short_reference                  | g      | std_res |
|----------------------------------|--------|---------|
| Kostrzewa-Nowak & Nowak (2020a)a | 0.785  | 0.590   |
| Kostrzewa-Nowak & Nowak (2020a)b | 0.454  | -0.123  |
| Kostrzewa-Nowak & Nowak (2020b)a | 0.442  | -0.144  |
| Kostrzewa-Nowak & Nowak (2020b)b | -0.051 | -1.292  |
| Kostrzewa-Nowak & Nowak (2020b)c | 0.685  | 0.350   |
| Kostrzewa-Nowak & Nowak (2020b)d | 0.609  | 0.196   |
| Kostrzewa-Nowak & Nowak (2020b)e | 1.141  | 1.291   |
| Steensberg et al. (2001)         | -0.854 | -3.100  |



**i** Note

No outlier.

## 12.4. Funnel Plot & Influential Cases



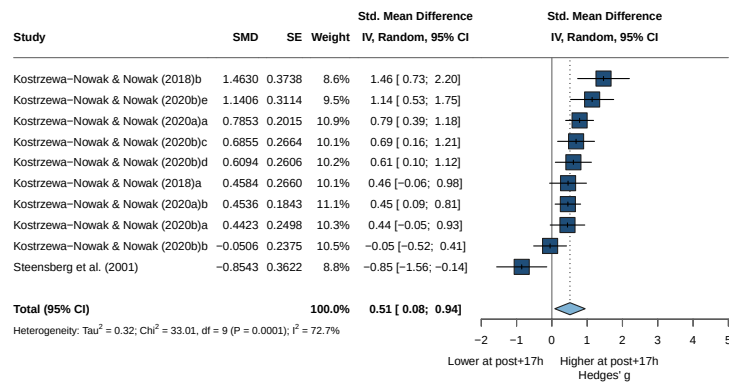
**Table 12.2.:** DFBETAS influence diagnostics

| Short_reference                    | g      | dfbetas     |
|------------------------------------|--------|-------------|
| Kostrzewska-Nowak & Nowak (2018)a  | 0.458  | -0.06601183 |
| Kostrzewska-Nowak & Nowak (2018)b  | 1.463  | 0.57178772  |
| Kostrzewska-Nowak & Nowak (2020a)a | 0.785  | 0.64171703  |
| Kostrzewska-Nowak & Nowak (2020a)b | 0.454  | -0.16812828 |
| Kostrzewska-Nowak & Nowak (2020b)a | 0.442  | -0.09893333 |
| Kostrzewska-Nowak & Nowak (2020b)b | -0.051 | -0.90092632 |
| Kostrzewska-Nowak & Nowak (2020b)c | 0.685  | 0.21588205  |
| Kostrzewska-Nowak & Nowak (2020b)d | 0.609  | 0.12756806  |
| Kostrzewska-Nowak & Nowak (2020b)e | 1.141  | 0.55656301  |
| Steensberg et al. (2001)           | -0.854 | -0.87675665 |

### **i** Note

No influential case.

## 12.5. Forest Plot



## 12.6. Asymmetry Test

Linear regression test of funnel plot asymmetry

Test result:  $t = 0.02$ ,  $df = 8$ ,  $p\text{-value} = 0.9847$

Bias estimate: 0.0606 (SE = 3.0606)

Details:

- multiplicative residual heterogeneity variance ( $\tau^2 = 4.1263$ )
- predictor: standard error
- weight: inverse variance
- reference: Egger et al. (1997), BMJ

### **i** Note

No asymmetry -> no trim-and-fill

## 12.7. Moderator Analyses

### 12.7.1. Risk of Bias

#### 12.7.1.1. Model

Number of studies:  $k = 10$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.3238$  [0.0746; 1.2164];  $\tau = 0.5690$  [0.2731; 1.1029]

$I^2 = 72.7\%$  [48.5%; 85.6%];  $H = 1.92$  [1.39; 2.63]

Test of heterogeneity:

| Q     | d.f. | p-value |
|-------|------|---------|
| 33.01 | 9    | 0.0001  |

Results for subgroups (random effects model (HK)):

|                | k | SMD    | 95% CI            | $\tau^2$ | $\tau$ |
|----------------|---|--------|-------------------|----------|--------|
| RoB = Moderate | 3 | 0.3550 | [-2.5088; 3.2188] | 1.2364   | 1.1119 |
| RoB = Serious  | 2 | 0.6095 | [-1.4940; 2.7129] | 0.0177   | 0.1331 |
| RoB = Low      | 5 | 0.5399 | [ 0.0105; 1.0692] | 0.1145   | 0.3384 |

|                | Q     | $I^2$ |
|----------------|-------|-------|
| RoB = Moderate | 20.08 | 90.0% |
| RoB = Serious  | 1.48  | 32.2% |
| RoB = Low      | 10.32 | 61.2% |

Test for subgroup differences (random effects model (HK)):

|                | Q    | d.f. | p-value |
|----------------|------|------|---------|
| Between groups | 0.19 | 2    | 0.9112  |

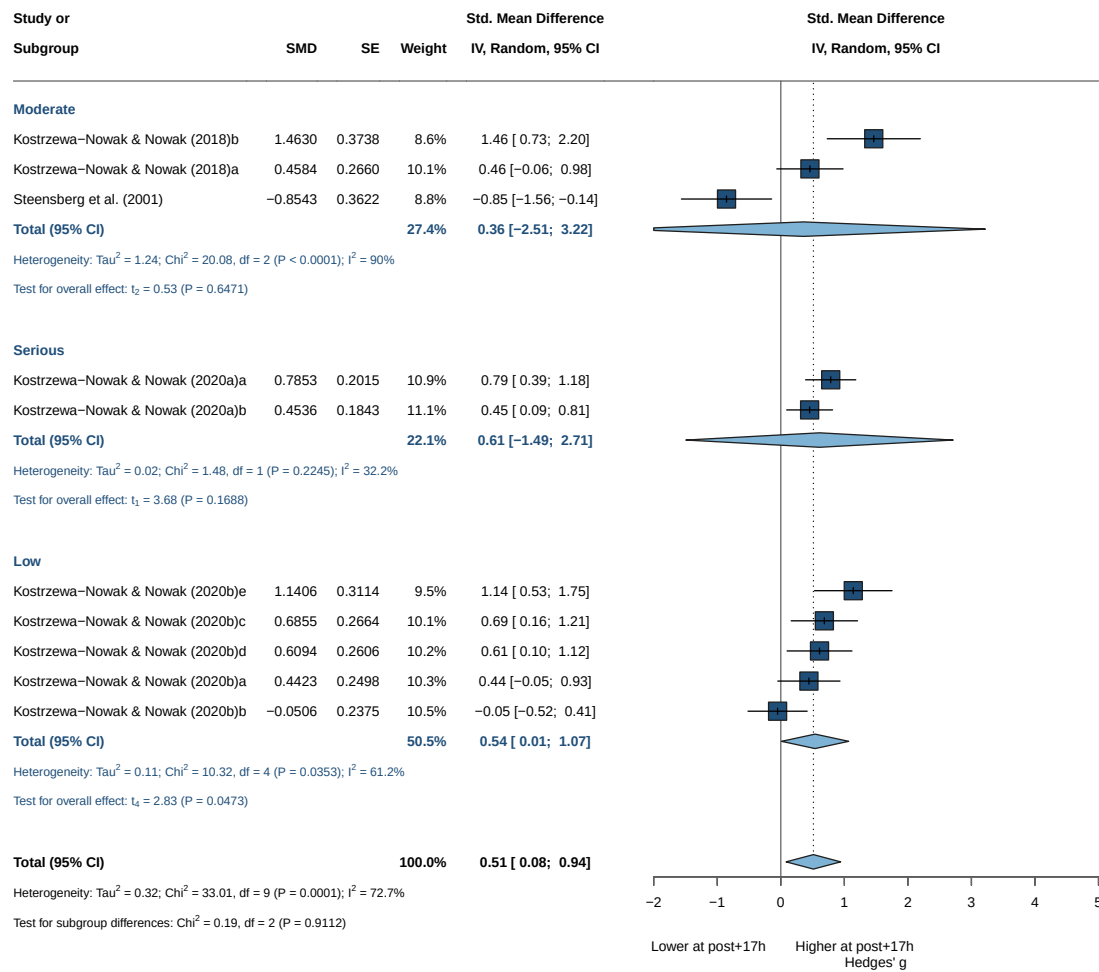
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

#### **i** Note

Not significant results

## 12.7.1.2. Forrest Plot



## 12.7.2. Intensity

## 12.7.2.1. Model

Number of studies:  $k = 10$

Quantifying heterogeneity (with 95% CIs):

$\tau^2 = 0.3238$  [0.0746; 1.2164];  $\tau = 0.5690$  [0.2731; 1.1029]

$I^2 = 72.7\%$  [48.5%; 85.6%];  $H = 1.92$  [1.39; 2.63]

Test of heterogeneity:

| Q     | d.f. | p-value |
|-------|------|---------|
| 33.01 | 9    | 0.0001  |

Results for subgroups (random effects model (HK)):

|                      | k      | SMD     | 95% CI             | $\tau^2$ |
|----------------------|--------|---------|--------------------|----------|
| intensity = Maximal  | 9      | 0.6254  | [ 0.3097; 0.9411]  | 0.1202   |
| intensity = Vigorous | 1      | -0.8543 | [-1.5641; -0.1444] | --       |
|                      | tau    | Q       | $I^2$              |          |
| intensity = Maximal  | 0.3466 | 18.06   | 55.7%              |          |

## 12. Th1/Th2 Ratio - 17 h Post-Exercise

intensity = Vigorous      -- 0.00      --

Test for subgroup differences (random effects model (HK)):

|                |       |      |         |
|----------------|-------|------|---------|
|                | Q     | d.f. | p-value |
| Between groups | 14.60 | 1    | 0.0001  |

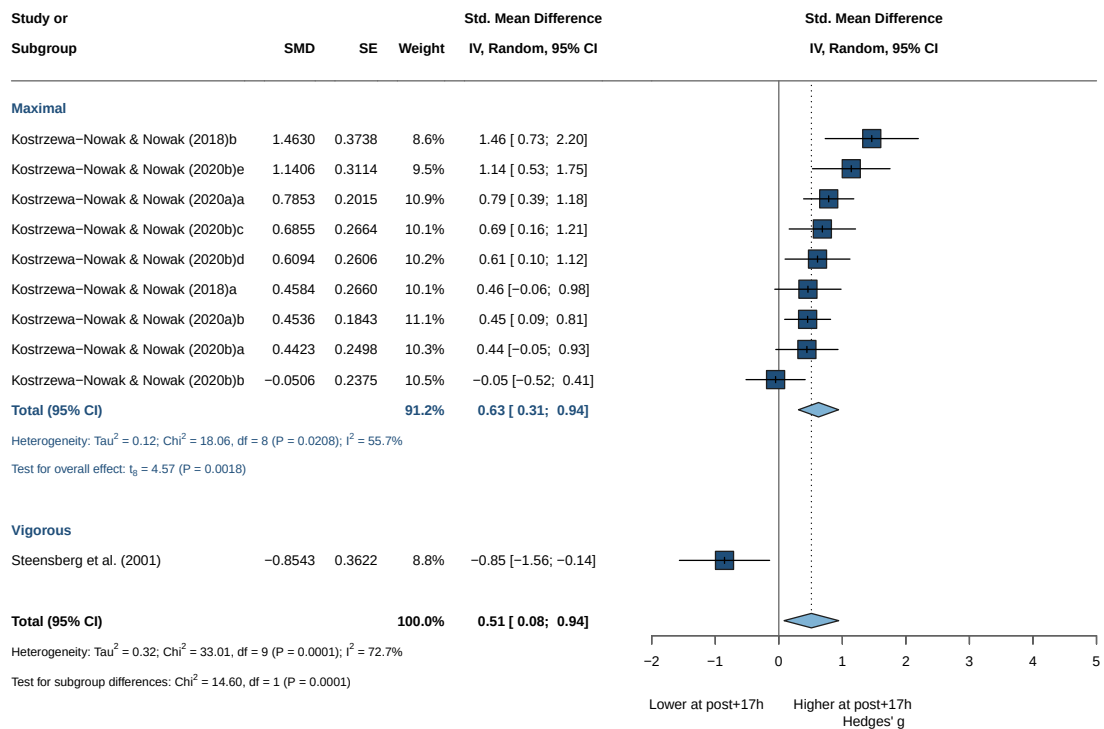
Details of meta-analysis methods:

- Inverse variance method
- Hedges estimator for  $\tau^2$
- Q-Profile method for confidence interval of  $\tau^2$  and  $\tau$
- Calculation of  $I^2$  based on Q
- Hartung-Knapp adjustment for random effects model
- Prediction interval based on t-distribution

### ! Important

Significant results but 10 vs 1

### 12.7.2.2. Forrest Plot



### 12.7.3. Continuous Moderators (Meta-Regression)

#### 12.7.3.1. Age

Mixed-Effects Model (k = 10; tau<sup>2</sup> estimator: HE)

|         |          |         |         |         |
|---------|----------|---------|---------|---------|
| logLik  | deviance | AIC     | BIC     | AICc    |
| -3.3225 | 14.8176  | 12.6449 | 13.5527 | 16.6449 |

|                                                                    |                      |
|--------------------------------------------------------------------|----------------------|
| tau <sup>2</sup> (estimated amount of residual heterogeneity):     | 0.0757 (SE = 0.0748) |
| tau (square root of estimated tau <sup>2</sup> value):             | 0.2751               |
| I <sup>2</sup> (residual heterogeneity / unaccounted variability): | 54.86%               |
| H <sup>2</sup> (unaccounted variability / sampling variability):   | 2.22                 |
| R <sup>2</sup> (amount of heterogeneity accounted for):            | 76.63%               |

Test for Residual Heterogeneity:

QE(df = 8) = 14.6115, p-val = 0.0672

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 8) = 12.7719, p-val = 0.0073

Model Results:

|         | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|---------|----|--------|---------|-------|
| intrcpt | 2.8912   | 0.6757 | 4.2791  | 8  | 0.0027 | 1.3331  |       |
| age     | -0.1264  | 0.0354 | -3.5738 | 8  | 0.0073 | -0.2079 |       |
| intrcpt | 4.4493   |        |         |    |        |         | **    |
| age     | -0.0448  |        |         |    |        |         | **    |

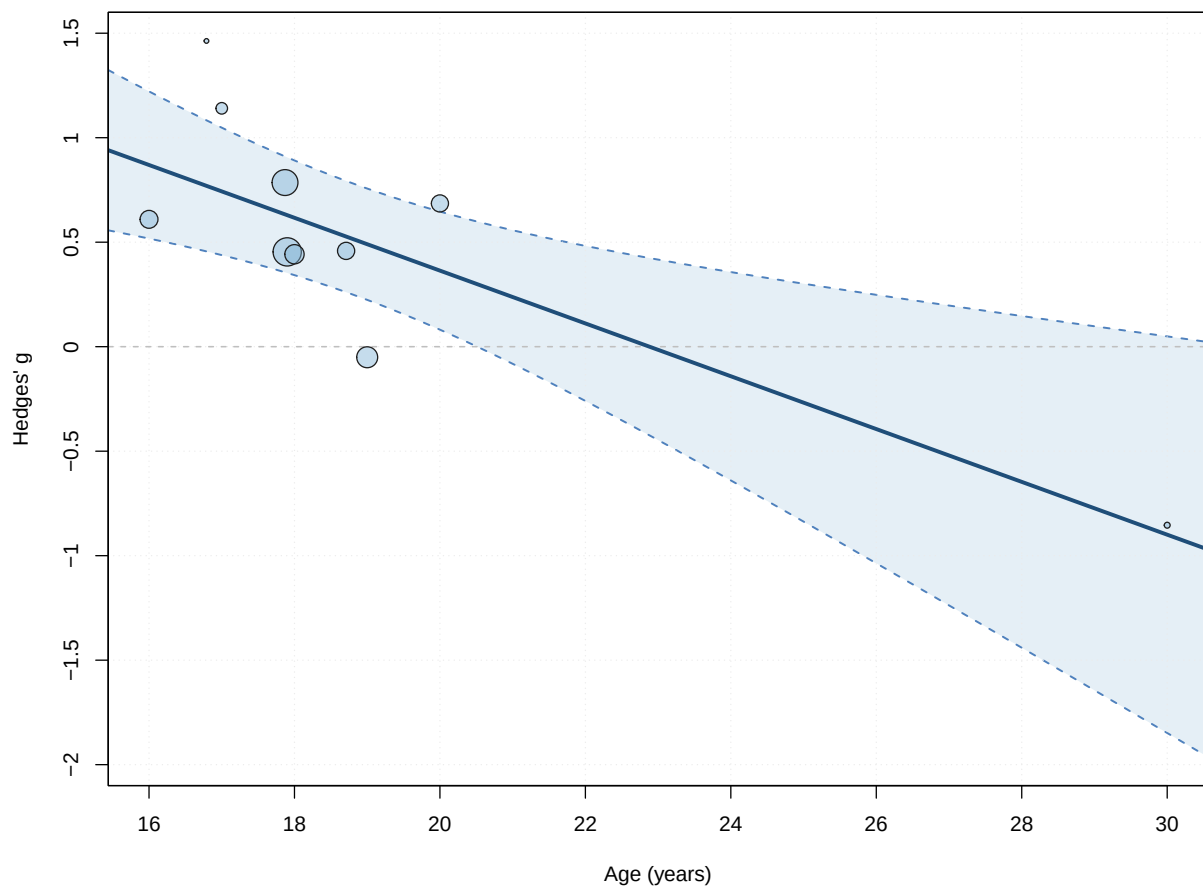
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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

## 12. Th1/Th2 Ratio - 17 h Post-Exercise



### 12.7.3.2. BMI

Mixed-Effects Model (k = 9; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -2.9847 | 13.9489  | 11.9695 | 12.5612 | 16.7695 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.0868 (SE = 0.0866)  
tau (square root of estimated tau<sup>2</sup> value): 0.2947  
I<sup>2</sup> (residual heterogeneity / unaccounted variability): 57.84%  
H<sup>2</sup> (unaccounted variability / sampling variability): 2.37  
R<sup>2</sup> (amount of heterogeneity accounted for): 27.72%

Test for Residual Heterogeneity:

QE(df = 7) = 12.7253, p-val = 0.0791

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 7) = 2.7310, p-val = 0.1424

Model Results:

|         | estimate | se     | tval    | df | pval   | ci.lb   |
|---------|----------|--------|---------|----|--------|---------|
| intrcpt | -3.4303  | 2.4532 | -1.3983 | 7  | 0.2047 | -9.2312 |
| bmi     | 0.1920   | 0.1162 | 1.6526  | 7  | 0.1424 | -0.0827 |
|         | ci.ub    |        |         |    |        |         |
| intrcpt | 2.3706   |        |         |    |        |         |
| bmi     | 0.4668   |        |         |    |        |         |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

**12.7.3.3. Duration**Mixed-Effects Model (k = 8; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -4.6365 | 14.5351  | 15.2729 | 15.5113 | 21.2729 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.1642 (SE = 0.1421)tau (square root of estimated tau<sup>2</sup> value): 0.4052I<sup>2</sup> (residual heterogeneity / unaccounted variability): 68.62%H<sup>2</sup> (unaccounted variability / sampling variability): 3.19R<sup>2</sup> (amount of heterogeneity accounted for): 60.58%

Test for Residual Heterogeneity:

QE(df = 6) = 16.5428, p-val = 0.0111

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 6) = 7.1231, p-val = 0.0371

Model Results:

|          | estimate | se     | tval    | df | pval   | ci.lb   |
|----------|----------|--------|---------|----|--------|---------|
| intrcpt  | 0.8663   | 0.2241 | 3.8659  | 6  | 0.0083 | 0.3180  |
| duration | -0.0115  | 0.0043 | -2.6689 | 6  | 0.0371 | -0.0220 |
|          | ci.ub    |        |         |    |        |         |
| intrcpt  | 1.4146   | **     |         |    |        |         |
| duration | -0.0010  | *      |         |    |        |         |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

**12.7.3.4. Dose**Mixed-Effects Model (k = 8; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC     | BIC     | AICc    |
|---------|----------|---------|---------|---------|
| -4.6365 | 14.5351  | 15.2729 | 15.5113 | 21.2729 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.1642 (SE = 0.1421)  
 tau (square root of estimated tau<sup>2</sup> value): 0.4052  
 I<sup>2</sup> (residual heterogeneity / unaccounted variability): 68.62%  
 H<sup>2</sup> (unaccounted variability / sampling variability): 3.19  
 R<sup>2</sup> (amount of heterogeneity accounted for): 60.58%

Test for Residual Heterogeneity:

QE(df = 6) = 16.5428, p-val = 0.0111

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 6) = 7.1231, p-val = 0.0371

Model Results:

|         | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|---------|----|--------|---------|-------|
| intrcpt | 0.9372   | 0.2421 | 3.8715  | 6  | 0.0083 | 0.3448  |       |
| dose    | -0.0017  | 0.0006 | -2.6689 | 6  | 0.0371 | -0.0033 |       |
| intrcpt | 1.5295   | **     |         |    |        |         |       |
| dose    | -0.0001  | *      |         |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

**12.7.3.5. MET****i** Note

Only modelled when the MET column is present; the current  $r = 0.5$  export does not include it, so this meta-regression is skipped.

**12.7.3.6. IFN**Mixed-Effects Model (k = 7; tau<sup>2</sup> estimator: HE)

| logLik | deviance | AIC | BIC | AICc |
|--------|----------|-----|-----|------|
|--------|----------|-----|-----|------|

-2.0723    11.1835    10.1447    9.9824    18.1447

tau<sup>2</sup> (estimated amount of residual heterogeneity):    0.0848 (SE = 0.0943)  
 tau (square root of estimated tau<sup>2</sup> value):    0.2913  
 I<sup>2</sup> (residual heterogeneity / unaccounted variability): 58.34%  
 H<sup>2</sup> (unaccounted variability / sampling variability):    2.40  
 R<sup>2</sup> (amount of heterogeneity accounted for):    0.00%

Test for Residual Heterogeneity:

QE(df = 5) = 10.9775, p-val = 0.0518

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.4201, p-val = 0.5455

Model Results:

|         | estimate | se     | tval    | df | pval   | ci.lb   |
|---------|----------|--------|---------|----|--------|---------|
| intrcpt | 0.8400   | 0.4512 | 1.8617  | 5  | 0.1217 | -0.3198 |
| IFN     | -0.0933  | 0.1440 | -0.6481 | 5  | 0.5455 | -0.4634 |
|         | ci.ub    |        |         |    |        |         |
| intrcpt | 1.9998   |        |         |    |        |         |
| IFN     | 0.2768   |        |         |    |        |         |

---

Signif. codes:    0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

#### Note

Not significant

### 12.7.3.7. IL\_4

Mixed-Effects Model (k = 7; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC    | BIC    | AICc    |
|---------|----------|--------|--------|---------|
| -1.4940 | 10.0268  | 8.9880 | 8.8258 | 16.9880 |

tau<sup>2</sup> (estimated amount of residual heterogeneity):    0.0657 (SE = 0.0832)  
 tau (square root of estimated tau<sup>2</sup> value):    0.2563  
 I<sup>2</sup> (residual heterogeneity / unaccounted variability): 52.41%  
 H<sup>2</sup> (unaccounted variability / sampling variability):    2.10  
 R<sup>2</sup> (amount of heterogeneity accounted for):    8.99%

Test for Residual Heterogeneity:

QE(df = 5) = 8.9170, p-val = 0.1124

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 1.4107, p-val = 0.2883

## 12. Th1/Th2 Ratio - 17 h Post-Exercise

Model Results:

|         | estimate | se     | tval   | df | pval   | ci.lb   |
|---------|----------|--------|--------|----|--------|---------|
| intrcpt | 0.4280   | 0.1697 | 2.5220 | 5  | 0.0530 | -0.0083 |
| IL_4    | 0.2553   | 0.2149 | 1.1877 | 5  | 0.2883 | -0.2972 |
|         | ci.ub    |        |        |    |        |         |
| intrcpt | 0.8643   | .      |        |    |        |         |
| IL_4    | 0.8077   |        |        |    |        |         |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

### **i** Note

Not significant

### 12.7.3.8. IL\_6

Mixed-Effects Model (k = 9; tau<sup>2</sup> estimator: HE)

| logLik  | deviance | AIC    | BIC    | AICc    |
|---------|----------|--------|--------|---------|
| -0.3049 | 8.5892   | 6.6098 | 7.2014 | 11.4098 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.0027 (SE = 0.0374)  
tau (square root of estimated tau<sup>2</sup> value): 0.0522  
I<sup>2</sup> (residual heterogeneity / unaccounted variability): 4.44%  
H<sup>2</sup> (unaccounted variability / sampling variability): 1.05  
R<sup>2</sup> (amount of heterogeneity accounted for): 97.74%

Test for Residual Heterogeneity:

QE(df = 7) = 8.6202, p-val = 0.2811

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 7) = 7.8742, p-val = 0.0263

Model Results:

|         | estimate | se     | tval   | df | pval   | ci.lb  |
|---------|----------|--------|--------|----|--------|--------|
| intrcpt | 0.3614   | 0.1208 | 2.9922 | 7  | 0.0202 | 0.0758 |
| IL_6    | 0.1441   | 0.0514 | 2.8061 | 7  | 0.0263 | 0.0227 |
|         | ci.ub    |        |        |    |        |        |
| intrcpt | 0.6469   | *      |        |    |        |        |
| IL_6    | 0.2655   | *      |        |    |        |        |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**!** Important

Significant

### 12.7.3.9. IL\_10

Mixed-Effects Model (k = 9; tau<sup>2</sup> estimator: HE)

|         |          |         |         |         |
|---------|----------|---------|---------|---------|
| logLik  | deviance | AIC     | BIC     | AICc    |
| -4.3135 | 16.6064  | 14.6270 | 15.2186 | 19.4270 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.1340 (SE = 0.1107)  
 tau (square root of estimated tau<sup>2</sup> value): 0.3661  
 I<sup>2</sup> (residual heterogeneity / unaccounted variability): 68.68%  
 H<sup>2</sup> (unaccounted variability / sampling variability): 3.19  
 R<sup>2</sup> (amount of heterogeneity accounted for): 0.00%

Test for Residual Heterogeneity:

QE(df = 7) = 17.5453, p-val = 0.0142

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 7) = 0.3559, p-val = 0.5696

Model Results:

|         | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|---------|----|--------|---------|-------|
| intrcpt | 0.7042   | 0.1924 | 3.6597  | 7  | 0.0081 | 0.2492  |       |
| IL_10   | -0.0754  | 0.1263 | -0.5966 | 7  | 0.5696 | -0.3741 |       |
| intrcpt | 1.1592   |        |         |    |        |         | **    |
| IL_10   | 0.2233   |        |         |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

### 12.7.3.10. IL\_2

Mixed-Effects Model (k = 7; tau<sup>2</sup> estimator: HE)

|         |          |         |         |         |
|---------|----------|---------|---------|---------|
| logLik  | deviance | AIC     | BIC     | AICc    |
| -2.1646 | 11.3680  | 10.3292 | 10.1669 | 18.3292 |

tau<sup>2</sup> (estimated amount of residual heterogeneity): 0.0869 (SE = 0.0971)

## 12. Th1/Th2 Ratio - 17 h Post-Exercise

```
tau (square root of estimated tau^2 value):      0.2947
I^2 (residual heterogeneity / unaccounted variability): 58.04%
H^2 (unaccounted variability / sampling variability): 2.38
R^2 (amount of heterogeneity accounted for):      0.00%
```

Test for Residual Heterogeneity:

QE(df = 5) = 11.4142, p-val = 0.0438

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 5) = 0.2537, p-val = 0.6359

Model Results:

|         | estimate | se     | tval    | df | pval   | ci.lb   | ci.ub |
|---------|----------|--------|---------|----|--------|---------|-------|
| intrcpt | 0.6698   | 0.2567 | 2.6093  | 5  | 0.0477 | 0.0099  |       |
| IL_2    | -0.0585  | 0.1162 | -0.5037 | 5  | 0.6359 | -0.3573 |       |
| intrcpt | 1.3296   |        |         |    |        |         | *     |
| IL_2    | 0.2402   |        |         |    |        |         |       |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

**i** Note

Not significant

## 12.8. Multivariate Models

Mixed-Effects Model (k = 8; tau^2 estimator: HE)

```
tau^2 (estimated amount of residual heterogeneity): 0.1642 (SE = 0.1421)
tau (square root of estimated tau^2 value):        0.4052
I^2 (residual heterogeneity / unaccounted variability): 68.62%
H^2 (unaccounted variability / sampling variability): 3.19
R^2 (amount of heterogeneity accounted for):        60.58%
```

Test for Residual Heterogeneity:

QE(df = 6) = 16.5428, p-val = 0.0111

Test of Moderators (coefficient 2):

F(df1 = 1, df2 = 6) = 7.1231, p-val = 0.0371

Model Results:

|                   | estimate | se     | tval    | df | pval   |
|-------------------|----------|--------|---------|----|--------|
| intrcpt           | 0.6369   | 0.1810 | 3.5193  | 6  | 0.0125 |
| intensityVigorous | -1.4912  | 0.5587 | -2.6689 | 6  | 0.0371 |

|                   | ci.lb   | ci.ub   |   |
|-------------------|---------|---------|---|
| intrcpt           | 0.1941  | 1.0797  | * |
| intensityVigorous | -2.8583 | -0.1240 | * |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

! Important

Model significant but no independent predictors



# References

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